



ANALYSIS AND DESIGN FOR ONLINE APPOINTMENT MANAGEMENT IN CAMARINES SUR POLYTECHNIC COLLEGES ADMISSION OFFICE

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College of Computer Studies
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for the degree Bachelor of Science in Information Systems

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APPROVAL SHEET

In partial fulfillment of the requirements for the degree of **Bachelor of Science in Information System**, this research entitled **Analysis and Design for Online Appointment Management in Camarines Sur Polytechnic Colleges Admission Office** prepared and submitted by **Renela Rose A. Garol, Mary Joy E. Tuyay, and Pablo R. Talagtag Jr.** has been examined and is recommended for approval and acceptance.

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DEDICATION

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ABSTRACT

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An Online Appointment Management is a software application to well organize the admission process for educational institutions such as schools, colleges, and universities. The Camarines Sur Polytechnic Colleges Admission Office already has an existing process where applicants will submit application form and schedule entrance examination. This study was conducted to analyze and design Online Appointment Management in Camarines Sur Polytechnic Colleges Admission Office. In general, the objective is to design user interface of online appointment management and database. To address the objectives of the study, the researchers used a mixed method research design and the design thinking framework. Data were obtained through interviews and survey questionnaires. The study respondents were the students of Camarines Sur Polytechnic Colleges and administrators in Admission Office. The data was evaluated and interpreted using a weighted mean. Due to problems encountered with the time consuming, duplication of applicant's name, lack of automation in appointment confirmations, delays and errors because of high volume of applicants and complex application process. To address this, the study designed the user interface and Database. Thus, the overall interpretation revealed that the designed user interface and database satisfied the respondents. As a result, designing of the User Interface and Database will speed up the appointment process, avoid duplication of applicant's name and provide better experience for applicants and admission staffs.



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Chapter 1

INTRODUCTION

This section covers the project context, purpose and description of the project, significance of the study, scope and limitation of the project, and the project dictionary which contains terms that will be used all throughout the study.

Project Context

In an era of technology innovation and digital transformation, educational institutions are embracing novel approaches to improve administrative procedures and provide seamless services to students and stakeholders. An educational institution's admission office is critical in administering the process of admitting new students. This procedure necessitates dealing with a huge number of prospective students, parents, and other stakeholders. Appointment management at admission offices has traditionally been a manual and time-consuming task, resulting in inefficiencies and potential schedule conflicts. The researchers want to design an online appointment management system to address these difficulties and improve the entire admission process.

An online appointment management is a software application designed to streamline the admission process for educational institutions such as schools, colleges, and universities. The system allows students and parents to schedule appointments with school staff and counselors for admission related queries and meetings.

Adoption of online appointment management has emerged as a possible alternative in response to these issues. An efficient online appointment management that leverages the power of technology can alter the way admission offices handle appointments, empowering both staff and prospective students.

The goal of this analysis and design project is to investigate the many aspects of adopting an online appointment management in an admission office. By identifying existing bottlenecks and imagining an ideal digital infrastructure, we hope to streamline the entire appointment process, resulting in a better student experience, increased staff productivity, and increased overall institutional efficiency.

The project's main objective is to do away with the outdated, time-consuming, and ineffective process of appointment booking approach. Educational institutions can improve the overall admission experience by establishing an online appointment management that provides students and parents with a more flexible and easy way to arrange appointments with staff members.

The basic features of an online appointment management include a central dashboard for staff to manage appointments, an online booking interface for students and parents to make appointments and the ability to reschedule or cancel appointments. On appointment data, including appointment volume, average wait time, and transaction numbers availability, the designed system can also offer analytics and reports. By using this information, admission procedures may be made better, bottlenecks can be found, and personnel levels can be made more effective.

In general, an online appointment management for admission can assist educational institutions in streamlining their admissions procedure, increasing client satisfaction, and lessening the administrative strain on staff.

Purpose and Description of the Project

The purpose of this project is to analyze and design an online appointment management in Camarines Sur Polytechnic Colleges Admission Office for incoming students. An Online Appointment Management could be a useful tool for managing appointments between students

and admission officers. Online appointment management for the Camarines Sur Polytechnic Colleges Admission Office project aims to facilitate appointments between incoming students and admission employees.

This project is to help the students and employees in Camarines Sur Polytechnic Colleges Admission Office by analyzing and designing an online appointment management, an admission office can streamline the appointment scheduling process and improve the overall efficiency of their operations. It can also provide a better experience for students by allowing them to book appointments at their convenience, and reducing the likelihood of missed appointments. The proposed low-fidelity prototype may incorporate various features including online appointment, calendar management for admission officers, and automated reminders for students on the waitlist, and reporting and analytics capabilities for generating appointment-related data.

The study's overarching goal is to organize the timing of the entrance exam for new students. The project has the following benefits: Accuracy, which enables data entry by the user; Error-prone data entry costs can be decreased; Convenience, which saves time and money by reducing the number of visits to the admission office; Better monitoring, which has real-time admission progress; Easy report generating enables managers to quickly gather data. The project's ability to adapt makes them more useful because it allows them to serve various clients. This project will aid the admission office in carrying out their duties effectively, decrease personnel costs, address pertinent scheduling process concerns, and produce precise and timely reports.

Objectives of the Project

The main objective of this project is to analyze and design Online Appointment Management in CSPC Admission Office to make it easy for the students who want to have transactions online.

Specifically, this project aims to achieve the following:



1. To determine the problems of the existing process in Online Application for all incoming students in terms of:
 - 1.1. Appointment
 - 1.2. Receiving
 - 1.3. Releasing of results
2. To design the structure of User Interface and database in terms of the following:
 - 2.1. Appointment
 - 2.2. Receiving
 - 2.3. Releasing of results
3. To evaluate the design of User Interface based on Nielsen's Heuristic Evaluation:
 - 3.1. Visibility of system status
 - 3.2. Match between system and the real world
 - 3.3. User control and freedom
 - 3.4. Consistency and standards
 - 3.5. Recognition rather than recall
 - 3.6. Flexibility and efficiency of use
 - 3.7. Aesthetic and minimalist design
4. To evaluate the Database design using:
 - 4.1. Data integrity
 - 4.2. Normalization
 - 4.3. Scalability

Significance of the Study

The project entitled Analysis and Design for Online Appointment Management in Camarines Sur Polytechnic Colleges Admission Office can be considered significant since this project can improve the admission process. The success of this planning can help to improve the

current process to be more flexible in terms of transaction changes. This study is through to benefits the following:

School Administration. The system helps the admission office in the existing problems in the processing of the online application.

Applicants. The applicant can easily go to the admission office and have transactions with their appointed numbers for them to easily assist the employees.

Researchers. The study shall provide extra new and additional knowledge that will guide the researchers in the Analysis and Design.

Future Researchers. This provides for future researchers with data, findings, and new inputs to develop future researchers of the related study.

Scope and Limitations of the Project

The study focused on analysis and design for online appointment management in Camarines Sur Polytechnic Colleges Admission Office. Furthermore, this study is limited based on specific objectives to determine the problems of existing process in Online Appointment Management for all incoming students in terms of Appointment, Receiving, and Releasing of results, to design the structure of User Interface and Database in terms of Appointment, Receiving, and Releasing of results and lastly is to evaluate the design of user interface and database. The scope of this study is to eliminate transactional conflicts in the admission of Camarines Sur Polytechnic Colleges, and its focus will be on online appointments made using online services. The suggested strategy will save people time and effort by eliminating excessive waiting times and providing the simplest means of protecting student personal information. As well as giving prospective students a quick and simple way to make appointments with the admissions personnel. Without having to phone or email the admissions office, this system would enable students to examine available appointment times and select a time that works for them.

This project is limited only in designing the structure of User Interface and Database. It will focus on scheduling an appointment in the Admission Office at Camarines Sur Polytechnic Colleges. The transaction of examination for job applicants is not included in this study. However, these transactions can serve as a foundation for potential improvements in the Student Testing and Admission Office. The analysis and design for online appointment management in the Camarines Sur Polytechnic Colleges Admission Office are determined by specific requirements and limitations.

Project Dictionary

This serves as a comprehensive reference, providing concise definitions and explanations of the key terms, concepts related to a capstone project.

Online Appointment Management. A type of digital solution that allows citizens to book, schedule, or cancel their appointments through web-based systems [1]. Students and staff can also use this in any devices to make it easy to monitor and manage appointments.

Admission Office. An office within a community college or State university responsible for recruiting and communicating with new and transfer students [2]. It is responsible for managing the applications, and enrollment process of the students.

Calendar Management. The act of assigning and balancing CLM activity, start and end dates and times [3]. It is a process of organizing and scheduling tasks, appointments within a specified timeframe using digital or physical tools.

Automated Reminders. A simple way to ease the overloaded to-do lists of people's busy lives. These scheduled electronic notifications arrive by e-mail, SMS (text message), phone or fax [4]. In this study, it refers to pre-scheduled that notify or message students to remind them about upcoming schedules.

Analytics. A field of computer science that uses math, statistics, and machine learning to find



meaningful patterns in data [5]. It is a systematic process that collects, interprets, processes and derives insights from data to gain understanding.

Technical issues. An issue arising regarding technical matters and, as used herein, can mean a Standard Issue, a High Priority Issue or a Critical Issue, as applicable [6]. It disrupts the intended operation of a system, often causing errors, delays, software malfunctions, bugs, connectivity problems or even failures in tasks or processes.

Privacy. The right to have some control over how your personal information is collected and used [7]. It involves the ability to maintain confidentiality and limiting their accessibility and visibility to authorized individuals.



Notes

- [1] Nicole Tennyson. 2023. The advantages of using online scheduling software for Government Offices. (February 2023). Retrieved October 3, 2023 from <https://qlless.com/the-advantages-of-using-online-scheduling-software-for-government-offices/>
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Chapter 2

REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter presents the studies and literature, both local and international that are related to the research. Related articles and ideas from other researchers are included in this chapter to guide the researchers to create ideas, procedures, and processes that might help the institution improve its existing system.

Related Literature and Systems

This chapter presents a review of related literature on Analysis and Design for Online Appointment Management in Camarines Sur Polytechnic Colleges Admission Office. The review of related literature is an essential component of academic research, providing a comprehensive overview of existing scholarly works relevant to the study's subject matter. The researchers will evaluate the important findings of research related to Online Appointment Management.

Online Appointment and Scheduling Management

Online appointment scheduling tools are growing common in educational institutions all around the world. With a host of advantages for both the organization and the users, these technologies have completely changed how appointments are scheduled and managed. In order to give a general overview of the online appointment management system for schools, this essay will integrate international and local literature.

Online appointment management solutions can increase the effectiveness and efficiency of appointment scheduling. According to the report, these solutions can cut down on the amount of time needed to plan an appointment and minimize any potential errors. The study also discovered that online appointment management solutions can improve parent participation by improving communication between schools and parents [9].

Online appointment management can make more appointment slots available and make the appointment booking procedure easier. According to the study, these solutions can lighten the

administrative staff's workload and increase convenience for parents by letting them make appointments on their smartphones or PCs [10].

Online appointment management solutions have become more common in the Philippines in recent years. The time and effort needed to plan appointments in schools can be greatly reduced by using online appointment management systems. According to the study, these solutions can increase appointment scheduling's general effectiveness, increasing both parents' and school staff members' happiness [11].

Online appointment management can boost parents' involvement in schools, according to a different study. The study found that these methods can give parents a more practical means of making appointments, leading to higher attendance rates for school gatherings and events [12].

Record Management

Record management in online appointments for schools is a vital step that assists schools in efficiently managing appointments. Several research have been conducted on this subject, providing insights on how schools might properly maintain their data in online appointment systems. Organizations are required by law to keep and manage records in line with applicable laws and regulations. Compliance with legal obligations, such as data protection, retention periods, and document deletion rules, is ensured through effective record management. The necessity of appropriate record keeping in meeting legal and regulatory duties, avoiding penalties, and safeguarding the organization's reputation is emphasized by researchers [20].

One study focused on the implementation of an online appointment system in a university setting. According to the study, the technology increased the efficiency of the appointment procedure and decreased the frequency of no-shows. The authors also proposed that the technology be used to manage records more effectively by allowing administrators to track appointments and provide reports [5].



Another study explored the implementation of an online appointment system in a school setting. The findings of the study, the approach improved communication between teachers and parents, cut appointment wait times, and improved the overall appointment experience. The researchers also mentioned that the technology may be used to manage records more effectively by providing a centralized database of appointments [6].

In 2020, during the COVID-19 pandemic, many schools implemented online appointment systems to manage appointments remotely. A study explored the impact of these systems on the appointment experience. The results of the study, online appointment systems are successful for managing appointments remotely, nevertheless there are some obstacles connected to technological issues and scheduling conflicts. The authors proposed that in the long run, schools should explore using online appointment systems to improve efficiency and convenience [7].

In a recent study, the researchers looked at how an online appointment system affected the appointment experience in a collegiate context. According to the study, the approach boosted appointment efficiency, reduced wait times, and raised student satisfaction. The authors also proposed that the technology be used to manage records more effectively by providing a consolidated database of appointments and allowing managers to track appointment histories [8].

Second, online appointment scheduling enhances communication between the admissions staff and prospective students. Appointment management systems can be used by admission offices to send automated emails and notifications to students, presenting them with vital information about the admission process and reminding them of impending appointments. This increases engagement and communication, both of which are critical for recruiting and maintaining students.

Lastly, it provides admission offices with a competitive advantage. Admission offices can separate themselves from other universities and improve their overall reputation by implementing

online appointment management solutions. This is especially significant in competitive markets where schools must stand out in order to attract the top pupils.

Efficient Appointment Scheduling

The article "Design and implementation of online lecturer appointment system for student projects" (n.d.) involves creating a digital platform that enables students to schedule appointments with their professors for guidance and support on their projects. This system streamlines the appointment booking process, enhances communication between students and lecturers, and facilitates effective project supervision. This concept is connected to the analysis and design for online appointment in the admission office in the sense that both projects focus on leveraging digital solutions to manage appointments efficiently within an educational institution. While the former targets student-lecturer interactions for project assistance, the latter aims to optimize appointment scheduling and management for school admissions, thereby enhancing overall administrative processes through online systems [1].

In the context of educational institutions, this innovative system addresses the challenges associated with traditional appointment booking methods. By introducing a digital solution, students can seamlessly schedule appointments with their professors, eliminating potential communication barriers and enhancing accessibility to guidance for their projects. The significance of this project is underscored by its potential to streamline administrative processes within academic environments. The implementation of an online appointment system contributes to the optimization of academic support services, specifically in the context of project supervision. This aligns with the broader trend of leveraging digital solutions to improve efficiency and effectiveness in educational institutions.

Drawing parallels with a related project focusing on the analysis and design of an online appointment system in the admission office, both initiatives share a common goal of harnessing



digital technologies to enhance administrative functions within educational settings. While the first project concentrates on student-lecturer interactions for project assistance, the latter aims to refine appointment scheduling and management in the context of school admissions. Both projects collectively contribute to the modernization of educational institutions through the integration of efficient and user-friendly online systems.

In addition, the article highlights the importance of digital platforms in improving student-lecturer interactions and overall administrative processes within educational institutions. The emphasis on online systems for appointment scheduling reflects a broader trend in the education sector towards adopting technology to enhance accessibility and streamline administrative workflows.

Admission Office

Admission offices are responsible for selecting students to attend an educational institution. They are the primary point of contact for prospective students and their families, and they provide information about the institution's academic programs, admission requirements, application procedures, financial aid, and student services. Admission offices play a critical role in shaping the institution's student body and ensuring that the admission process is fair and transparent [13]. The admissions office is critical to the school enrollment process. Schools are implementing creative tactics and technologies to improve productivity and the overall experience for prospective students and their families. Streamlining the admission office enables schools to implement a simplified application process. By utilizing online application platforms, applicants can conveniently submit their forms and required documents electronically. A study found that schools in the Philippines that implemented an online application system experienced a significant reduction in paperwork and manual data entry, leading to a more efficient and streamlined admission process [17]. Effective communication between prospective students, their families, and school management is facilitated by efficient admission office operations. Using

digital communication channels such as email and instant messaging enables for quick responses to enquiries and admission status updates. According to the findings of the study, streamlined communication in the admissions process has a beneficial impact on applicant satisfaction and enrollment rates [18]. Admissions offices require online appointment management for a variety of reasons. For starters, online appointment management streamlines and improves the admissions process. Prospective students can simply book appointments at their leisure using an online appointment system, eliminating the need to call or visit the admissions office. This decreases staff members' workload and allows them to focus on other critical responsibilities.

Improved User Experience for Admission Office

In the article "Appointment system", the idea and practice of effectively organizing appointments, which could be useful in a variety of circumstances. It can involve looking into how an online appointment system speeds the procedure for setting up appointments for potential students who are applying for admission, improving the general effectiveness and user experience of the school's admission office. The article can go into detail about how such a system might be used in the admissions process of a school, including its requirements, benefits, and design considerations [2]. A school's online appointment management often includes a user-friendly interface through which parents or students may log in, browse available time slots, and schedule appointments with instructors or other school personnel. Administrators may also be able to specify particular availability for each teacher, determine appointment durations, and manage the overall scheduling process through the system. Automated email or SMS notifications to remind participants of upcoming appointments, and the option to provide additional information or specific requirements when booking appointments are all features commonly found in a school appointment management. The system can improve communication and collaboration among parents, students, and school staff by simplifying meeting scheduling and ensuring that everyone is aware of their impending appointments [14]. Online appointment management solutions in

schools provide a centralized platform for students, parents, and teachers to schedule appointments. By eliminating the requirement for human appointment booking, these solutions reduce paperwork and administrative burdens. The use of an online appointment management in schools greatly decreased administrative workload, allowing administrators to better allocate their time and resources [15]. Traditional appointment scheduling methods usually include numerous phone calls and in-person contacts, which can lead to misunderstanding and confusion. Online appointment management solutions provide stakeholders with a constant communication channel. According to one study, online appointment management systems allowed clear and effective communication between parents, teachers, and administrators, reducing misunderstandings and ensuring all parties were notified about scheduled meetings [16].

User Experience (UX) and User Interface (UI) Design Relationship

The article titled "UX/UI Design of Online Learning Platforms and Their Impact on Learning" by Miya et al. [2022] likely delves into the relationship between user experience (UX) and user interface (UI) design and their influence on the efficacy of online learning platforms. It probably examines how well-designed interfaces, intuitive navigation, and engaging visual elements contribute to enhancing the learning experience and outcomes for users. When connecting this article to the context of "Analysis and Design for Online Appointment Management in Camarines Sur Polytechnic Colleges Admission Office," one could consider applying the insights from the UX/UI design article to optimize the appointment management system's user interface. By incorporating user-centered design principles and focusing on creating a seamless and user-friendly interface, the school's admission office could potentially improve the efficiency of the appointment scheduling process. This alignment between effective design and functional usability could result in a more positive interaction for students and parents engaging with the appointment management system, ultimately leading to enhanced user satisfaction and a more streamlined appointment management process [3].

Effective User Interface (UI) Design Principles

According to the article "Designing the User Interference" [2016], designing the User Interface focuses on principles and guidelines for creating effective and user-friendly interfaces in software applications. It addresses aspects such as user-centered design, visual design, usability, and interaction design. The principles from the user interface design article can be applied to ensure a seamless and intuitive experience for users interacting with the online appointment management system. This involves designing clear navigation, user-friendly forms, intuitive buttons, and visually appealing layouts to optimize the user experience during the school admission process. By incorporating these UI design principles, the online appointment management system can enhance user satisfaction, streamline processes, and improve overall efficiency in the school admission office [4].

Well Structured Database

According to Qaffas et al. [2012], the database would store student information, appointment schedules, available slots, and historical data, enabling efficient retrieval and analysis. By implementing a well-structured database, the school admission office can ensure accurate record-keeping, seamless appointment allocation, and insightful data-driven decision-making, contributing to a streamlined and effective admission process for prospective students [19]. Creating a well-structured database for online appointment management in an admission office is critical for successfully handling the influx of potential students and guaranteeing a smooth admissions process. The database should be structured by defining tables for various entities such as Students, Staff, Appointments, and Communications. To maintain data integrity and rapid retrieval, each table should have proper fields and associations. The students table, for example, should include fields such as student ID, name, contact information, and application status. To link each appointment to the associated individuals, the Appointments table should include fields for date, time, student ID, and staff ID. A Communications table, in addition, can

be used to manage any communication between the admissions office and students, letting staff keep a record of encounters, emails, or texts. On top of that, the database should have a user-friendly interface so that staff members can simply input and retrieve information. Through an easy-to-use interface, employees should be able to schedule appointments, examine student profiles, and issue notifications. Integrating tools such as email notifications and automatic reminders can improve appointment management efficiency.

Heuristic-based User Interface Evaluation

Heuristic evaluation is a technique that uses accepted principles to evaluate a user interface's (UI) usability; one well-known set of heuristics is Nielsen's [22]. These guiding concepts include important elements such as error prevention, user control, consistency, visibility of system state, alignment with the actual world, recognition over recollection, ease of use, attractive design, and assistance with error recovery and documentation. Ensuring the user interface provides consistent and precise feedback, as well as informing users about system status, is a crucial aspect of heuristic evaluation.

This builds user confidence. To further facilitate understanding and use, a user-friendly user interface should match users' mental models. As UI design has evolved, inclusive design principles have become increasingly important. The goal is to build interfaces that are usable by people with a variety of needs, including those who are disabled. Another interesting trend is using machine learning to customize user interfaces (UIs) according to user behavior and preferences, making the experience unique to the user.

Synthesis of the State-of-the-Art

Due to its reliance on manual procedures, the current Student Testing and Admission system is inefficient and causes delays. Through a review of local and worldwide literature on the advantages of online appointment scheduling tools, the suggested online appointment management system seeks to address these problems. These solutions improve communication

and lessen administrative burden, which makes the appointment process more effective and comfortable for both parents and school personnel.

It highlighted how crucial record keeping is to maintaining compliance with regulatory requirements, underscoring the need of record management in online appointments. Numerous studies show that putting online appointment systems in place during a pandemic like COVID-19 increases productivity, decreases wait times, and boosts user satisfaction in educational settings. Emphasis on Admission Offices highlights how important they are in forming an institution's student body. It has been demonstrated that putting online application tools in place greatly streamlines the admissions process by reducing paperwork and human data entry. Adding online appointment scheduling to the admissions process improves user experience overall, communication, and efficiency. In the context of online learning platforms, the relationship between User Interface (UI) and User Experience (UX) design is examined. By incorporating knowledge from this area into the online appointment management system's design, the user experience may be optimized, improving communication between parents and students.

Efficient User Interface (UI) Design Principles are recognized as essential for establishing a smooth and user-friendly online appointment scheduling system experience. The significance of having a well-organized database is stressed, with an emphasis on keeping track of appointments, student information, and historical data. Accurate record-keeping, effective appointment scheduling, and data-driven decision-making throughout the admissions process all depend on a well-designed database.

In addition, the synthesis emphasizes the shift from outdated process to an effective online appointment management, referencing both domestic and foreign research to bolster the system's advantages in terms of record management, communication, efficiency, and user experience in the context of educational institutions, especially admission offices.



Gap Bridged by the Study

The literature highlights several flaws in the current student testing and admission procedure, many of which can be avoided by installing an online appointment management at the Admission Office of Camarines Sur Polytechnic Colleges. Manual and time-consuming procedures, insufficient communication, the need for an improved user experience, a lack of efficiency and streamlined processes, the need for effective record management, adaptability to contemporary challenges, and the potential for gaining a competitive edge are all examples of gaps. Moreover, the proposed online appointment management system shows promise in filling these significant gaps in the literature. It is capable of automating and simplifying numerous operations, improving communication channels, improving user experience, and ensuring efficient and accurate record-keeping. Finally, these enhancements will help both the admissions office and prospective students.

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Chapter 3

TECHNICAL BACKGROUND

This chapter describes the research design, method, and statistical tools that will be used for the study. The framework for evaluating information systems used to complete the project will also be defined and illustrated.

Design Thinking Framework

In terms of system analysis and evaluation, this study will follow the framework illustrated in Figure 1.

Design thinking is a problem-solving technique that includes brainstorming, prototyping, and testing, as well as a thorough grasp of the user's wants and desires. It is a person-centered approach that values empathy, experimentation, and iteration [1]. The benefit of applying design thinking in the Analysis and Design for Online Appointment Management in Camarines Sur Polytechnic Colleges Admission Office is that it puts the needs of the user first. It is critical to understand the needs of the students, parents, and staff who will use an online appointment management while building one. This can be accomplished by user interviews, monitoring users in their environment, or developing personas. You may begin to develop a system that satisfies the needs of the user once you have a thorough grasp of them.

Figure 1, further presents components of the framework to be adopted.

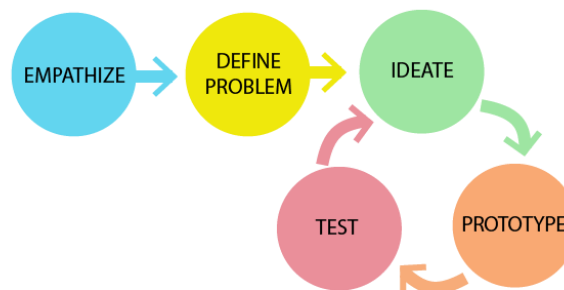


Figure 1: **Design Thinking Frameworks**

Empathize. Conduct research to gain insight into what your users do, say, think, and feel. The idea is to collect enough data to actually begin to sympathize with your users and their points of view [2]. Designers can obtain a deeper knowledge of the system's users' requirements, challenges, and expectations by empathizing with them, especially students and admission personnel. This method can assist guarantee that the strategic plan is oriented on the users and tailored to the specific needs and experiences of admissions personnel and students, resulting in a more successful and effective system.

Define. Combine all of your findings and look for areas where your users are having issues. Begin to emphasize chances for innovation as you identify your users' demands. To gain insights, use the data acquired during the empathy phase [2]. The defined phase of design thinking plays an important role in the Analysis and Design for Online Appointment Management because it entails specifying precise goals and objectives for the plan. This step entails conducting research and obtaining data on the current process, its strengths and limitations, and the demands of clients, which may include conducting face to face interview, questionnaires, or chat conversations with students and admission personnel to identify particular areas for improvement.

Ideate. Brainstorm a variety of wild, innovative solutions to satisfy the unmet user needs found during the define phase. Allow yourself and your team complete freedom; no concept is too far-fetched, and quantity trumps quality [2]. The ideate stage plays a vital role in attaining the objectives of the Analysis and Design for Online Appointment Management because it allows members to produce multiple solutions according to the demands and experiences of customers, allowing for creative thinking and the generation of multiple ideas. This stage allows for a wide range of potential solutions to be explored before proceeding on to the prototyping phase, making it an important step in the design thinking process.

Prototype. Create physical, tactile representations of a subset of your ideas. The purpose of this phase is to figure out which parts of your ideas work and which don't. Through feedback on your

prototypes, you begin to analyze the effect vs. feasibility of your ideas in this phase [2].

Prototyping has significance in the Analysis and Design for Online Appointment Management because it allows for early testing and refinement of the recommended solution, reducing both resources and time. It also enables the creation of numerous solutions for testing at the same time, giving stakeholders a greater idea of how the solution will work in practice and improving their support for its deployment.

Test. Test your prototypes with real people to gather feedback and uncover any issues [2].

Testing ensures that the solution that was created through design thinking is successful and fits the requirements of the customers. It enables criticism and every iteration, resulting in the final solution that is more probable to achieve the objectives of the Analysis and Design for Online Appointment Management in CSPC Admission Office

Material and Statistical Tools

Detailed information on the research design and research method in reference to the objectives of the study will be discussed in this section.

Research Design

Anchored from the general and specific objectives of this study, the researchers will employ mix method research design. As defined by Tegan George [2021] Mixed methods research combines elements of quantitative research and qualitative research in order to answer the research question, Mixed methods can help gain a more complete picture than a standalone quantitative or qualitative study, as it integrates benefits of both methods. Mixed methods research is often used in the behavioral, health, and social sciences, especially in multidisciplinary settings and complex situational or societal research. This research design is applicable to this study as it specifies objectives that suggest both descriptive and quantitative forms of research. The study will also try to assess the level of performance of existing and newly designed systems in the record section in terms of speed and accuracy which are both quantitative and descriptive

data. Further, this study sought to compare the existing and new system in the record section and determine its significant difference in terms of speed and accuracy.

To implement this mix method approach, the researcher will conduct an interview to the students and staff of the admission office on the problems encountered in terms of document storage and retrieval, and on the evaluation of the existing online application on its level of performance in terms of speed and accuracy. An interview is employed on objectives 1, to determine the problems of the existing process of Online Appointment process for incoming students, the on objective 2, to design the structure of user interface and database. A researcher's questionnaire will be prepared for this purpose. After the development of the document repository system, another set of researcher-made questionnaires will be given to the students and staff of the admission office to evaluate the design of user interface and database. Lastly, the result on the level of performance of the existing and new document repository system will be analyzed statistically to determine its significant difference.

Research Method

This section will discuss the detailed implementation of the mix method approach to realize the objectives of this study.

Instrument

This subsection, introduces the instrument to be used by the researchers.

Interview Guide. A structured set of questions or topics designed to help interviewers gather specific information from interviewees. It serves as a framework for conducting interviews in a systematic and consistent manner. The goal of assessing the existing appointment management process in Admission office.

Survey Questionnaire. The survey questionnaire will be used to address the third specific objectives of this study to evaluate the low-fidelity prototype of user interface using Visibility of system status, Match between system and the real world, User control and freedom, Consistency

and standards, Recognition rather than recall, Flexibility and efficiency of use, Aesthetic and minimalist design, and for Database Design using Data integrity, Normalization and Scalability

Likert Scale. Researchers also used a Likert scale type of questionnaire in this study. Likert scale involves presenting a series of statements or questions and asking respondents to indicate their level of agreement or disagreement on a predefined scale. The Likert scale typically ranges from "Strongly Disagree" to "Strongly Agree," with several intermediate points, allowing for a nuanced assessment of respondents' attitudes.

The table 1 shows the description of the 5 Point Likert Scale and the interpretation of the user's level of agreement of the User Interface and Database Design. This table 1 shows the corresponding description from the highest to lowest level of agreement.

Table 1

5-POINT LIKERT SCALE FOR EVALUATING USER INTERFACE AND DATABASE DESIGN

Scale	Range	Level of Agreement	Description
5	4.21 - 5.00	Strongly Agree	The participant strongly supports or agrees with the statement that the User Interface and Database is well-designed
4	3.21 - 4.20	Agree	Implies a positive stance toward the statement or question. The respondent agrees with the presented User Interface and Database Design.
3	2.61 - 3.20	Neutral	The respondent has neither a positive response nor a negative response, but undecided denotes a state of confusion of the respondent.
2	1.81 - 2.60	Disagree	Suggests a level of disagreement with the statement or question, but not as strong as Strongly Disagree.
1	1.00 - 1.80	Strongly Disagree	Indicates a strong and definitive disagreement with the statement or question. The respondent strongly opposes or disagrees with the presented User Interface and Database Design.

The 5-point Likert Scale that was used by respondents who were students and staff members in Admission Office is shown in Table 1. This will be used to evaluate the objective 2 which is to design the user interface and database design. After evaluation, calculated weighted mean will demonstrate whether the designed prototype.

Data Gathering Procedure

This section introduces the data gathering procedure that will be conducted by the researchers.

Interview. An interview is a question-and-answer session where one person asks questions, and the other person answers those questions. It can be a one-on-one, two-way conversation, or there can be more than one interviewer and more than one participant. The interview is the most important part of the whole selection process. It is used to decide if a person should be interviewed further, hired, or taken out of consideration. It is the main way to learn more about applicants and the basis for judging their job-related knowledge, skills, and abilities. [3]. Researchers can collect enormous amounts of data from a diverse population via interview in a relatively short period of time. Researchers can gather in-depth and complicated information about staff members' and potential students' experiences and perspectives, including their perceptions of the advantages and disadvantages of using the current process or online application, as well as their suggestions for improvement, through interviews.

When conducting interviews, it's important to ensure that the interview process is conducted in a professional and ethical manner, with informed consent obtained from all participants and confidentiality and privacy maintained throughout the data collection and analysis process. The interview questions should be carefully designed to explore the research questions and objectives, and they should be phrased in a way that is clear and understandable to all participants.

Survey Testing. Survey testing is the context of software testing. The data will be collected from the users or customers who have interacted with the current process and this will be used to answer the third specific objective of the study. This includes questions about their experience and opinions related to the software's several aspects such as functionality, performance, user interface, and more.

Survey testing is an important tool in the analysis and design of online appointment management. It provides direct input from potential users to developers and designers, allowing them to adjust the system to user preferences, tweak features, and assure optimal performance. The resulting online appointment management is more likely to give a seamless and gratifying experience by incorporating user opinions. As technology advances, incorporating survey testing into the design process will remain a critical technique for developing online appointment management that effectively caters to user demands and contributes to enhanced service delivery across industries.

Sampling Technique

A convenience sample strategy will be utilized to collect comments and insights for the Analysis and Design at Student Testing and Admission. Convenience Sampling is a non-probability sampling strategy in which units are chosen for inclusion in the sample because they are the most accessible to the researchers. This could be because of geographical proximity, availability at a specific time, or willingness to engage in the research. Convenience sampling, sometimes known as accidental sampling, is a sort of non-random sampling [4]. This strategy provides quick access to participants within the organization, such as students and administrative staff. Data will be gathered through interviews. The researchers will choose 25 students in Camarines Sur Polytechnic Colleges and 5 staff of the Admission Office based on their availability and desire to engage, ensuring that the method is realistic and practicable. To find patterns and themes in the collected data, thematic analysis and weighted mean will be

performed. While convenience sampling has limits, it is a convenient and fast technique to collect vital information for enhancing the strategic plan.

Statistical Test

The researchers will utilize the following statistical measures to analyze the gathered data:

Thematic Analysis. is an approach to examining qualitative data where the researcher actively engages with the data, seeking to identify meaningful patterns and themes. This method places the researcher's personal perspective and interpretation at the forefront of the data analysis process, emphasizing their subjective involvement in making sense of the information [5].

Using thematic analysis helps the researchers to know the patterns in the data that are important or interesting, provides ample room for flexibility in interpreting data and offers a convenient way to manage extensive datasets by categorizing them into overarching themes.

Weighted Mean. This method was used to get the average response of the respondents to the evaluated significant difference in acceptability between the proposed and existing appointment management. To determine measure statistical software would be used.

$$WM = \frac{FW}{n}$$

Where:

WM = Weighted Mean

W = Weighted Category of Response

F = Frequency of number of respondents

n = number of responses

Notes

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Chapter 4

RESULTS AND DISCUSSIONS

This section shows the discussion of results according to identified specific objectives, integrated in each phase in accordance with the selected framework, the Design Thinking Framework. Further, it covers different phases as illustrations to support the entire process of the designed system. It will also cover the screen captured photos of user interface and database design.

Empathize

The researchers examine the present practices and systems in-depth during this phase, emphasizing the importance of understanding the complexities and pain points of the current admissions process. Researchers strive to comprehend and appreciate the difficulties that the admissions office staff and the candidates experience, and this compassionate approach informs all we do. Researchers' dedication to changing the admissions process into one that is more focused on the needs of the students and less taxing for the committed staff is driven by this empathic viewpoint. In order to improve the entire experience for each candidate and make their road to higher education easier to navigate, it is not just our intention to spot inefficiencies but also to proactively suggest remedies.

Researchers acknowledge the critical significance of empathy for the people participating in the admission process at the core of this research and design phase. This stage offers a priceless chance to actively listen to the worries, annoyances, and requirements of the admission office staff, who work tirelessly to assist potential students, as well as the students themselves, who have goals to secure their future via education. The goal is to develop a solution that fills the gaps, speeds the procedure, and eventually promotes a more amicable connection between these two crucial groups. Researchers are intimately aware of the stresses experienced by both sides. The unwavering dedication to empathy in this project ensures that this study not only addresses

the technical issues but also the human side of the application process, all the while working to develop a solution that enhances the lives of those involved and helps to build a more inclusive and accessible educational system.

Gantt Chart

One of the most widely used and effective tools for displaying activities (tasks or events) against time is a Gantt chart, which is frequently used in project management. A list of the activities is located on the left side of the chart, and an appropriate time scale is located along the top. Every activity is symbolized by a bar, and the position and length of the bar indicate the activity's beginning, middle, and finish dates.

Figure 2 shows the graphical presentation of the schedule time and duration for each task to be done. Gantt chart used as a references for scheduling tasks that must be completed and submitted on time. It depicts the researcher's objectives for the period allocated which must be met to meet the criteria.

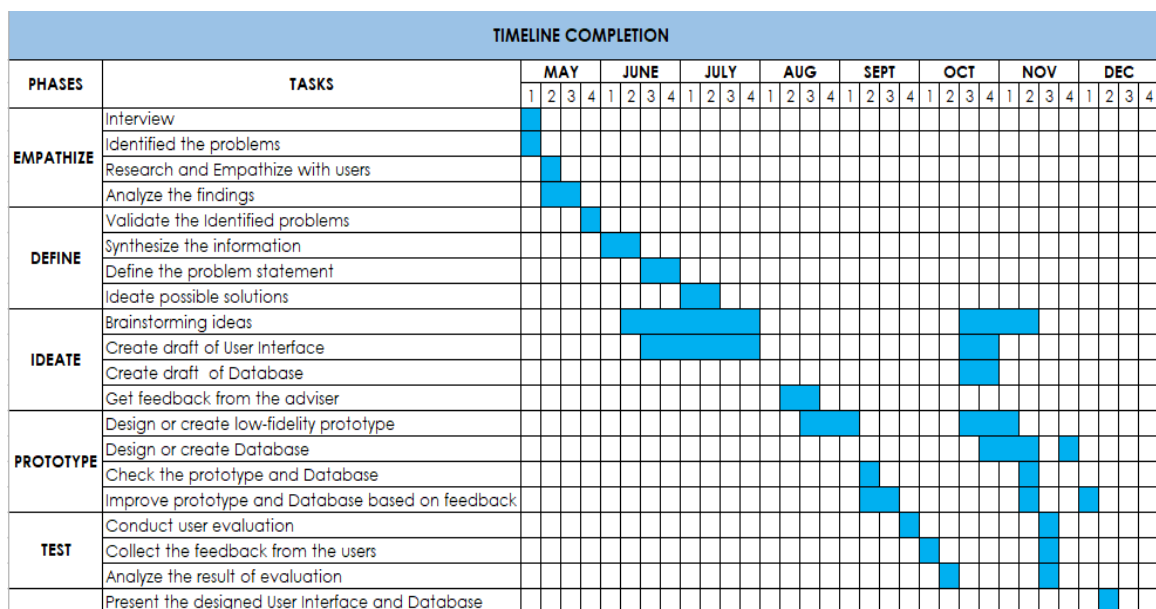


Figure 2: **Gantt Chart**

The researchers use a Gantt chart as a vital tool to guarantee the careful administration and prompt completion of their projects. This graphic depiction functions as a schedule and

framework for actions that are essential to the effective completion of the research projects, in addition to acting as a roadmap. The Gantt chart offers a thorough summary of the researcher's goal, providing a precise timetable for the allotted duration and outlining certain checkpoints that need to be met in order to comply with the online appointment management requirements.

The Gantt chart provides a dynamic view of the project's path with its visual representation of tasks and deadlines. It is useful not only for monitoring advancement but also for anticipating problems and proactively identifying bottlenecks that could disrupt the normal course of research operations. The Gantt chart transforms into a dynamic and adaptive tool that enables researchers to make well-informed judgments, modify timetables as needed, and take the initiative in project management by giving a visual depiction of the complexities of the project.

The Gantt chart also acts as a communication tool for stakeholders, collaborators, and team members. It is an efficient way to communicate to all parties involved the project's scope, timetable, and important milestones because of its clarity and comprehensibility. This encourages team members to work together and create synergy by fostering a common understanding and commitment to the project's objectives.

Define

In this phase, it shows the table and interpretation of the identified problems that the researchers gathered during the interview to achieve the specific objective 1: To determine the problems of the existing process in Online Application for all incoming students. These are the identified problems and also the proposed solutions in every identified problem that the respondents encountered.

Table 2 presents the identified problems that the researchers gathered during the interview in Admission Office, this table also presents the description of the identified problems and the proposed solutions in all the identified problems in Camarines Sur Polytechnic Colleges Admission Office.

Table 2

IDENTIFIED PROBLEMS AND PROPOSED SOLUTIONS

Identified Problems	Description	Proposed Solutions
Time consuming	Delays in appointment scheduling, efficiency reduced	Design an easy-to-use online appointment scheduling system that simplifies the booking process for applicants and automates scheduling tasks.
Duplication of Applicant's Name	Data inconsistencies, potential confusion	To resolve the issue of duplicate applicant data, we will set up a centralized database. This database will securely store applicant information and eliminate any unnecessary duplication.
Lack of automation in appointment confirmations	The risk of missed appointments arises when students are uncertain about their schedules due to a delay in administrators confirming them promptly.	Integrate email or SMS reminders into the system to reduce the need for manual follow-ups and confirmations.
Handling a High Volume of Applicants	The existing manual appointment process faces challenges in handling a high volume of applicants, leading to delays and the possibility of errors in the scheduling of appointments.	Design an automated appointment scheduling system capable of effectively handling a substantial influx of applicants. This system will assign appointment slots by considering both staff availability and applicant preferences, thereby reducing scheduling conflicts and minimizing delays.
Complex Application Process	The current application process is intricate and burdensome, resulting in confusion among applicants and potentially causing some to abandon the process.	Simplify the application process by designing a user-friendly online portal that features straightforward instructions and easy-to-navigate menus. Offer real-time assistance to applicants, incorporate a progress tracking feature, and ensure seamless uploading and verification of necessary documents directly within the system.

Table 1 highlights significant problems within the present appointment scheduling and application procedure that can have a negative impact on system efficiency and effectiveness. The first issue, time-consuming appointment scheduling that causes delays and lower efficiency, can be solved by using an easy-to-use online appointment scheduling system. This proposed solution

intends to streamline the booking process for applicants and automate scheduling chores, hence speeding up the process and lowering the possibility of human error.

Another major issue is applicant name duplication, which can result in data inconsistencies and misunderstanding. The proposed method entails creating a centralized database to securely store applicant information, hence eliminating duplicate data entries. This consolidated database not only increases data integrity, but it also simplifies applicant monitoring and management.

Furthermore, by integrating email or SMS reminders into the system, the absence of automation in appointment confirmations, which raises the danger of missed appointments and administrative delays, can be minimized. This automation eliminates the need for manual follow-ups and confirmations, ensuring that appointments are confirmed on time and reducing the likelihood of missed appointments.

The figure below shows the fishbone diagram, this figure shows the identified problems of the existing process in the Admission Office.

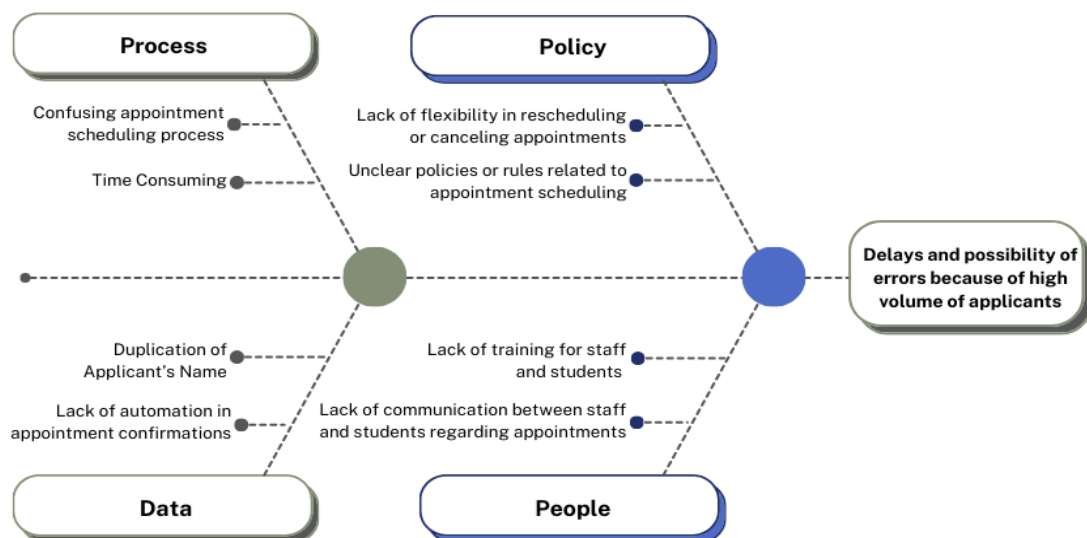


Figure 3: **Fishbone Diagram**

The fishbone diagram presented focuses on systematically identifying and categorizing challenges associated with managing a high volume of applicants that caused delays and errors within an online appointment management. Within the outlined categories, specific issues are brought to light. In the field of Process, issues include a Confusing Appointment Scheduling Process, where the lack of clarity leads to confusion among applicants, potentially resulting in errors, missed appointments, or delays. Additionally, the process is noted to be Time Consuming, posing a risk of frustration for both applicants and staff due to prolonged procedures.

Within the People category, challenges emerge from a Lack of Training for Staff and Students, emphasizing the importance of comprehensive training to mitigate errors, misunderstandings, and enhance overall efficiency. Another highlighted concern is the Lack of Communication between Staff and Students Regarding Appointments, indicating potential problems stemming from ineffective communication, such as missed appointments and dissatisfaction among applicants.

In terms of Policy, the identified problems involve Unclear Policies or Rules Related to Appointment Scheduling, introducing the risk of inconsistencies, disputes, and challenges in managing a substantial number of applicants. Additionally, the system's Lack of Flexibility in Rescheduling or Canceling Appointments is recognized as a potential source of frustration, hindering applicants' ability to manage their appointments effectively.

The Data category exposes issues like the Duplication of Applicant's Name, emphasizing the significance of data integrity to prevent confusion and maintain accurate record-keeping. Moreover, the Lack of Automation in Appointment Confirmations is highlighted as a challenge, suggesting that automating confirmation processes could enhance efficiency, alleviate staff burdens, and ensure timely and accurate communication. In an overall interpretation, the fishbone diagram serves as a visual representation of various challenges across the dimensions of process, people, policy, and data within the appointment management.

Ideate

The researchers used the design thinking framework in the early stages of the project to pioneer and create a low-fidelity prototype for an online appointment management specific to the demands of the Admission Office. The trip of the researchers began with a thorough examination of the needs of both Admission Office employees and potential users, which included students and administrators. The researchers got great insights into the issues people encountered and their objectives by conducting face to face interviews and chat conversations with administrators.

The figure below shows the flowchart of the current process for the admin interface of online application in Admission Office. This figure shows the step-by-step process from submitting the application form until releasing the results.

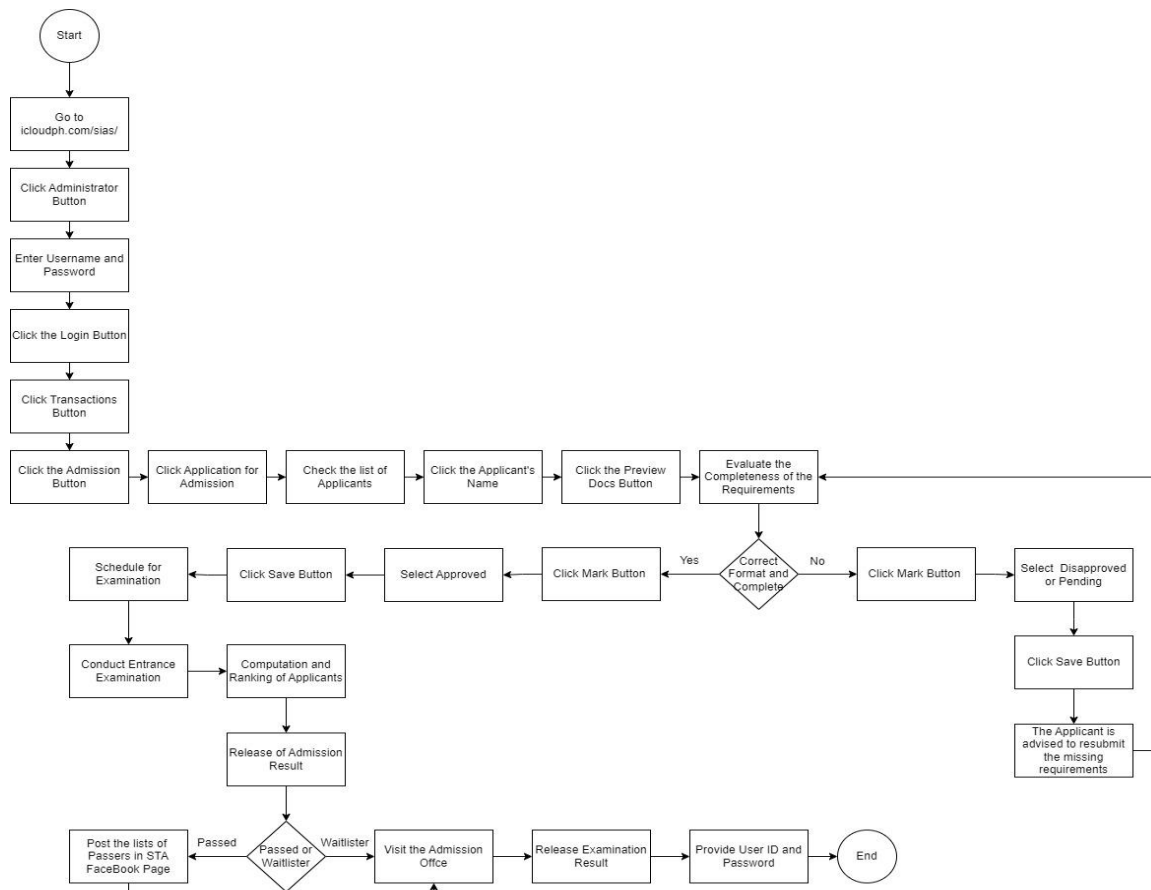


Figure 4: **Flowchart of Existing Process (Admin)**

Figure 4 shows the flowchart of the existing process for staff in the admission office. The outlined flowchart describes a systematic and structured admission process for prospective staff.

To begin the process, navigate the icloudph.com/sias/ online portal, a dedicated platform designed to streamline the admission procedures. Start by clicking on the administrator button, which will prompt you to enter your designated username and password for authentication. Once successfully logged in, proceed by clicking on the transactions button and then select the admission option.

Within the admission tab, you'll find the application for admission button. Click on it to access a comprehensive list of all submitted applications. Here, you have the opportunity to click on individual applicants' names to delve into the details of their submitted documents. For a closer inspection of each document, utilize the preview documents button.

After thoroughly reviewing all submitted documents, the next step is to mark the applications. Click on the mark button and choose from the options approved, disapproved, or pending to indicate your decision regarding each application. In the case of a disapproved or pending application, click the save button and applicants will receive guidance to resubmit the missing requirement to ensure a comprehensive and accurate evaluation.

If the applicants is approved, proceed to click the save button. The system will then automatically generate a schedule for the entrance examination, providing a fair and standardized means of assessing applicants. Following the entrance examination, the system will rank applicants based on their scores, facilitating a transparent and merit-based selection process.

After the examination results are compiled, administrators can post the lists of successful candidates on the official STA Facebook page, fostering transparency and keeping stakeholders informed. Successful applicants and waitlisters will be notified to visit the Admission Office to claim their results. During this visit, they will be required to provide their user ID and password for verification purposes, adding an extra layer of security to the process.

The figure below shows the flowchart of the current process for the applicant interface of online application in Admission Office. This figure shows the step-by-step process.

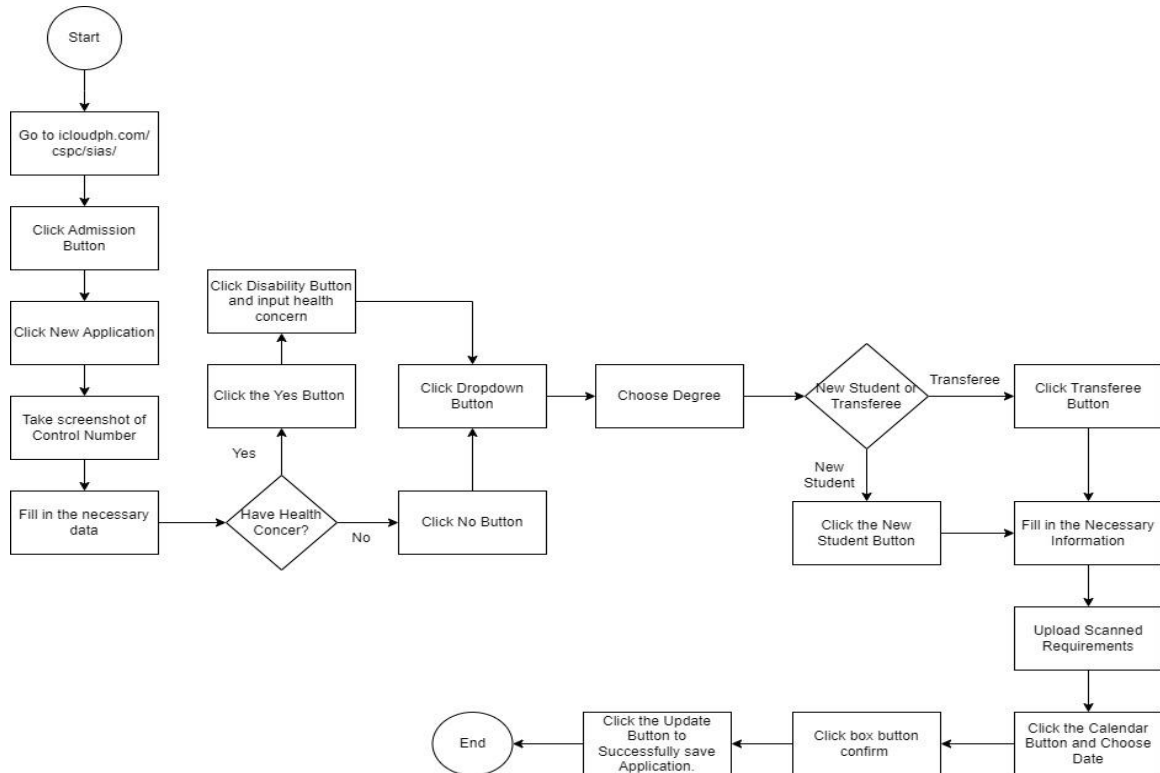


Figure 5: **Flowchart of Existing Process (Applicant)**

Figure 5 shows the flowchart of the existing process for student in the admission office. To begin the process, go to icloudph.com/cspc/sias/, prompting them to access the dedicated online portal. The applicant is guided to click admission button and then proceed to click new application, the user is then advised to take screenshot of control number and proceed to fill in the necessary data. For those with health concerns, click the yes button then click disability button. If the user does have a disability, they are further instructed to input health concern for proper accommodation and support. Alternatively, if there are no health concerns, the user is directed to click no button and click dropdown button to choose their desired degree. If the student is a new student, then click the new student button then fill in the necessary information. Subsequently, if the user is a transferee, then click transferee button and select click yes button to indicate their

status as a transferee student. Followed by to confirm their transferee status. Following this, the user is prompted to fill in the necessary information comprehensively. Once all required information is provided, they are instructed to upload scanned requirements for a thorough application. The user should click the calendar button to choose a suitable date for their entrance examination. After confirming the examination date, they should click the update button to lock in their selection, marking the conclusion of the application process and click save button.

. The next figure represents the flowchart of the proposed online appointment management process in CSPC Admission Office

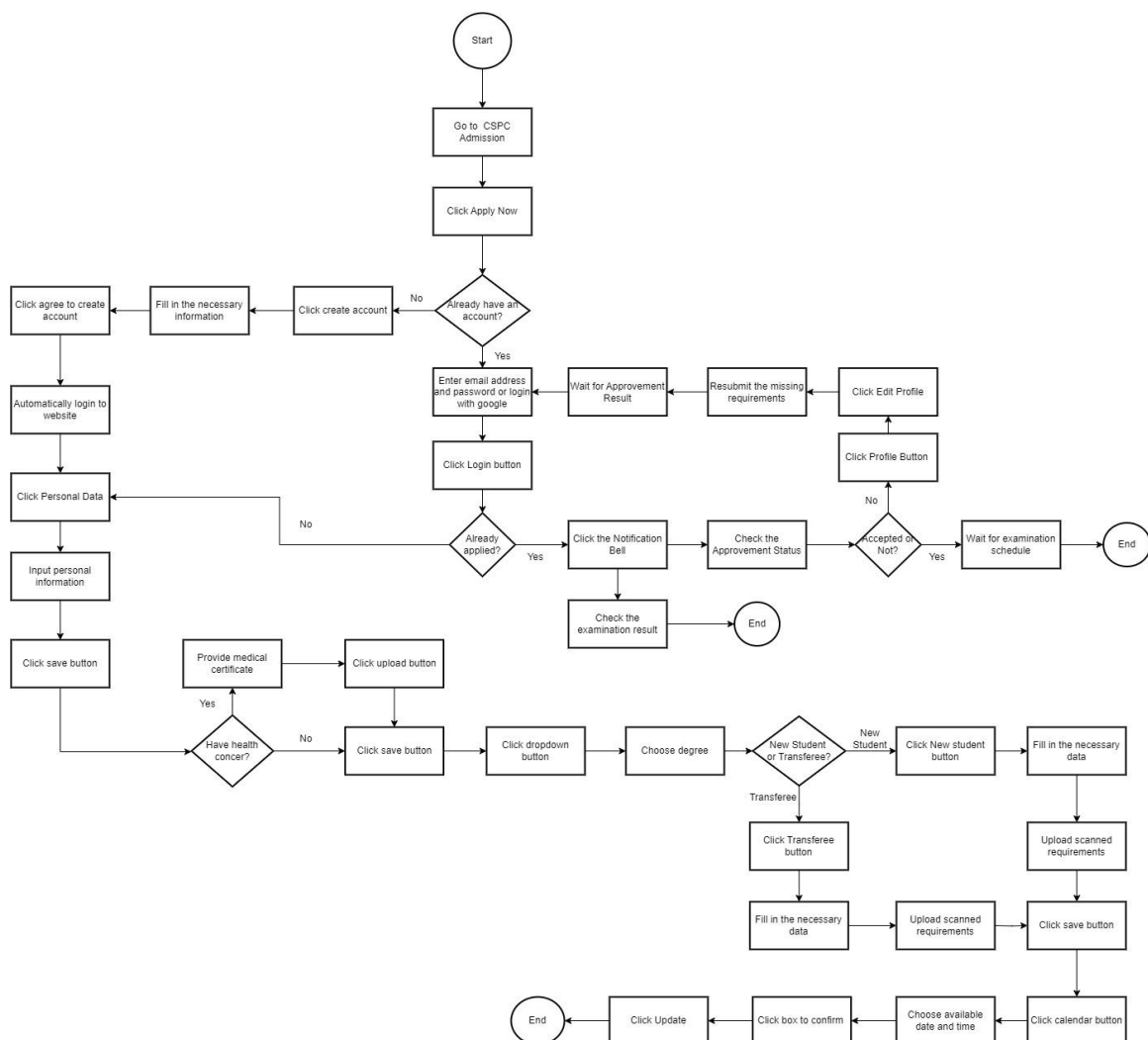


Figure 6: **Flowchart of Online Appointment Process (Applicant)**

Figure 6 show the process of students for Online Appointment Management for Camarines Sur Polytechnic Colleges wherein the applicants start by going to CSPC admission website then click apply now. If the applicant is new, click create account, and fill the necessary information, click agree to create account and it will automatically log in to website, click the personal data to fill in the personal information and click save button. For those with health concerns, provide a medical certificate and click upload button then save it. Alternatively, if there are no health concerns, the user is directed click drop down to choose their desired degree. If the student is a new, then click the new student button then fill in the necessary information. Subsequently, if the user is a transferee, then click transferee button and select click yes button to indicate their status as a transferee then fill in the necessary information. Once all required information is provided, they are instructed to upload scanned requirements for a thorough application. Continuing the process, the user should click the calendar button to choose a suitable date for their entrance examination. After confirming the examination date, they should click the update button to lock in their selection, marking the conclusion of the application process. To finalize the application, the user should click the update button to successfully save the application.

But, if the applicants already have an account, enter email address and password or log in with google. Once successfully logged in, if the applicants haven't applied yet click the personal data to fill in the personal information needed. Alternatively, if the applicants already applied proceed by clicking the notification bell to check the examination status or the approvment status if accepted, wait for the examination schedule. If not, click profile button then click edit profile and resubmit the missing requirements. Once all required missing requirements is provided, wait for the approvment result.

The next figure represents the flowchart of the proposed online appointment management in process in Camarines Sur Polytechnic Colleges Admission Office. This flowchart is for the staffs' interface, this is separated to the student interface.

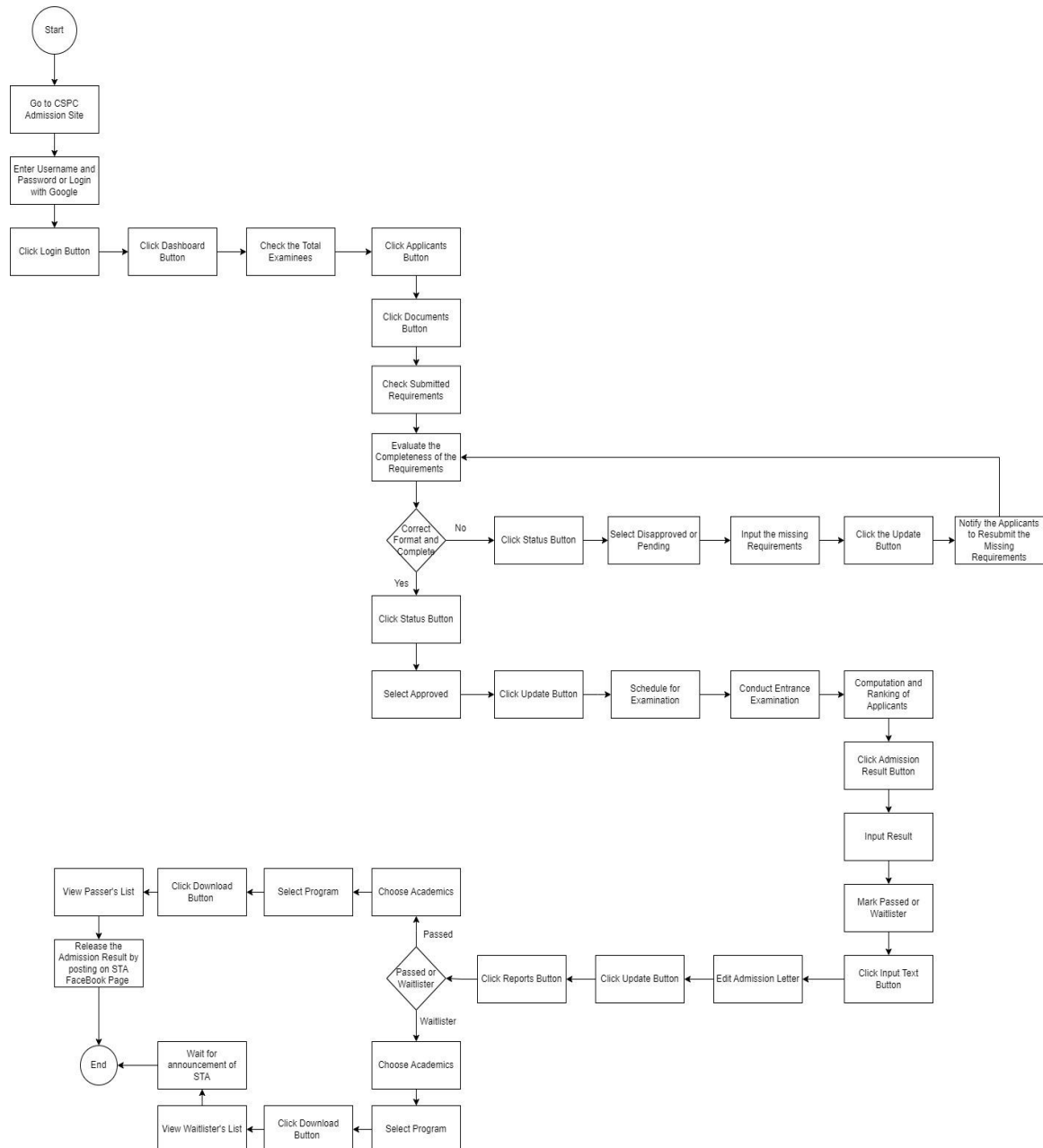


Figure 7: **Flowchart of Online Appointment Process (Admin)**

In this figure, it shows the flowchart of the proposed online appointment management in Camarines Sur Polytechnic Colleges Admission Office. This flowchart is for admin, this flowchart shows the step-by-step process of the design user interface of online appointment management. To begin the process, the staff logging into the CSPC Admission. The staff then enters their username and password using their CSPC account. Once the staff is logged in, they click on the Dashboard button. On the Dashboard, the staff can view admission status, check the total number of examinees, examinees for the month, and the incoming examinees. The staff will then click applicant button, then click on the Documents button to check the applicants' submitted documents and check if they are complete.

If the student's requirements are complete, the admission staff will approve their admission. Click the status button and select approved, click update button to schedule the student for examination, then the admission staff will conduct entrance exam for the approved applicants. Staff will start the computation and ranking of the applicants, then click the admission result button, Click the result button and input exam result, if the applicants is passed click dropdown under mark and click passed click update button, click the input text button, edit the admission letter and click send, admission will send admission letter, after that click the reports button to view the list of passers and waitlisters, if the admin click the passed button it will show the academics then choose what department and choose what program, the admin can also download the list by clicking download button. Then release the admission result by posting on STA Facebook page. The waitlisted applicants, will wait for the STA announcement. If the student's requirements are incomplete, the admission staff will disapprove their admission and notify the student to resubmit their missing requirements.

Researchers engaged in brainstorming sessions as they proceeded through the ideation phase. After narrowing down a number of interesting ideas, researchers moved on to creating simple prototypes with an emphasis on simplicity. To achieve the objectives, researchers will

engage in comprehensive research to identify the specific requirements of the admission office, including aspects like appointment scheduling, transaction numbers availability, and user interface specifications. Through interviews and chat conversation with the staff, researchers will gather crucial insights to ensure that the solution effectively addresses the primary challenges at hand. By the end of this ideation stage, researchers aim is to have a well-defined project scope, complete with clear objectives and an initial list of features.

Subsequent to the ideation phase, the next step entails the creation of a low-fidelity prototype using Wix. This prototype will act as a visual representation of the envisioned online appointment management. Researchers focus will be on generating wireframes and basic mock-ups that outline the user interface and navigation flow. This low-fidelity prototype will enable us to present design concepts to stakeholders and collect feedback for further refinement. This iterative process is pivotal in ensuring that our final solution is user-friendly and in alignment with the specific requirements of the admission office.

The ideation stage of this project centers on extensive research and collaboration with stakeholders to establish project objectives and features. Following this, the development of a low-fidelity prototype using Wix will provide a tangible visualization of the design concept, thereby facilitating valuable feedback and guaranteeing the creation of a user-centric solution for online appointment management in the admission office. The ultimate design would be a user-centric and effective solution targeted at improving the appointment management experience within the Admission Office. This was ensured by using these low-fidelity prototypes as the basis for gathering input and carrying out additional refining.

Prototype

This section provides the photos of the designed system. It shows the process or steps for Online Appointment in the Admission Office. This is an online appointment that allows the student applicant to file an appointment for entrance examination. The following figures present

the design and user interface, for each figure presents the process and steps indicated in the designed system. These are the photos of the landing page, login page, student personal data, medical information, student application form, appointment selection and the student's profile. For admin, it shows the landing page, login page, dashboard, applicant list and examinees information.

The figure below shows the landing page of online appointment management. In this figure it will show the apply button, but if the applicants wants to see the instruction first they can scroll down to see the instructions, for them to be familiar with the online appointment management.



Figure 8: **Landing Page (Student)**

Figure 8 presents the landing page of Online Appointment Management in Camarines Sur Polytechnic Colleges Admission Office for student interface. If the user opens the site this will be the first to appear in the website. This is an online appointment that allows the student applicant to file an appointment for entrance examination, this page contains the login button that the students click to proceed to the next interface.

The next figure shows the instruction for the applicants who will use the online appointment management. In this figure it will help the applicants to be familiar to the online appointment management.

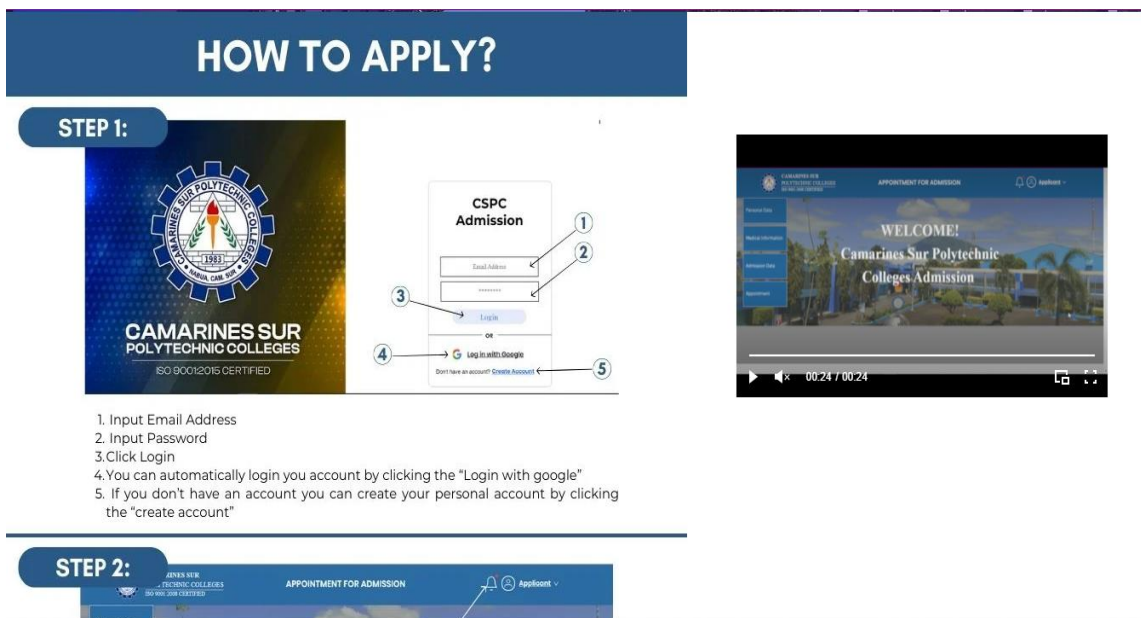


Figure 9: **Instructions for Online Appointment Management**

The figure 9 shows the instruction for the applicants who will use the online appointment management in CSPC, these instructions that included in the site will help the applicants to be familiar with the online appointment management, and in this figure it also included the video on how to use the online appointment management. If the applicants scroll down, they will see the instructions on how they will be using the online appointment management. After the applicants see the instructions they can click the “Apply Now” button to automatically directed to the login page.

The figure below show the log in page for the students, this is the first step that the students need to fill up to proceed in to the next step, if the students click the login button, this figure below shows the login page for students. This login page serves as a security to all the

users who use Online Appointment Management. They are required to fill out the username and password when they already have an account. This figure is the first step that needs to be done by the students for them to access the online appointment management in Camarines Sur Polytechnic Colleges Admission Office.

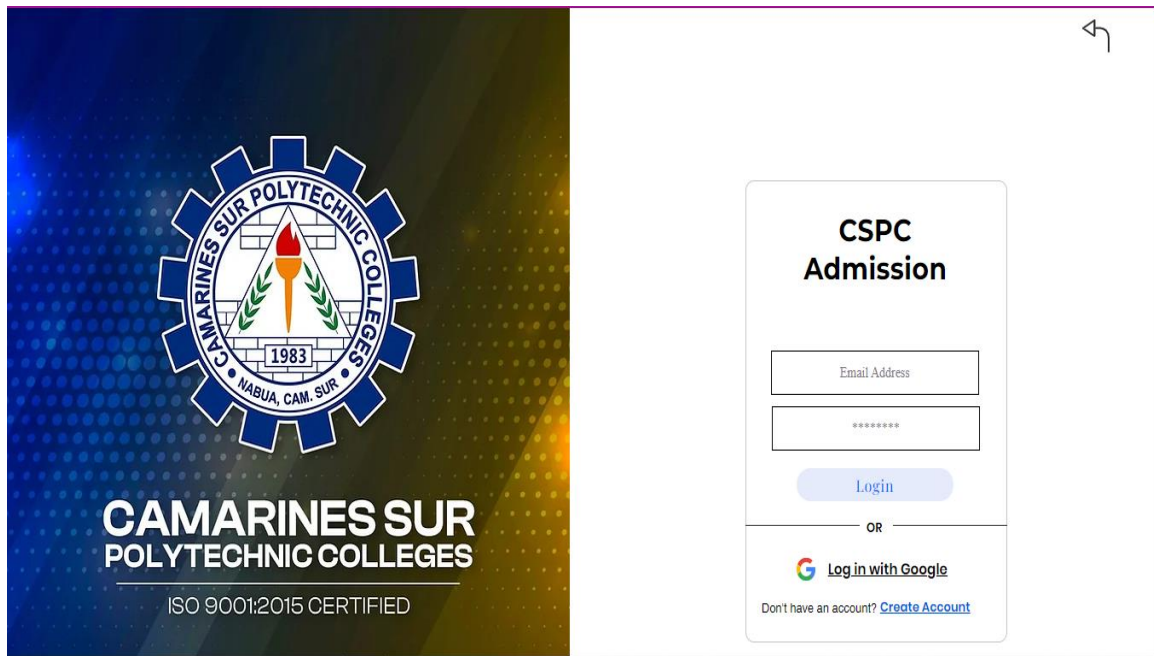


Figure 10: **Login Page (Student)**

Figure 10 presents the login page of Online Appointment Management in Camarines Sur Polytechnic Colleges Admission Office, this is the next interface when the users click the login button in the landing page. They are required to fill out the username and password if they already have an account for them to access the Online Appointment Management, it is the first step that needs to be done by the students. The students must provide the needed information for them to access the Online Appointment Management. They can also click the login with google button if they already have an account in google for them to easily access the online appointment management.

The next figure will show the main page of the Online Appointment Management in Camarines Sur Polytechnic Colleges Admission Office. This is the next figure after the students click the login button, this welcome interface will appear in the main page. In this interface the students will see the main menu which is the personal information, medical information, admission data and appointment.

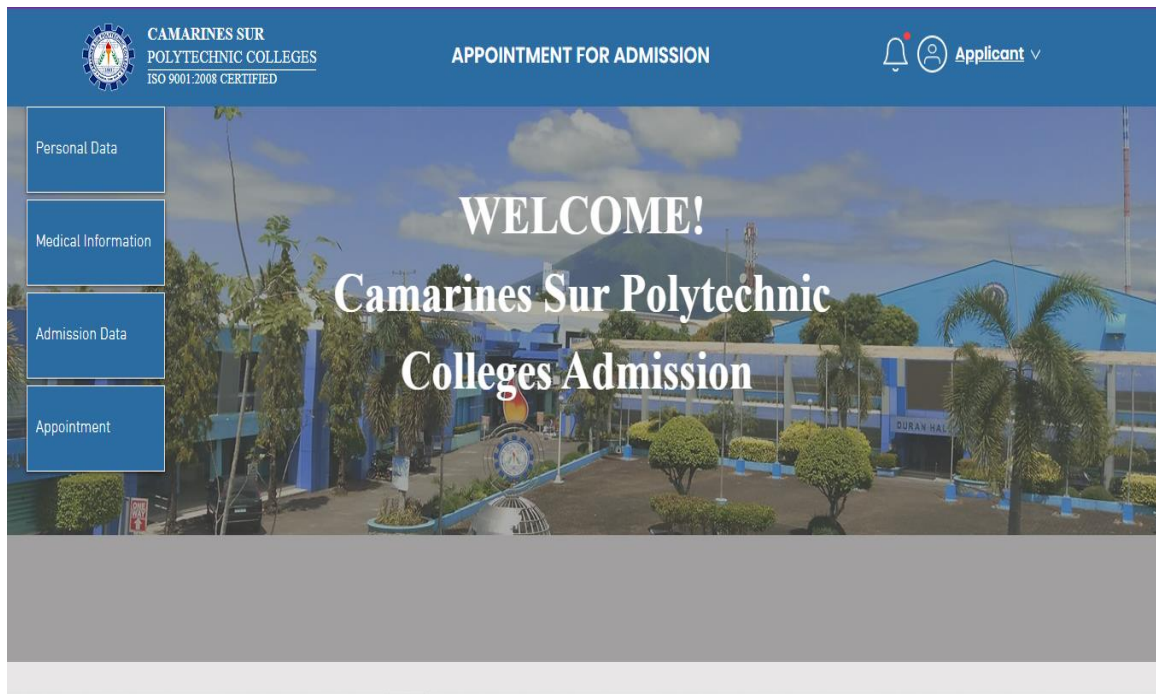
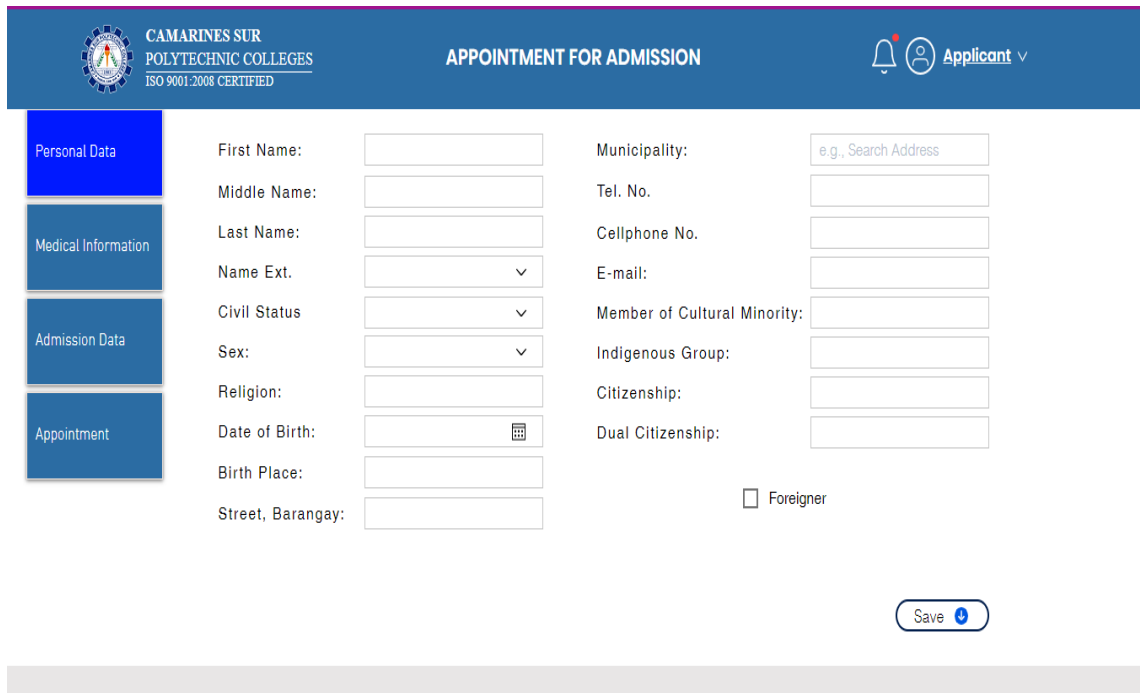


Figure 11: **Main Page**

Figure 11 shows the main page after clicking the login button this will appear, this is the welcome page where the users will see the main menu which is the personal information, medical information, student's application form or admission data and last is the appointment scheduling. The students can click the menu button if they want to, they can access which button they want to open and also in this interface has profile button in the right side which they can click the arrow button to manage their accounts and also it will show the logout button if they want to logout in Online Appointment Management. They can also click the Camarines Sur Polytechnic Colleges Logo to go back to the login page.

The next figure below presents as the personal data, wherein the students will provide the

needed information for the admission, this is necessary to fill out the form and complete to serve as the personal profile of the students and will help the admissions committee evaluate their personal data, wherein it will be submitted to the administrator and verify the information. After that the administrator will inform the student through a notification.



The screenshot displays the 'APPOINTMENT FOR ADMISSION' form. The header includes the Camarines Sur Polytechnic Colleges logo and name, the title 'APPOINTMENT FOR ADMISSION', and a user profile icon labeled 'Applicant'. The left sidebar contains four menu items: 'Personal Data' (highlighted), 'Medical Information', 'Admission Data', and 'Appointment'. The main form area contains the following fields:

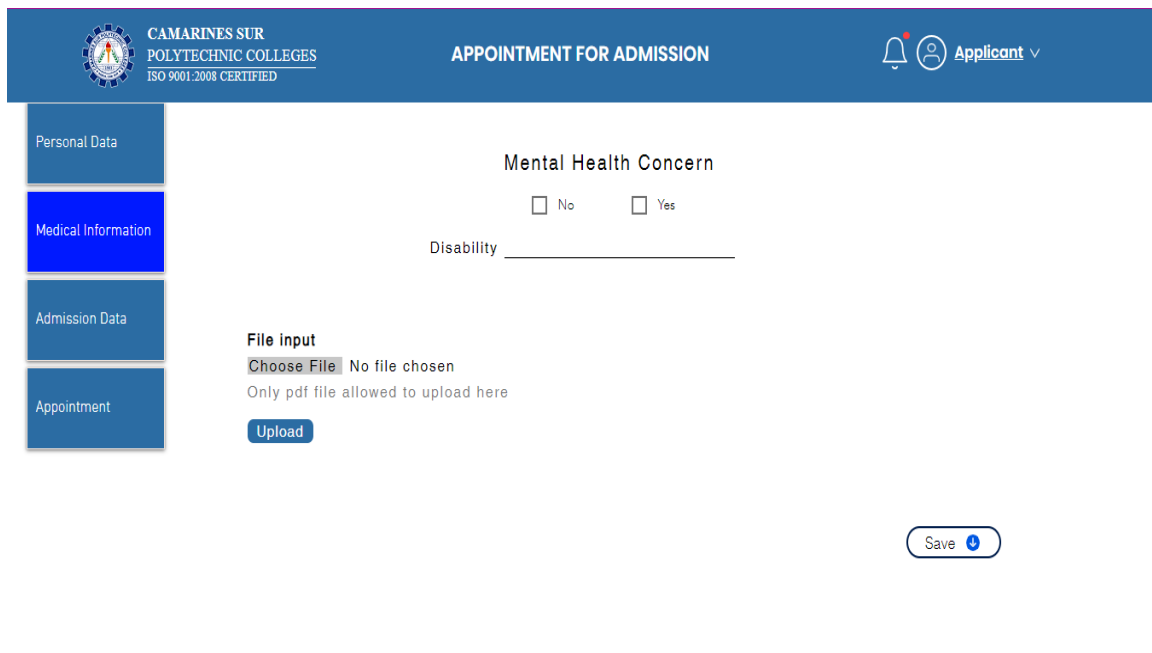
- First Name:
- Middle Name:
- Last Name:
- Name Ext.:
- Civil Status:
- Sex:
- Religion:
- Date of Birth:
- Birth Place:
- Street, Barangay:
- Municipality:
- Tel. No.:
- Cellphone No.:
- E-mail:
- Member of Cultural Minority:
- Indigenous Group:
- Citizenship:
- Dual Citizenship:
- ☐ Foreigner

A 'Save' button is located at the bottom right of the form.

Figure 12: **Personal Data**

The figure 12 shows the various input fields that the students needed to fill out for the admission, this figure will be the first to appear when they successfully log in using their account, the student must provide the personal information; first name, middle name, last name, name extension, the civil status, sex, religion, date of birth, birth place, street, barangay, municipality, telephone number, cellphone number, email address, member of cultural minority, indigenous group, citizenship, dual citizenship and foreigner, after the student provide the necessary information, click the “save” button to admit.

The figure below shows the medical information, this figure will appear after the students fill out their information in figure 12, after they fill out the personal information, they will click the save button and it will automatically redirect in this interface. The medical information form is not mandatory to fill out, if the student doesn't have a medical issues or certificate they can also proceed to the next step of the admission.



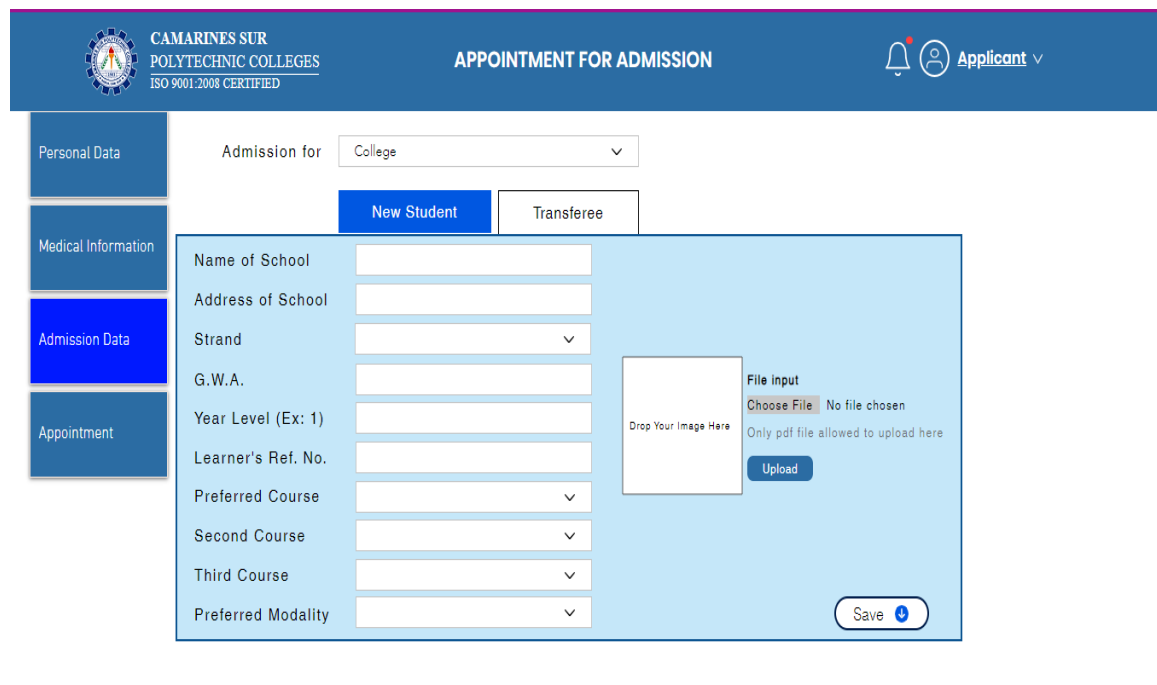
The screenshot displays the 'Medical Information' form within the 'APPOINTMENT FOR ADMISSION' interface. The sidebar on the left shows the navigation menu with 'Medical Information' selected. The main content area includes a 'Mental Health Concern' section with 'No' and 'Yes' checkboxes, a 'Disability' text input field, a 'File Input' section with a 'Choose File' button and 'No file chosen' status, and an 'Upload' button. A 'Save' button is located at the bottom right.

Figure 13: **Medical Information**

Figure 13 shows the Medical Information, this contains the information of the students about their medical history, the students will select if they have health concern or clarification if they have disability. After the students click the save button in the personal information interface this figure will automatically appear. In this interface it is for the students who have health problems and for the students with disabilities. It is important to note that the specific requirements can vary between different medical schools and healthcare programs and also the admin will carefully review the admission requirements of the students in specific institutions or programs to which they plan to apply. In this figure, they can also upload their medical certificates in this interface, they will choose their files and the files are only in pdf. Through this

the staff can easily determine if the students have health concerns or are disabled, this can also help the students to be the priority on entrance examination day. The students will click the upload button to save their medical information and also their medical requirements.

This figure below shows the admission data that the students must need to fill out all the information that is needed. This will be their academic information during their last school year.



The screenshot shows the 'APPOINTMENT FOR ADMISSION' form for 'College'. The 'New Student' tab is selected. The form includes a sidebar with navigation links: Personal Data, Medical Information, Admission Data (highlighted), and Appointment. The main form area contains the following fields:

- Admission for: College (dropdown)
- Name of School: Text input
- Address of School: Text input
- Strand: Dropdown menu
- G.W.A.: Text input
- Year Level (Ex: 1): Text input
- Learner's Ref. No.: Text input
- Preferred Course: Dropdown menu
- Second Course: Dropdown menu
- Third Course: Dropdown menu
- Preferred Modality: Dropdown menu

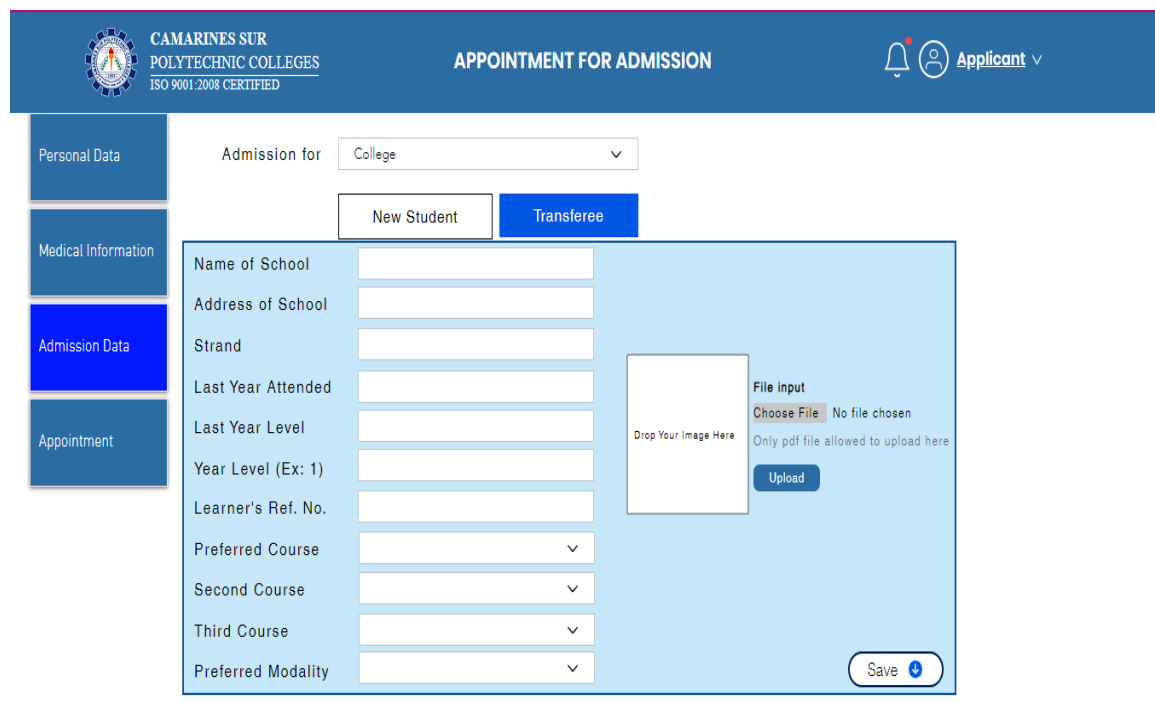
There is a 'Drop Your Image Here' section with a 'Choose File' button and an 'Upload' button. A note states: 'Only pdf file allowed to upload here'. A 'Save' button is located at the bottom right of the form.

Figure 14: **Admission Data for New Students**

In this figure, it shows the online application form the students will click the dropdown button in admission for and select if college, masteral or doctoral. After that, they will click if they are new students or transferee, if they click the new student this interface will appear and the students must provide the information needed in the application form which is the name of school, address of school, in strand button they can click the dropdown button to pick their strand during senior high school, general weighted average, year level, learner's reference number, preferred course, second course, third course there are dropdown button to easily pick their

preferred courses and for preferred modality there are also a dropdown button which is face to face, online class, blended learning and modular. After completing the information, the students must be advised to upload their requirements because the staff will be evaluating their requirements for them to be qualified for entrance examination, after that click the save button.

The next figure is for transferees. This interface will not appear in the new student button, their interface is separated to avoid confusions to the students, and this interface is the continuation of admission data, but in this figure is applicable only for transferee students.



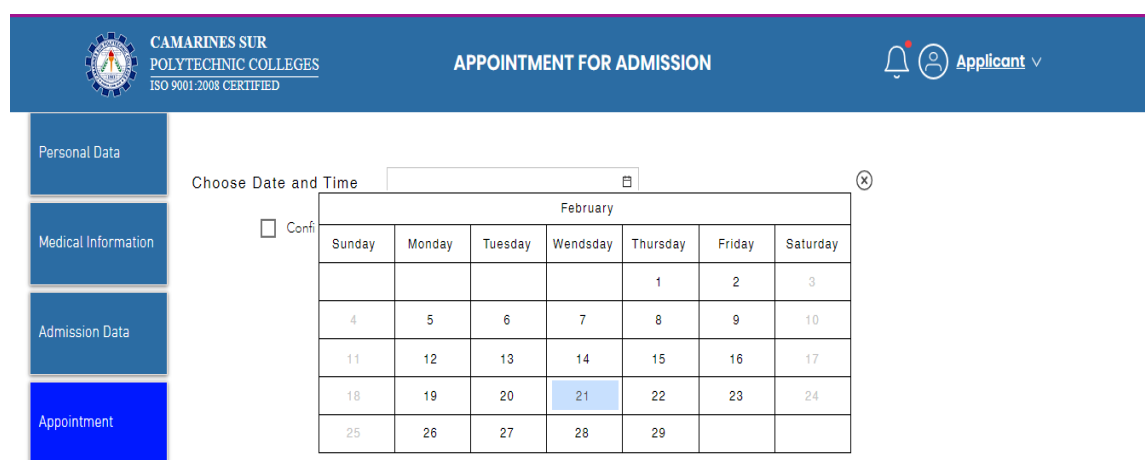
The screenshot shows the 'APPOINTMENT FOR ADMISSION' interface for Camarines Sur Polytechnic Colleges. The header includes the college logo, name, and ISO 9001:2008 certification. The user is logged in as an 'Applicant'. The left sidebar contains navigation tabs: 'Personal Data', 'Medical Information', 'Admission Data' (selected), and 'Appointment'. The main form area is titled 'Admission for' with a dropdown set to 'College'. Below this are two buttons: 'New Student' and 'Transferee' (selected). The form fields include: 'Name of School', 'Address of School', 'Strand', 'Last Year Attended', 'Last Year Level', 'Year Level (Ex: 1)', 'Learner's Ref. No.', 'Preferred Course', 'Second Course', 'Third Course', and 'Preferred Modality'. There is a file upload section with a 'Drop Your Image Here' box, a 'File input' section with 'Choose File' and 'No file chosen' options, and an 'Upload' button. A 'Save' button is located at the bottom right of the form.

Figure 15: **Admission Data for Transferees**

Figure 15 is the continuation of figure 7 which is the admission data for transferees. This figure will show the application form that needs to be filled out of all the transferees including the name of school, address of school, strand last year attended, last year level, year level, learner's reference number, preferred courses and preferred modality. After filling out all the information needed, the student needs to upload their requirements in pdf format and also their 2x2 picture.

This is advisable because the staff will evaluate the requirements of the students and after evaluation of the requirements, they will send letters if the student is accepted or rejected. After uploading the requirements click the save button.

The figure 16 presents the appointment of the Online Appointment Management for Camarines Sur Polytechnic Colleges. It includes the date and time that are unavailable and available and the remaining slot for the entrance examination



CAMARINES SUR POLYTECHNIC COLLEGES
 ISO 9001:2008 CERTIFIED

APPOINTMENT FOR ADMISSION

🔔 👤 Applicant ▾

Personal Data

Medical Information

Admission Data

Appointment

Choose Date and Time

☐ Confirm

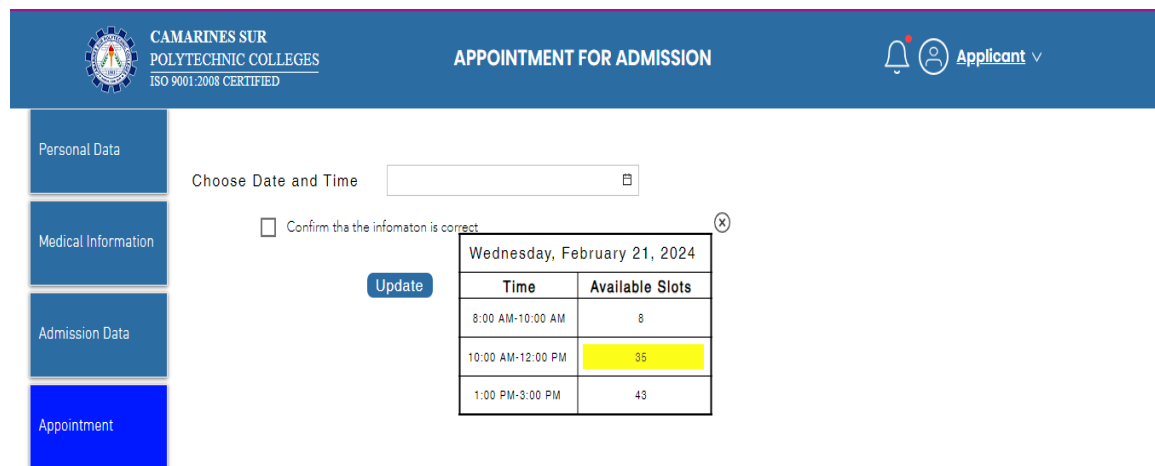
February						
Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29		

Figure 16: **Appointment Selection**

This figure illustrates the appointment scheduling process. The student must require submitting their specified deadline it is provided by the institutions to understand the relevant dates for their appointment to ensure that the application process proceeds and that the students have a chance of being considered for admission. When the student clicks the calendar button, a calendar will appear, allowing the student to choose the preferred exam schedule. If the selected

day turns red, it indicates that it is no longer available. But, if the selected day turns blue, it signifies that the schedule is still available.

The next figure is the continuity of the figure above, this figure will appear after the selection of the day for the examination schedule. This interface is the time selection wherein the students will choose their preferred time that aligns with their personal preferences for examination.



The screenshot shows the 'APPOINTMENT FOR ADMISSION' interface. On the left is a sidebar with navigation links: Personal Data, Medical Information, Admission Data, and Appointment. The main area has a header with the college logo and name, and a user profile labeled 'Applicant'. Below the header, there is a 'Choose Date and Time' dropdown menu. A checkbox labeled 'Confirm that the information is correct' is present. An 'Update' button is located below the checkbox. A table displays the available slots for Wednesday, February 21, 2024.

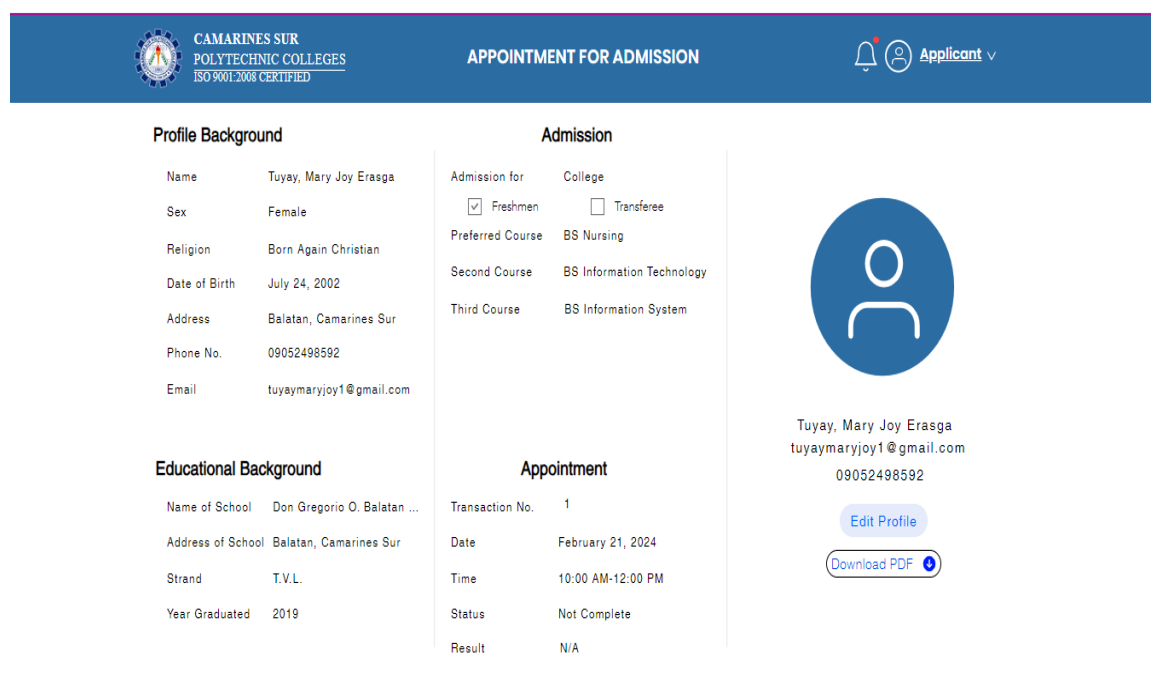
Time	Available Slots
8:00 AM-10:00 AM	8
10:00 AM-12:00 PM	35
1:00 PM-3:00 PM	43

Figure 17: **Time Selection**

Figure 17 shows the time selection after selecting the preferred day for examination. It will automatically directly in this interface wherein the students will choose preferred time for examination. Students have three choices, and in this interface, they can determine if the preferred time is available. When they click on the time, available slots will be displayed, color-coded as red, yellow, or green. Red indicates that the chosen time has only a few available slots, yellow suggests it is about to run out of slots, and green means there are still many remaining

slots. After that, the date and time for examination will be displayed and the students will click the update button to save their appointment schedule.

The next figure shows the profile of the students after they fill out all the requirements needed such as, personal data, medical information, admission data, and last appointment scheduling. This interface will also appear on the admin interface.



The screenshot displays the 'APPOINTMENT FOR ADMISSION' interface for a student. The header includes the Camarines Sur Polytechnic Colleges logo, the title 'APPOINTMENT FOR ADMISSION', and a user profile dropdown labeled 'Applicant'. The main content is divided into four sections: Profile Background, Admission, Educational Background, and Appointment. The Profile Background section lists personal details for Mary Joy Erasga. The Admission section shows her preferred and second courses. The Educational Background section lists her school and graduation year. The Appointment section shows her transaction number, date, time, and status. A profile picture placeholder and contact information are on the right, along with 'Edit Profile' and 'Download PDF' buttons.

Profile Background		Admission	
Name	Tuyay, Mary Joy Erasga	Admission for	College
Sex	Female	☒ Freshmen	☐ Transferee
Religion	Born Again Christian	Preferred Course	BS Nursing
Date of Birth	July 24, 2002	Second Course	BS Information Technology
Address	Balatan, Camarines Sur	Third Course	BS Information System
Phone No.	09052498592		
Email	tuyaymaryjoy1@gmail.com		

Educational Background		Appointment	
Name of School	Don Gregorio O. Balatan ...	Transaction No.	1
Address of School	Balatan, Camarines Sur	Date	February 21, 2024
Strand	T.V.L.	Time	10:00 AM-12:00 PM
Year Graduated	2019	Status	Not Complete
		Result	N/A

Student Information:
 Tuyay, Mary Joy Erasga
 tuyaymaryjoy1@gmail.com
 09052498592

[Edit Profile](#)
[Download PDF](#)

Figure 18: **Student's Profile**

In this figure, after students have completed the necessary information and when they have a schedule for the entrance examination, their profile background, educational background, admission and appointment will appear in the student's profile. Additionally, the student's profile picture is positioned on the right side. Students can edit their previously filled-out information by clicking on 'Edit Profile.' Once their application is approved, students can download their appointment slip by clicking the download PDF button.

The figure below shows the appointment slip of the students after successfully appoint a schedule, the provided application slip to the student is a unique identifier for the appointment

once the student goes to the exact appointment date and time the student needs to give the appointment slip to the management to verify the details.


CAMARINES SUR
 POLYTECHNIC COLLEGES
 ISO 9001:2008 CERTIFIED

APPOINTMENT SLIP FOR ADMISSION

	Name: Tuyay, Mary Joy Erasga	Transaction No. 1
	Sex: Female	Appointment Date: February 21, 2024
	Address: Balatan, Camarines Sur	Appointment Time: 10:00 AM-12:00 PM
	Contact No. 09052498592	
	Preferred Course: BS Nursing Second Course: BS Information Technology Third Course: BS Information System	


CAMARINES SUR
 POLYTECHNIC COLLEGES
 ISO 9001:2008 CERTIFIED

APPOINTMENT SLIP FOR ADMISSION

Profile Background <table border="0" style="width: 100%;"> <tr><td style="width: 20%;">Name</td><td>Tuyay, Mary Joy Erasga</td></tr> <tr><td>Sex</td><td>Female</td></tr> <tr><td>Religion</td><td>Born Again Christian</td></tr> <tr><td>Date of Birth</td><td>July 24, 2002</td></tr> <tr><td>Address</td><td>Balatan, Camarines Sur</td></tr> <tr><td>Phone No.</td><td>09052498592</td></tr> <tr><td>Email:</td><td>tuyaymaryjoy1@gmail.com</td></tr> </table>	Name	Tuyay, Mary Joy Erasga	Sex	Female	Religion	Born Again Christian	Date of Birth	July 24, 2002	Address	Balatan, Camarines Sur	Phone No.	09052498592	Email:	tuyaymaryjoy1@gmail.com	Admission <table border="0" style="width: 100%;"> <tr> <td style="width: 30%;">Admission for</td> <td>College</td> </tr> <tr> <td>☒ Freshmen</td> <td>☐ Transferee</td> </tr> <tr><td>Preferred Course</td><td>BS Nursing</td></tr> <tr><td>Second Course</td><td>BS Information Technology</td></tr> <tr><td>Third Course</td><td>BS Information System</td></tr> </table>	Admission for	College	☒ Freshmen	☐ Transferee	Preferred Course	BS Nursing	Second Course	BS Information Technology	Third Course	BS Information System
Name	Tuyay, Mary Joy Erasga																								
Sex	Female																								
Religion	Born Again Christian																								
Date of Birth	July 24, 2002																								
Address	Balatan, Camarines Sur																								
Phone No.	09052498592																								
Email:	tuyaymaryjoy1@gmail.com																								
Admission for	College																								
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Preferred Course	BS Nursing																								
Second Course	BS Information Technology																								
Third Course	BS Information System																								

Educational Background <table border="0" style="width: 100%;"> <tr><td style="width: 20%;">Name of School</td><td>Don Gregorio O. Balatan Institute</td></tr> <tr><td>Address of School</td><td>Balatan, Camarines Sur</td></tr> <tr><td>Strand</td><td>T.V.L</td></tr> <tr><td>Year Graduated</td><td>2019</td></tr> </table>	Name of School	Don Gregorio O. Balatan Institute	Address of School	Balatan, Camarines Sur	Strand	T.V.L	Year Graduated	2019	Appointment <table border="0" style="width: 100%;"> <tr><td style="width: 20%;">Transaction No.</td><td>1</td></tr> <tr><td>Date</td><td>February 21, 2024</td></tr> <tr><td>Time</td><td>10:00 AM – 12:00 PM</td></tr> <tr><td>Status</td><td>Not Complete</td></tr> <tr><td>Result</td><td>N/A</td></tr> </table>	Transaction No.	1	Date	February 21, 2024	Time	10:00 AM – 12:00 PM	Status	Not Complete	Result	N/A
Name of School	Don Gregorio O. Balatan Institute																		
Address of School	Balatan, Camarines Sur																		
Strand	T.V.L																		
Year Graduated	2019																		
Transaction No.	1																		
Date	February 21, 2024																		
Time	10:00 AM – 12:00 PM																		
Status	Not Complete																		
Result	N/A																		

Figure 19: Appointment Slip

Figure 19 shows about the appointment slip of the students, the students will receive this once they completed the scheduling process and most likely receive via email, this is advisable to save or print the application slip for the records or download the pdf in the website this can be

find and download in their profile and it is necessary to have the appointment slip on the schedule date and time. This application slip serves as proof that the students have successfully registered for the exam and may include important details related to their application.

The figure below shows the notification for admission result, in this figure the applicants will be notified through message from STA by clicking the notification bell, this message will be a big help for the applicants to be updated about their admission result. This will be used as communication link between the staff of Student Testing and Admission and the applicants.

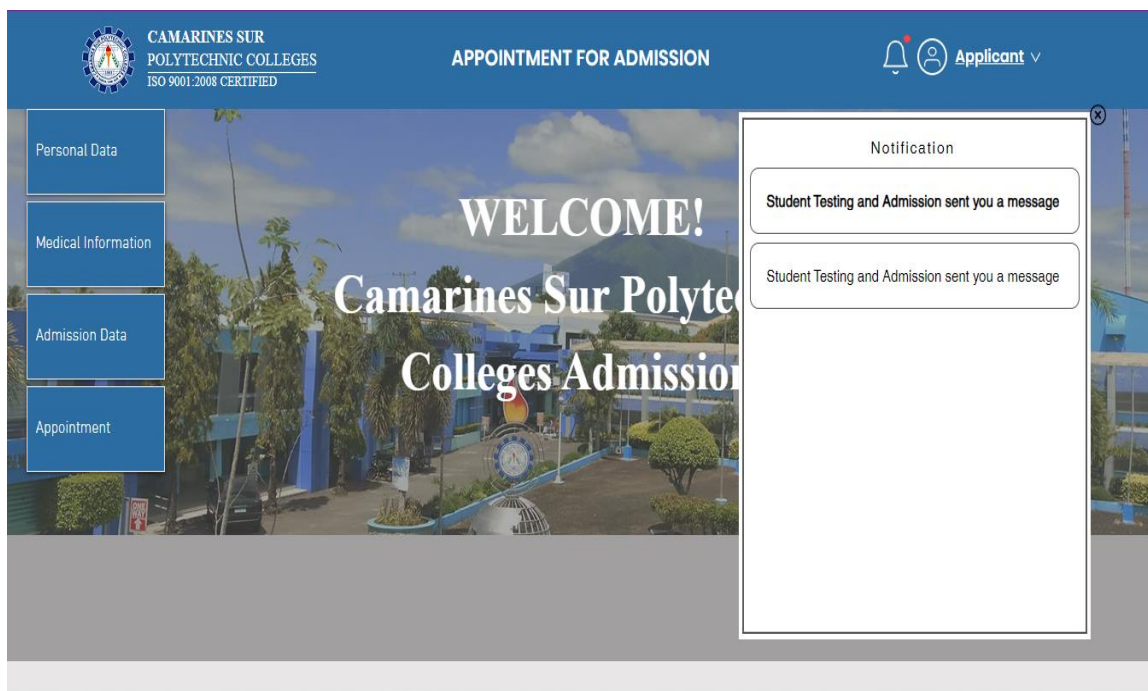


Figure 20: **Notification for Admission Result**

Figure 20 shows the notification for admission result of the applicants, when the admin already compute the admission results of the applicants they will send a letter that contains greetings to officially welcome the applicants for being one of the students in Camarines Sur Polytechnic Colleges and admission result, this will be serves as the feedback and will provide the students with a clear understanding of their academic performance. After of successfully take the entrance examination, they can open again the Camarines Sur Polytechnic Colleges

Admission Site and after successfully login in the site they can click the notification bell at the right side and beside of applicant's profile to check the messages from Student Testing and Admission for their admission result, this will help the applicants to be updated about the current status or result of their admission and to help the applicants to prepare for the next process they need to do.

Figure 21 shows the total examinees of the students throughout the admission date and the percentage of the program that the students applied, this will show the difference of number of the students who applied for the admission for the upcoming entrance examination within the organizations.

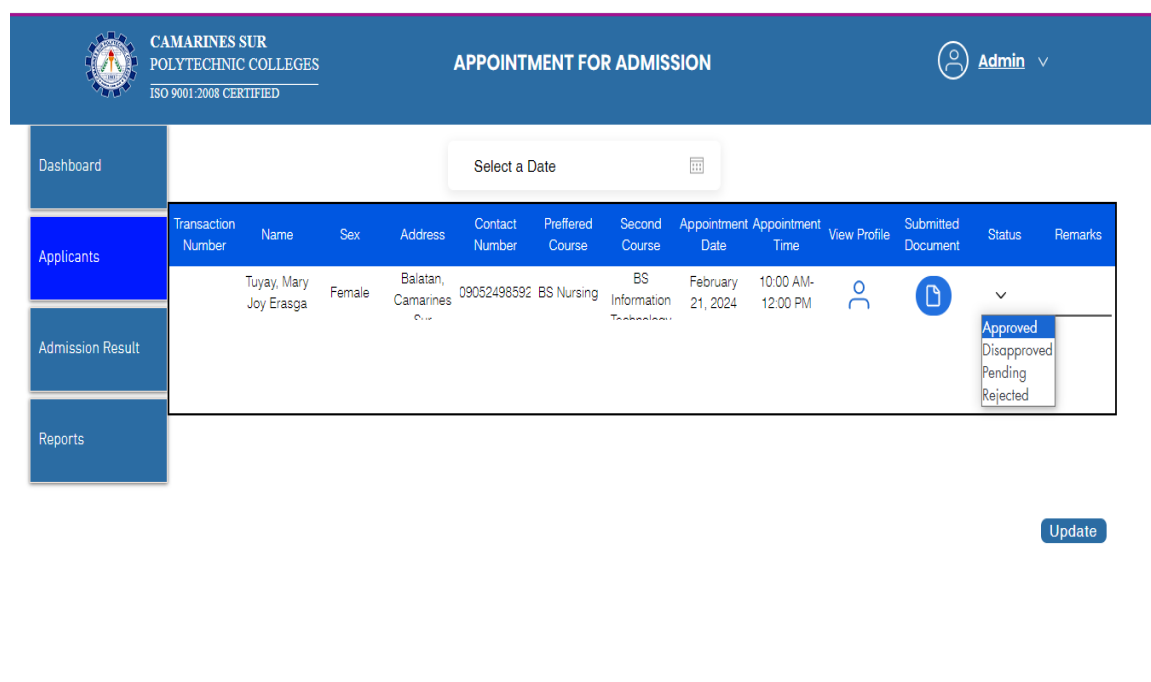


Figure 21: **Admin Dashboard**

Figure 21 show the admin dashboard, the admin can view on how many students totally applied for the admission, This shows the monitoring the number of the students who applied an appointment in Camarines Sur Polytechnic Colleges, it is divided into 3 wherein the admin can see the total examines, examinees for this month, and incoming examinees into this 3 the admin will know the total of the overall students who applied for the appointment from the start to finish

by selecting the date they want. Figure shown below is a pie chart that shows the percentage of each program wherein it signifies the number of students who applied to the programs available in the Camarines Sur Polytechnic Colleges.

The figure below shows the applicant's list, wherein the admin can monitor, recording, change the status of the students and also evaluate the requirements of the applicant, this will endure to address reducing downtime and improving overall user satisfaction.



Transaction Number	Name	Sex	Address	Contact Number	Preferred Course	Second Course	Appointment Date	Appointment Time	View Profile	Submitted Document	Status	Remarks
	Tuyay, Mary Joy Erasga	Female	Balatan, Camarines Sur	09052498592	BS Nursing	BS Information Technology	February 21, 2024	10:00 AM-12:00 PM			Approved Disapproved Pending Rejected	

Update

Figure 22: **Applicants Management**

This figure 22 shows the list of applicants who have been appointed and submitted their application form for the entrance examination. The admin has the ability to see the number of applicants who submitted their application form and schedule appointment, here they can also see the transaction number of the applicants, name and other information of the applicant, also see here the preferred course, appointment date, appointment time and the admin can also see the background profile of the applicants based on the information filled in. Also in this interface, the admin will be able to evaluate the requirements passed by the applicants if it is complete or if there is something missing, and last, the admin will put a remark on what is the status of the

students if the student is accepted or is there something missing or wrong format in passed requirements and also if the students is not completed the admin will put comments about the missing requirements of the students so that the student can easily identify what requirements needed to resubmit, when a remark has been made, the admin will click the update button so that the student's status will be automatically updated on their profile.

The next figure shows the admission result, the figure shows the list of applicants. The admin can manage the examination result of the applicant who has already take the entrance examination. This is the last figure for admin interface.

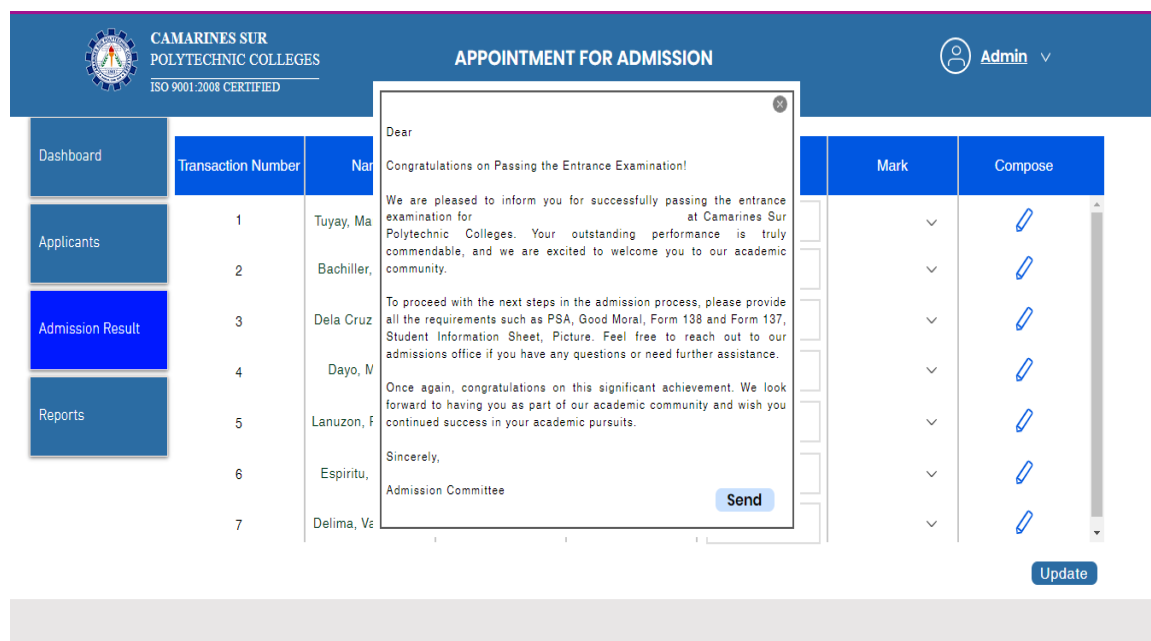
 CAMARINES SUR POLYTECHNIC COLLEGES ISO 9001:2008 CERTIFIED APPOINTMENT FOR ADMISSION  Admin ▾							
Dashboard	Transaction Number	Name	Contact Number	Preferred Program	Result	Mark	Compose
Applicants	1	Tuyay, Mary Joy E.	09456732753	BS Nursing		▾	
Admission Result	2	Bachiller, Jose B.	09325823648	BS Computer Science		▾	
Reports	3	Dela Cruz, Mark L.	09273682364	BS Mechanical Engineering		▾	
	4	Dayo, Mark S.	09237682637	BS Electrical Engineering		▾	
	5	Lanuzon, Patrick O.	09142353452	BS Nursing		▾	
	6	Espiritu, Jean S.	09838726321	BS Information Technology		▾	
	7	Delima, Vanessa P.	09213652143	BS Information System		▾	
Update							

Figure 23: **Admission Result**

Figure 23 shows the admission result, this page will show the list of applicants. In this figure the admin can view the list of applicants that has already taken the examination, transaction number, contact number and preferred program. The admin can also input the grades of the applicants and also, the admin will click compose button to edit the admission letter and send the letter to the applicants, they can view the profile of the applicant if they want to see the updated profile of the applicants by clicking the name of applicants. The admin must click the save button

to automatically update the students in their profile, and also after inputting the grades of the students the admin will also update the students.

The figure below shows the admission letter that the admin will send to the applicants for them to be notified regarding their examination result if there passed the entrance examination or they are belong to waitlisted and also indicated in the letter all the requirements that needed to submit in STA offices.



Transaction Number	Name	Mark	Compose
1	Tuyay, Ma	✓	
2	Bachiller,	✓	
3	Dela Cruz	✓	
4	Dayo, M	✓	
5	Lanuzon, F	✓	
6	Espiritu,	✓	
7	Delima, V	✓	

Dear

Congratulations on Passing the Entrance Examination!

We are pleased to inform you for successfully passing the entrance examination for at Camarines Sur Polytechnic Colleges. Your outstanding performance is truly commendable, and we are excited to welcome you to our academic community.

To proceed with the next steps in the admission process, please provide all the requirements such as PSA, Good Moral, Form 138 and Form 137, Student Information Sheet, Picture. Feel free to reach out to our admissions office if you have any questions or need further assistance.

Once again, congratulations on this significant achievement. We look forward to having you as part of our academic community and wish you continued success in your academic pursuits.

Sincerely,
 Admission Committee

Send

Update

Figure 24: **Admission Letter**

Figure 24 shows the admission letter, when the admin input the result of the applicants examination they will click the pen icon and edit admission letter, the letter contains the status of the applicants based on their result, the description of the letter are already permanent once the admin edit the admission letter, they only need to change is the name, and the program they applied, it also contains the required documents they will need to submit to the STA office such as: original PSA, good moral, form 137 and form 138, student information sheet and a picture, this will serves as an official and formal confirmation for the applicant to know the status of their appointment and for the student's academic journey at Camarines Sur Polytechnic Colleges.

The figure below shows the admission reports that the admin can view the list of the applicants who already taken the entrance examination, this figure shows the Camarines Sur Polytechnic Colleges academic and the programs.

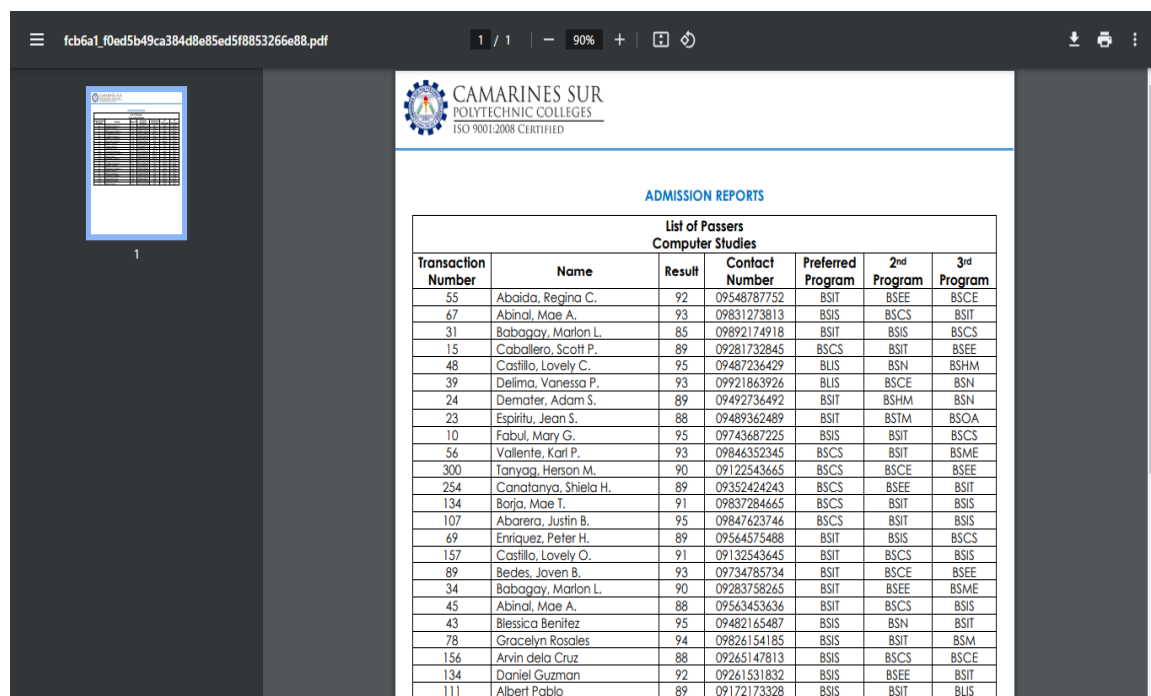


Transaction Number	Name	Result	Contact Number	Preferred Program	2nd Program	3rd Program
55	Fabul, Mary G.	92	09548787752	BSIT	BSEE	BSCE
67	Espiritu, Jean S.	93	09831273813	BSIS	BSCS	BSIT
31	Demater, Adam S.	85	09892174918	BSIT	BSIS	BSCS
15	Delima, Vanessa P.	89	09281732845	BSCS	BSIT	BSEE
48	Castillo, Lovely C.	95	09487236429	BIIS	BSIN	BSHM

Figure 25: **Admission Reports**

Figure 25 shows the admission reports, when the admin input the result of the applicant's examination they can also click the reports button. In this figure, the will click the passed or waitlist button and the colleges in CSPC will appear, if the admin click the CCS, all the list of passers under the CCS Department will appear, and if the admin click the Bachelor of Science in Computer Science all the list of passers under the BS Computer Science will appear. The admin also click the download button to download the PDF File of admission reports. If the admin click the CCS and click the download button all the passers under CCS Department is included. But, if the admin click the program like information technology, all the passers under Information Technology is included.

The next figure will shows the PDF File of Admission Reports that contains the transaction number of the applicants, name, result, contact number, preferred program, 2nd program and 3rd program. This file will the list of applicants who passed the entrance examination and the admin will release this list by posting in Student Testing and Admission Facebook Page.



Transaction Number	Name	Result	Contact Number	Preferred Program	2 nd Program	3 rd Program
55	Abaida, Regina C.	92	09548787752	BSIT	BSEE	BSCE
67	Abinal, Mae A.	93	09831273813	BSIS	BSCS	BSIT
31	Babagay, Marlon L.	85	09892174918	BSIT	BSIS	BSCS
15	Caballero, Scott P.	89	09281732845	BSCS	BSIT	BSEE
48	Castillo, Lovely C.	95	09487236429	BLIS	BSN	BSHM
39	Delima, Vanessa P.	93	09921863926	BLIS	BSCE	BSN
24	Demater, Adam S.	89	09492736492	BSIT	BSHM	BSN
23	Espiritu, Jean S.	88	09489362489	BSIT	BSTM	BSOA
10	Fabul, Mary G.	95	09743687225	BSIS	BSIT	BSCS
56	Vallente, Karl P.	93	09846352345	BSCS	BSIT	BSME
300	Tanyag, Herson M.	90	09122543665	BSCS	BSCE	BSEE
254	Canatanya, Shiela H.	89	09352424243	BSCS	BSEE	BSIT
134	Borja, Mae T.	91	09837284665	BSCS	BSIT	BSIS
107	Abarera, Justin B.	95	09847623746	BSCS	BSIT	BSIS
69	Enriquez, Peter H.	89	09564575488	BSIT	BSIS	BSCS
157	Castillo, Lovely O.	91	09132543645	BSIT	BSCS	BSIS
89	Bedes, Joven B.	93	09734785734	BSIT	BSCE	BSEE
34	Babagay, Marlon L.	90	09283758265	BSIT	BSEE	BSME
45	Abinal, Mae A.	88	09563453636	BSIT	BSCS	BSIS
43	Blessica Benitez	95	09482165487	BSIS	BSN	BSIT
78	Gracelyn Rosales	94	09826154185	BSIS	BSIT	BSM
156	Arvin dela Cruz	88	09265147813	BSIS	BSCS	BSCE
134	Daniel Guzman	92	09261531832	BSIS	BSEE	BSIT
111	Albert Pablo	89	09172173328	BSIS	BSIT	BLIS

Figure 26: Admission Reports (PDF File)

Figure 26 shows the PDF File of Admission Reports that contains the list of the applicants who passed the entrance examination, including in this file the transaction number of the applicants, name, result, preferred program, 2nd program and 3rd program. Also the admin can download the PDF File of admission report in specific programs. The admin can print this file or this will be the list of passers that the admin will be posting on Student Testing and Admission FaceBook Page.

Data Design

The arrangement of data according to a database model is known as data design. The designers decide what data should be kept and how the various data pieces should interact. They may start fitting the data to the database model using this information.

Entity Relationship Diagram

An entity relationship diagram, or ERD, shows the connections between entity sets that are kept in a database. It serves as a high-level logical data model and is used to develop conceptual designs for databases. Each row in a relational database table represents an entity, which is an autonomous, physical thing or idea. One of an entity's attributes is a particular property that serves as an identifier, and a relationship is an association that explains the interactions between two entities. The term "cardinality" refers to the quantity of one entity instances that can or need to be connected to each instance of another entity. Relationships can be one-to-one, one-to-many, or many-to-many.

One needs to use an entity-relationship diagram, or ER diagram, to model the data in a database. It acts as the cornerstone around which a database is built. Entity relationships are depicted in ER diagrams. Features, which are the kinds of information that we'll save. They also show how various objects interact with one another. Entity relationship diagrams provide a database with a visual starting point. Architecture and can also be applied to help determine the information systems requirements of a company. The method utilized to visually represent the system's needs is graphical in nature, hence they are simpler to understand.

An E-R model facilitates comprehension by offering a diagrammatic and graphical representation of many entities, their properties, and their relationships with each other. Duplication and other problems are thereby decreased, and the data structure is made simpler to understand.

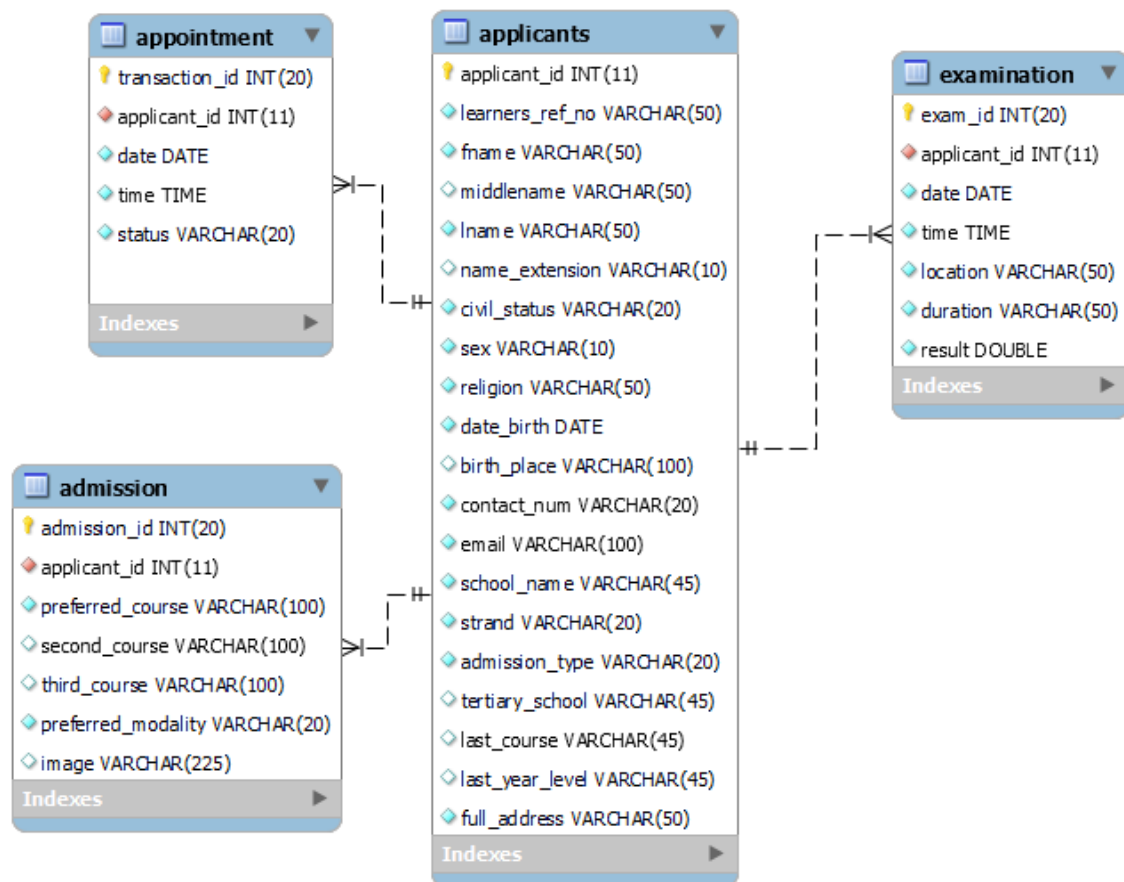


Figure 27: **Entity Relationship Diagram**

Figure 27 shows the entity relationship diagram which shows the relationship or entities that can help to visualize the structure and logic of a database of the low-fidelity prototype. Entities, attributes, and relationship are the three main notions that entity relationship diagram is built on. Appointment, Applicants, Admission and Examination are the main entities of the database. Under the applicant entity, the **applicant_id** is the primary key, which means that each student has a unique ID. The date and time columns store the date and time of the applicant's admission. The status column stores the status of the applicant's admission, such as pending, approved, or rejected. The result column stores the result of the applicant's admission, such as passed or failed. The religion, date of birth, preferred course, second course, third course, preferred modality, and image columns store additional information about the applicant. The religion column stores the applicant's religion. The date of birth column stores the applicant's date

of birth. The preferred course column stores the applicant's preferred course of study. The second course and third course columns store the applicant's second and third preferred courses of study, respectively. The preferred modality column stores the applicant's preferred mode of study, such as online, face-to-face or blended learning. The image column stores the applicant's profile image.

The database also includes a number of indexes. Indexes are used to speed up database queries. The admission_id index is likely used to speed up queries that filter or sort the table by admission ID. The applicant_id index is likely used to speed up queries that filter or sort the table by applicant ID. The date and time indexes are likely used to speed up queries that filter or sort the table by date or time.

The cardinalities of the relationships between the tables are, admission to transaction: One-to-many. Each applicant can have multiple transactions associated with it, such as entrance exam or document submissions. Admission to applicants: One-to-many. Each admission can have multiple applicants. Admission to examination: One-to-one. Each student is associated with exactly one entrance examination.

Data Dictionary

Data dictionary in the online appointment management database, this provides information about the data used in a database or information system to serves as a comprehensive reference for understanding the structure. An E-R model facilitates comprehension by offering a diagrammatic and graphical representation of many entities, their properties, and their relationships with each other. Duplication and other problems are thereby decreased, and the data structure is made simpler to understand. Each dataset is described in the online appointment management database's data dictionary. Table name, field name, description, field type, field size, and constraints. The entity's attributes are contained in each table and can be utilized as a reference by upcoming developers. There are four tables in the database.

Table 3

APPLICANT DATA DICTIONARY

FIELD NAME	DESCRIPTION	FIELD TYPE	FIELD SIZE	CONSTRAINTS
applicant_id	Primary Key. Stores the ID assigned for every student	int	11	Null
learners_ref_no	Stores the LRN of the student	varchar	50	Null
fname	Stores the first name of the student	varchar	50	Not null
middlename	Stores the middle name of the student	varchar	50	Not null
lname	Stores the last name of the student	varchar	50	Not null
name_extension	Stores the extension name of the student	varchar	10	Null
civil_status	Stores the civil status of the student	varchar	20	Not null
sex	Stores the sex of the student	varchar	10	Not null
religion	Stores the religion of the student	varchar	50	Not null
date_birth	Stores the date of birth of the student	date		Not null
birth_place	Stores the birth place of the student	varchar	100	Not null
contact_num	Stores the contact number of the student	varchar	20	Unique key
email	Stores the email of the student	varchar	100	Unique key
school_name	Stores the school name of the student	varchar	45	Not null
strand	Stores the strand of the student	varchar	20	Not null
admission_type	Stores the admission type of the student	varchar	20	Not null
tertiary_school	Stores the tertiary of the student	varchar	45	Not null
last_course	Stores the last course of the student	varchar	45	Not null
last_year_level	Stores the last year level of the student	varchar	45	Not null
full_address	Stores the address of the student	varchar	50	Not null

The applicant data table data dictionary was shown in Table 3. The table serves a distinct purpose in organizing and managing comprehensive information related to applicants. It has a

primary key of the applicant id. This applicant data table will be significant since it will be responsible for storing all the data of the applicants, including learner's reference number, first name, middle name, last name, name extension, civil status, sex, religion, date of birth, place of birth, contact number, and email, school name, strand, admission type, tertiary school, last course, last year level, and full address.

The next table shows the data dictionary of appointment. Each dataset is described in the online appointment management database's data dictionary. Table name, field name, description, field type, field size, and constraints.

Table 4

APPOINTMENT DATA DICTIONARY

FIELD NAME	DESCRIPTION	FIELD TYPE	FIELD SIZE	CONSTRAINTS
transaction_id	Primary Key. Stores the transaction ID	Int	20	Not null
applicant_id	Foreign Key. Stores the student ID	Int	11	Not null
date	Stores the date of appointment	Date		Not null
time	Stores the time of appointment	time		Not null
status	Stores the status of appointment	Varchar	20	Not null

The appointment table data dictionary was shown in Table 4. The table serves a distinct purpose in organizing and managing comprehensive information related to appointment. It has a primary key the transaction id and foreign key of applicant id. The data dictionary for appointment has significant to because it is responsible for storing all the data in appointment of the applicants. This appointment table data dictionary includes date of appointment, time of appointment, and status.

The next table shows the data dictionary of appointment. Each dataset is described in the online appointment management database's data dictionary. Table name, field name, description, field type, field size, and constraints.

Table 5

EXAMINATION DATA DICTIONARY

FIELD NAME	DESCRIPTION	FIELD TYPE	FIELD SIZE	CONSTRAINTS
exam_id	Primary Key. Stores the exam ID	Int	20	Not null
applicant_id	Foreign Key. Stores the student ID	Int	11	Not null
date	Stores the date of examination	Date		Not null
time	Stores the time of examination	time		Not null
location	Stores the location of examination	Varchar	50	Not null
duration	Stores the duration of examination	varchar	50	Not null
result	Stores the result of examination	double		Not null

The appointment table data dictionary was shown in Table 5. The table serves a distinct purpose in organizing and managing comprehensive information related to examination. The primary key in this table is the exam id and the foreign key is the applicant id, in this table the entities included are date, time, location, duration of examination and result.

The next table shows the data dictionary of appointment. Each dataset is described in the online appointment management database's data dictionary. Table name, field name, description, field type, field size, and constraints.

Table 6

ADMISSION DATA DICTIONARY

FIELD NAME	DESCRIPTION	FIELD TYPE	FIELD SIZE	CONSTRAINTS
admission_id	Primary Key. Stores the admission ID	Int	20	Not null
applicant_id	Foreign Key. Stores the student ID	Int	20	Not null
preferred_course	Stores the preferred course of the student	Varchar	100	Not null
second_course	Stores the second course of the student	Varchar	100	Not null
third_course	Stores the third course of the student	Varchar	100	Not null
preferred_modality	Stores the preferred modality of the student	varchar	20	Not null
image	Stores the image of the student	varchar	225	Not null

The admission table data dictionary was shown in Table 6. The table serves a distinct purpose in organizing and managing comprehensive information related to admission. The primary key is the admission id and the foreign key is the student id, this data dictionary for admission consists of preferred course, second course, third course and preferred modality and image of the applicant.

Test phase

In the project's testing phase, a pivotal step involved evaluating a low-fidelity prototype. To achieve this, an extensive survey was undertaken, engaging 30 participants. The prototype was thoughtfully crafted to mirror the core functionalities and user interface envisioned for appointment management, intentionally maintaining a simplified design. Participants were presented with this prototype and requested to offer feedback and insights based on their interaction with it.

The table below shows the result of evaluation for User Interface, it shows the level of agreement of the users in terms of visibility of the system status. This table also shows the weighted mean and verbal interpretation of the prototype of the user interface.

Table 7

USER INTERFACE EVALUATION RESULT		
STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
I. USER INTERFACE		
A. Visibility of system status		
1. The prototype effectively conveys the simulated status and feedback of the visualize system.	4.3	Strongly Agree
2. I can easily interpret and understand the status indicators presented in the prototype.	4.73	Strongly Agree
3. The prototype provides clear and concise information about the simulated processes and activities.	4.2	Agree
4. The status updates within the prototype align well with what I would expect from a fully developed system.	4.3	Strongly Agree
5. Even in this prototype stage, the system status is visibly represented, contributing to a sense of user awareness.	4.16	Agree
OVERALL MEAN	4.33	Strongly Agree

Table 7 shows the result of evaluation for user interface in terms of visibility of system status. The study of the prototype's system status visibility yielded very favorable results regarding user response. Statements about how well simulated status and feedback were conveyed, how simple it was to understand status indicators, and how succinctly and clearly information about simulated processes and activities was provided were all overwhelmingly strongly agreed upon by participants. There was a great consensus over the alignment of status updates in the prototype with what is expected of a fully built system, which strengthened trust in the prototype's representation. The prototype was successful in conveying feedback and system status, as evidenced by the overall mean rating of 4.33, which strongly agrees. This confirms the prototype's potential as an easy-to-use and intuitive interface.

The next table shows the result of evaluation for User Interface, it shows the level of agreement of the users in terms of match between system and the real world. This table also shows the weighted mean and verbal interpretation of the prototype of the user interface.

Table 8

USER INTERFACE EVALUATION RESULT

STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
B. Match between system and the real world		
1. The prototype uses terminology that aligns well with real-world language and concepts.	4.56	Strongly Agree
2. The actions and interactions in the prototype are closely similar to real-world scenarios.	4.36	Strongly Agree
3. The prototype's interface effectively copies real-world objects or processes, creating a sense of familiarity.	4.56	Strongly Agree
4. I find that the prototype's commands and functionalities are consistent with what I would expect in a real-world context.	4.6	Strongly Agree
5. Interacting with the prototype feels natural, and it accurately reflects how I would perform tasks in the real world.	4.3	Strongly Agree
OVERALL MEAN	4.47	Strongly Agree

Table 8 shows the result of evaluation for user interface in terms of match between system and the real world. The results of the evaluation of how well the system matches the real-world point to a very positive user experience. Participants regularly indicated that they strongly agreed with claims made regarding how well the prototype employs terminology and real-world language. Users also reported feeling a sense of familiarity because the prototype's interactions and behaviors were quite similar to actual events. They also said that the prototype's functions and instructions met their expectations for real-world scenarios. The prototype's great match between the system and real-world experiences is highlighted by its impressive overall mean of 4.47, which indicates a strongly agree rating. This indicates that the prototype has the ability to offer users an intuitive and user-friendly interface.

The table below shows the result of evaluation for User Interface, it shows the level of agreement of the users in terms of user control and freedom. This table also shows the weighted mean and verbal interpretation of the prototype of the user interface.

Table 9

USER INTERFACE EVALUATION RESULT		
STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
C. User control and freedom		
1. The prototype provides clear options and pathways, giving me a sense of control over my interactions.	4.16	Agree
2. I feel that the prototype allows me the freedom to navigate and explore features according to my preferences.	4.33	Strongly Agree
3. The prototype offers sufficient user guidance and feedback, empowering me to make informed choices.	3.9	Agree
4. I find it easy to backtrack actions within the prototype, enhancing my perceived control and freedom.	4.6	Strongly Agree
5. I feel a sense of freedom and flexibility in how I can interact with the simulated system.	4.53	Strongly Agree
OVERALL MEAN	4.30	Strongly Agree

Table 9 shows the result of evaluation for user interface in terms of user control and freedom. An evaluation of the prototype's user freedom and control indicates a satisfying experience for the user. Due to the prototype's clear options and pathways, participants reported feeling quite in control of their interactions. A particularly strong consensus was reached about the freedom of exploration afforded by the prototype, with users feeling empowered to explore and navigate features in accordance with their preferences. Although the provision of user help and feedback received a somewhat lower rating, users still felt that it was sufficient to make well-informed decisions. One of the prototype's best features was how simple it was to go back and change one's mind, giving consumers a greater sense of power and autonomy. The prototype was successful in giving users a sense of autonomy, flexibility, and control over their interactions with the simulated system, as evidenced by the overall mean rating of 4.30, which highly agrees.

The table below shows the result of evaluation for User Interface, it shows the level of agreement of the users in terms of consistency and standard. This table also shows the weighted mean and verbal interpretation of the prototype of the user interface.

Table 10

USER INTERFACE EVALUATION RESULT		
STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
D. Consistency and Standards		
1. The elements and interfaces within the prototype follow consistent design patterns and standards.	4.6	Strongly Agree
2. I find that the prototype sticks to established conventions, making it easy to predict the behavior of different components.	4.16	Agree
3. The use of terminology and symbols in the prototype aligns with industry or user interface standards.	4.5	Strongly Agree
4. The interactions and workflows within the prototype maintain a high level of consistency and conformity.	4.3	Strongly Agree
5. The design and structure demonstrate a commitment to consistency and adherence to established standards.	4.1	Agree
OVERALL MEAN	4.33	Strongly Agree

Table 10 shows the result of evaluation for user interface in terms of consistency and Standards. The results of the evaluation of the prototype's consistency and standards show that it adheres closely to accepted design patterns and industry standards. Users overwhelmingly agreed that the prototype's parts and interfaces adhered to standardized design principles, promoting predictability in the actions of various components. Strong confirmation that the terminology and symbols used adhere to industry or user interface standards was also received. A strong dedication to uniformity and adherence to set standards in the general design and structure was also observed by participants, as was the high degree of consistency and conformity in interactions and workflows. The prototype has succeeded in keeping a consistent and standards-

compliant user interface, which promises a predictable and familiar user experience, as seen by the strong overall mean of 4.33, which indicates a strongly agree rating.

The table below shows the result of evaluation for User Interface, it shows the level of agreement of the users in terms of recognition rather than recall. This table also shows the weighted mean and verbal interpretation of the prototype of the user interface.

Table 11

USER INTERFACE EVALUATION RESULT

STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
E. Recognition rather than Recall		
1. The prototype presents information in a way that allows me to recognize and select options without having to recall specific details.	4.16	Agree
2. I find that the prototype's interface includes visible signs that aid in recognizing available actions or features.	4.23	Strongly Agree
3. The prototype's design minimizes the need for me to remember complex sequences of steps; instead, options are easily recognizable.	4.36	Strongly Agree
4. I can easily identify and understand icons, labels, or symbols within the prototype, reducing the cognitive load on recall.	4.6	Strongly Agree
5. The prototype's layout and structure promote recognition of relevant information, minimizing the need for users to remember specific details.	4.3	Strongly Agree
OVERALL MEAN	3.48	Agree

Table 11 shows the result of evaluation for user interface in terms of recognition rather than Recall. Users generally agreed on the effectiveness of the interface in promoting recognition, according to the findings from the assessment of recognition rather than recall in the prototype, indicating a favorable user experience. Strong support was given to the addition of obvious signals that make accessible actions or features easier to recognize. Users commented that the architecture of the prototype effectively reduced the need to memorize intricate step sequences,

with options being readily identifiable. Furthermore, users reported that the prototype's images, labels, and symbols were simple to recognize and comprehend, which helped to lessen the cognitive burden related to memory. The prototype can be further optimized for improved identification and decreased dependency on recollection for an even more efficient user experience, even though the overall mean of 3.48 indicates an agreement rating.

The table below shows the result of evaluation for User Interface, it shows the level of agreement of the users in terms of flexibility and efficiency of use. This table also shows the weighted mean and verbal interpretation of the prototype of the user interface.

Table 12

USER INTERFACE EVALUATION RESULT

STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
F. Flexibility and efficiency of use		
1. The prototype allows for customization and adapts well to different user preferences, enhancing flexibility.	4.63	Strongly Agree
2. I feel that the prototype's design enables me to perform tasks efficiently, with minimal unnecessary steps or interactions.	4.16	Agree
3. The prototype provides shortcuts or advanced features for experienced users, contributing to efficiency.	4.66	Strongly Agree
4. I find that the prototype supports a range of user abilities and skill levels, offering a flexible and inclusive experience.	4.13	Agree
5. This prototype stage, I recognize the balance between flexibility and efficiency, making it adaptable to various user needs.	4.23	Strongly Agree
OVERALL MEAN	4.36	Strongly Agree

Table 12 shows the result of evaluation for user interface in terms of flexibility and Efficiency of use. The results of the assessment of the prototype's adaptability and usability point to a very favorable user impression. The prototype's ability to be customized, effectively accommodating varying user preferences and augmenting overall flexibility was universally acknowledged by the participants. The design of the prototype allowed users to complete tasks

with the least number of needless steps or interactions, and they reported a positive experience in terms of task efficiency. Strong support was given to the idea of offering advanced features or shortcuts to more seasoned users, as this boosted productivity. Participants also concurred that the prototype provides a flexible and inclusive experience by accommodating a variety of user abilities and skill levels. The total mean of 4.36, which denotes a strongly agree rating, highlights how well the prototype meets users' wants and preferences by offering an adaptable and effective interface.

The table below shows the result of evaluation for User Interface, it shows the level of agreement of the users in terms of aesthetic and minimalist design. This table also shows the weighted mean and verbal interpretation of the prototype of the user interface.

Table 13

USER INTERFACE EVALUATION RESULT

STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
G. Aesthetic and Minimalist design		
1. The prototype exhibits a visually appealing and aesthetically pleasing design.	4.53	Strongly Agree
2. I appreciate the use of minimalist elements in the prototype, contributing to a clean and uncluttered interface.	4.6	Strongly Agree
3. The color scheme and visual elements in the prototype are thoughtfully chosen, enhancing overall aesthetics.	4.6	Strongly Agree
4. The prototype's design prioritizes simplicity and clarity, aligning with a minimalist approach.	4.56	Strongly Agree
5. I find the design to be both visually pleasing and functionally minimalist.	4.5	Strongly Agree
OVERALL MEAN	4.55	Strongly Agree

Table 13 shows the result of evaluation for user interface in terms of aesthetic and Minimalist design. The results of the evaluation of the prototype's minimalistic and beautiful design show that users have responded very well to it. Participants were largely in strong

agreement that the prototype had an attractive and well-designed appearance. Strong approval was given to the prototype's usage of minimalist components, which helped create a clear and uncluttered interface. The deliberate selection of the color palette and other visual components, which improved the design's overall aesthetics, was also praised by users. Participants praised the prototype's dedication to clarity and simplicity, which is consistent with a minimalist approach. They also found the design to be both aesthetically beautiful and practically minimalist. The prototype was successful in striking a balance between aesthetics and minimalist design, as evidenced by the strong overall mean of 4.55, which indicates a strongly agree rating. This suggests that the interface will be both aesthetically pleasing and easy to use.

The next table shows the total average of the evaluation of the level of agreement of the user interface in terms of visibility of the system status, match between the system and the real world, user control and freedom, consistency and standard, recognition rather than recall, flexibility and efficiency of use, and aesthetic and minimalist design.

Table 14

SUMMARY OF THE EVALUATION RESULT OF USER INTERFACE

STATEMENTS	OVERALL WEIGHTED MEAN	VERBAL INTERPRETATION
1. Visibility of the system status	4.33	Strongly Agree
2. Match between the system and the real world	4.47	Strongly Agree
3. User control and freedom	4.30	Strongly Agree
4. Consistency and standard	4.33	Strongly Agree
5. Recognition rather than recall	3.48	Agree
6. Flexibility and efficiency of use	4.36	Strongly Agree
7. Aesthetic and minimalist design	4.55	Strongly Agree
OVERALL MEAN	4.26	Strongly Agree

Presented in table 14, the summary of results gathered upon the evaluation of students and admission staff in Camarines Sur Polytechnic Colleges Admission Office on the User Interface of Online appointment management. The overall result of the agreement as shown in table 14 is strongly agree (4.26) with the following weighted mean 4.33, 4.47, 4.30, 4.33, 3.48,

4.36, 4.55 for visibility of the system status, match between the system and the real world, user control and freedom, consistency and standard, recognition rather than recall, flexibility and efficiency of use, and aesthetic and minimalist design.

This result shows that in overall, Online Appointment Management in CSPC Admission Office is strongly agree to the users. In terms of visibility of the system status the designed user interface gained a strongly level of agreement. In terms of match between system and the real word the designed user interface gained strongly level of agreement to the users. In terms of user control and freedom the designed user interface gained a strongly level of agreement. In terms of consistency and standard the designed user interface gained a strongly level of agreement. In terms of recognition rather than recall the designed user interface gained agree in level of agreement. In terms of flexibility and efficiency of use the designed user interface gained a strongly level of agreement. In terms of aesthetic and minimalist deign the designed user interface gained a strongly level of agreement.

The previously mentioned results were attained as a result of the researchers' meticulous examination of the admissions office's appointment management low-fidelity prototype. Users also appear to think that the prototype is dependable, orderly, and ready to accommodate future expansion. This is encouraging for the prototype long-term survival and utility. Richard Harper, Abigail Sellen, Stuart K. Card, Steven M. Easterly, and Jacqueline A. Sedmak, who noted that "the importance of designing interfaces that give users a sense of agency and control over their interactions" support this interpretation. In fact, the appointment management low-fidelity prototype achieved a very respectable outcome.

It simply means that all of the needs expressed by users may be satisfied by the low-fidelity prototype. Therefore, in order to get a positive review from people, it is imperative that you always consider satisfying their needs.

However, it is crucial to acknowledge that the favorable opinions conveyed might not fully account for possible drawbacks. A full assessment ought to consider variables including potential biases, contextual limitations, and the particular application scope in order to ensure a thorough comprehension of the system's capabilities and potential downsides.

The table below shows the result of evaluation for Database, it shows the level of agreement of the users in terms of data integrity. This table also shows the weighted mean and verbal interpretation of the designed database.

Table 15

DATABASE EVALUATION RESULT		
STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
II. DATABASE		
A. Data Integrity		
8. The database design includes proper mechanisms to ensure data integrity (e.g., unique keys, foreign keys).	4.53	Strongly Agree
9. Data integrity constraints are clearly defined and well-implemented in the database.	4.2	Agree
10. Data accuracy and consistency are maintained effectively in the database.	4.2	Agree
OVERALL MEAN	4.31	Strongly Agree

Table 15 shows the result of evaluation for database in terms of data integrity. The results of the assessment of the database design's data integrity show that users have a high degree of confidence. The majority of participants strongly agreed that appropriate data integrity-ensuring techniques, including unique keys and foreign keys, should be included in database designs. Users generally thought that the database effectively preserved data accuracy and consistency, even if there was a somewhat lower but still good agreement with the clarity and implementation of data integrity constraints. The database design was successful in giving users trust about the integrity of the data, as seen by the aggregate mean rating of 4.31, which indicates a highly agree

rating. The sturdy basis for the overall reliability of the database system is facilitated by the appropriate implementation of key procedures and the efficient maintenance of data accuracy.

The table below shows the result of evaluation for Database, it shows the level of agreement of the users in terms of normalization. This table also shows the weighted mean and verbal interpretation of the designed database.

Table 16

DATABASE EVALUATION RESULT

STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
B. Normalization		
1. The database design applies appropriate normalization techniques to minimize data redundancy.	4.36	Strongly Agree
2. Normalization has been performed without sacrificing data usability.	4.03	Agree
3. The level of normalization applied is suitable for the database's purpose and performance requirements.	4.2	Agree
OVERALL MEAN	4.19	Agree

Table 16 shows the result of evaluation for database in terms of normalization. Positive user perception is indicated by the evaluation of normalization in the database design findings. Participants overwhelmingly agreed that proper normalization procedures were used to reduce redundant data, indicating a dedication to effective and well-organized data organization. Although there was a little reduced but still favorable consensus about normalizing without compromising data usability, users believed that the degree of normalization implemented was appropriate for the database's intended usage and performance needs. With an overall mean of 4.19, which denotes an agree rating, it appears that consumers understand how normalization can reduce redundancy while preserving usability, which helps to create an organized and effective database architecture.



The table below shows the result of evaluation for Database, it shows the level of agreement of the users in terms of scalability. This table also shows the weighted mean and verbal interpretation of the designed database.

Table 17

DATABASE EVALUATION RESULT		
STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
C. Scalability		
1. The prototype provides clear options and pathways, giving me a sense of control over my interactions.	3.9	Agree
2. I feel that the prototype allows me the freedom to navigate and explore features according to my preferences.	4.1	Agree
3. The prototype offers sufficient user guidance and feedback, empowering me to make informed choices.	4.3	Strongly Agree
OVERALL MEAN	4.1	Agree

Table 17 shows the result of evaluation for database in terms of scalability. Participants rated user assistance and feedback slightly higher than other features, but overall, they agreed that the prototype gives users a sense of power and freedom in their interactions. Although the context is not in line with a system's scalability, the overall mean of 4.1, or an agree rating, indicates that users found the prototype to be generally supportive in terms of control and direction and easy to use.

The next table shows the total average of the evaluation of the level of agreement of the designed database in terms of data integrity, normalization, and scalability.

Table 18

SUMMARY OF THE EVALUATION RESULT OF DATABASE		
STATEMENTS	OVERALL WEIGHTED MEAN	VERBAL INTERPRETATION
1. Data integrity	4.31	Strongly Agree
2. Normalization	4.19	Agree
3. Scalability	4.1	Agree
OVERALL MEAN	4.2	Agree

Presented in table 18, the summary of results gathered upon the evaluation of students and admission staff in Camarines Sur Polytechnic Colleges Admission Office on the Database of Online appointment management. The overall result of the agreement as shown in table 18 is Agree (4.2) with the following weighted mean 4.31, 4.19, 4.1, for data integrity, normalization, and scalability.

This result shows that in overall, Online Appointment Management in CSPC Admission Office is agree to the users. In terms of data integrity the designed database gained a strongly level of agreement. In terms of normalization the designed database gained agrees to the users. In terms of scalability the designed database gained agrees in the level of agreement.

The previously mentioned results were attained as a result of the researchers' meticulous examination of the admissions office's appointment procedure database. Users also appear to think that the system is dependable, orderly, and ready to accommodate future expansion. This is encouraging for the system's long-term survival and utility. Ralph H. Sprague, Judith S. Carlson, Silberschatz, Korth, Sudarshan, and Martin Fowler, who noted that "the importance of normalization and scalability is efficient for data storage, retrieval, and manipulation" and "the impact of data integrity on organizational success and user satisfaction," support this interpretation. In fact, the appointment management database achieved a very respectable outcome. It merely implies that the database is sufficient to meet every need that users have indicated. As a result, it is essential to always take into account meeting the wants of consumers in order to receive a highly favorable evaluation from them.

However, it is crucial to acknowledge that the favorable opinions conveyed might not fully account for possible drawbacks. A full assessment ought to consider variables including potential biases, contextual limitations, and the particular application scope in order to ensure a thorough comprehension of the system's capabilities and potential downsides.



Thus, in terms of data integrity, standardization, and scalability, the prototype finds the database to be extremely acceptable as it satisfies the functional requirements required by the appointment management system.

Notes

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Chapter 5

SUMMARY OF FINDINGS, CONCLUSIONS AND RECOMMENDATIONS

In this chapter, summary of the study, findings, conclusions and recommendations were presented it also includes the restructuring of database and redesigning the graphic user interface for the new version of appointment process. Conclusions and recommendations were obtained based from the data gathered by the researchers.

Summary

The goal of the project, "Analysis and Design of Online Appointment Management in Camarines Sur Polytechnic Colleges Admission Office," was to enhance the user interface design of the appointment process currently in used by the CSPC Admission Office. Its specific goal was to identify and evaluate the issues with the appointment process's database performance that the students faced, as well as the issues that users had with the design of the user interface. In light of this, the main goals of this study are to analyze, evaluate, and design the Camarines Sur Polytechnic Colleges Admission Office appointment process.

Its specific goal was to determine the problem of User Interface based on the Heuristic Evaluation Criteria as well as its visibility of design status, match between design and the real world, User control and freedom, Consistency and Standards, Recognition rather than Recall, Flexibility and Efficiency of use, and Aesthetic and Minimalist design. In addition to analyzing the database and user interface of the current appointment process, the study will identify any problems or concerns with the database and user interface (UI). Lastly, before the online appointment management is analyzed and designed, modifications and enhancements will be suggested.

The study's primary focus is on database analysis and assessment as well as user interface evaluation and design. As a result, only those procedures and problems that interfere with specific database performance and user satisfaction with graphic user interfaces will be examined.

This research utilized the mixed-method approach. The respondents were the faculty of the admission office and the students of Camarines Sur Polytechnic Colleges. After the collection of the data through an interview guide for S01 and survey questionnaire for S03, analysis was done using the thematic analysis and weighted mean.

To produce a quality design, the researchers follow the phases of the Design thinking framework methodically when designing the online appointment management.

Findings

The following conclusions were reached when the data was analyzed:

1. Based from the conducted interview to the students of CSPC and staff members of the Admission Office, these are the following problems encountered in current online admission process in Student Testing and Admission: Time Consuming, Duplication of Applicant's Name, Lack of automation in appointment confirmations, Lack of automation in appointment confirmations, Handling a High Volume of Applicants and Complex Application Process.
2. Design Thinking Framework guided the researchers to propose the User Interface and Database Design. Further, the researchers used Wix to design the User Interface and MySQL as database.
3. Upon the analysis of the result of evaluation in user interface, researchers came up with the findings:

3.1. Visibility of system status. The study of the prototype's system status visibility yielded very favorable results regarding user response. Statements about how well simulated status and feedback were conveyed, how simple it was to understand status indicators, and how succinctly and clearly information about simulated processes and activities was provided were all overwhelmingly strongly agreed upon by participants. There was a great consensus over the alignment of status updates in the prototype with what is expected of a fully built system, which strengthened trust in



the prototype's representation. The users also remarked that the system's state was clearly shown, which greatly enhanced their awareness of it even at this early level. The prototype was successful in conveying feedback and system status, as evidenced by the overall mean rating of 4.33, which strongly agrees. This confirms the prototype's potential as an easy-to-use and intuitive interface.

3.2. Match between system and the real world. The results of the evaluation of how well the system matches the real-world point to a very positive user experience. Participants regularly indicated that they strongly agreed with claims made regarding how well the prototype employs terminology and real-world language. Users also reported feeling a sense of familiarity because the prototype's interactions and behaviors were quite similar to actual events. Strong support was given to the interface's capacity to imitate real-world items or procedures, which added to its overall coherence and consistency. Users said that engaging with the prototype was natural and accurately reflected how they would complete tasks in the real world. They also said that the prototype's functions and instructions met their expectations for real-world scenarios. The prototype's great match between the system and real-world experiences is highlighted by its impressive overall mean of 4.47, which indicates a strongly agree rating. This indicates that the prototype has the ability to offer users an intuitive and user-friendly interface.

3.3. User control and freedom. An evaluation of the prototype's user freedom and control indicates a satisfying experience for the user. Due to the prototype's clear options and pathways, participants reported feeling quite in control of their interactions. A particularly strong consensus was reached about the freedom of exploration afforded by the prototype, with users feeling empowered to explore and navigate features in accordance with their preferences. Although the provision of user help and feedback

received a somewhat lower rating, users still felt that it was sufficient to make well-informed decisions. One of the prototype's best features was how simple it was to go back and change one's mind, giving consumers a greater sense of power and autonomy. The prototype was successful in giving users a sense of autonomy, flexibility, and control over their interactions with the simulated system, as evidenced by the overall mean rating of 4.30, which highly agrees.

3.4. Consistency and Standards. The results of the evaluation of the prototype's consistency and standards show that it adheres closely to accepted design patterns and industry standards. Users overwhelmingly agreed that the prototype's parts and interfaces adhered to standardized design principles, promoting predictability in the actions of various components. Strong confirmation that the terminology and symbols used adhere to industry or user interface standards was also received. A strong dedication to uniformity and adherence to set standards in the general design and structure was also observed by participants, as was the high degree of consistency and conformity in interactions and workflows. The prototype has succeeded in keeping a consistent and standards-compliant user interface, which promises a predictable and familiar user experience, as seen by the strong overall mean of 4.33, which indicates a strongly agree rating.

3.5. Recognition rather than Recall. Users generally agreed on the effectiveness of the interface in promoting recognition, according to the findings from the assessment of recognition rather than recall in the prototype, indicating a favorable user experience. The prototype, according to the participants, presented information in a way that made it possible for them to identify and choose possibilities without having to remember precise specifics. Strong support was given to the addition of obvious signals that make accessible actions or features easier to recognize. Users commented

that the architecture of the prototype effectively reduced the need to memorize intricate step sequences, with options being readily identifiable. Furthermore, users reported that the prototype's images, labels, and symbols were simple to recognize and comprehend, which helped to lessen the cognitive burden related to memory. The prototype can be further optimized for improved identification and decreased dependency on recollection for an even more efficient user experience, even though the overall mean of 3.48 indicates an agreement rating.

3.6. Flexibility and Efficiency of use. The results of the assessment of the prototype's adaptability and usability point to a very favorable user impression. The prototype's ability to be customized, effectively accommodating varying user preferences and augmenting overall flexibility was universally acknowledged by the participants. The design of the prototype allowed users to complete tasks with the least number of needless steps or interactions, and they reported a positive experience in terms of task efficiency. Strong support was given to the idea of offering advanced features or shortcuts to more seasoned users, as this boosted productivity. Participants also concurred that the prototype provides a flexible and inclusive experience by accommodating a variety of user abilities and skill levels. The total mean of 4.36, which denotes a strongly agree rating, highlights how well the prototype meets users' wants and preferences by offering an adaptable and effective interface.

3.7. Aesthetic and Minimalist design. The results of the evaluation of the prototype's minimalistic and beautiful design show that users have responded very well to it. Participants were largely in strong agreement that the prototype had an attractive and well-designed appearance. Strong approval was given to the prototype's usage of minimalist components, which helped create a clear and uncluttered interface. The deliberate selection of the color palette and other visual components, which improved

the design's overall aesthetics, was also praised by users. Participants praised the prototype's dedication to clarity and simplicity, which is consistent with a minimalist approach. They also found the design to be both aesthetically beautiful and practically minimalist. The prototype was successful in striking a balance between aesthetics and minimalist design, as evidenced by the strong overall mean of 4.55, which indicates a strongly agree rating. This suggests that the interface will be both aesthetically pleasing and easy to use.

4. Upon the analysis of the result of evaluation in database, researchers came up with the findings:

- 4.1.Data Integrity. The results of the assessment of the database design's data integrity show that users have a high degree of confidence. The majority of participants strongly agreed that appropriate data integrity-ensuring techniques, including unique keys and foreign keys, should be included in database designs. Users generally thought that the database effectively preserved data accuracy and consistency, even if there was a somewhat lower but still good agreement with the clarity and implementation of data integrity constraints. The database design was successful in giving users trust about the integrity of the data, as seen by the aggregate mean rating of 4.31, which indicates a highly agree rating. The sturdy basis for the overall reliability of the database system is facilitated by the appropriate implementation of key procedures and the efficient maintenance of data accuracy.

- 4.2.Normalization. Positive user perception is indicated by the evaluation of normalization in the database design findings. Participants overwhelmingly agreed that proper normalization procedures were used to reduce redundant data, indicating a dedication to effective and well-organized data organization. Although there was a little reduced but still favorable consensus about normalizing without compromising

data usability, users believed that the degree of normalization implemented was appropriate for the database's intended usage and performance needs. With an overall mean of 4.19, which denotes an agree rating, it appears that consumers understand how normalization can reduce redundancy while preserving usability, which helps to create an organized and effective database architecture.

4.3. Scalability. Participants rated user assistance and feedback slightly higher than other features, but overall, they agreed that the prototype gives users a sense of power and freedom in their interactions. Although the context is not in line with a system's scalability, the overall mean of 4.1, or an agree rating, indicates that users found the prototype to be generally supportive in terms of control and direction and easy to use.

Conclusions

The researchers reached the following conclusions considering all of the findings:

1. Due to problems encountered with the highest frequency of duplication of applicant's name or application form, Lack of automation in appointment confirmations and Handling a High Volume of Applicants in Admission Process in CSPC Admission Office. Therefore, if an Online Appointment Management can be develop in Admission Office the staff can avoid the duplication of the applicant's name or application form, also the staff can easily update or notified the students about their admission results. And also, the applicants can avoid the redundancy of submitting their application form and requirements.
2. The designed User Interface and Database in Admission Office represents a comprehensive approach to enhance the User Interface and to satisfy the users. By focusing on appointment scheduling for entrance examination, this study aims to streamline processes, reduce staff workload, avoid duplication of applicant's name, to easily update or notified the applicants about their admission results. The researchers designed Database such as Entity Relationship Diagram might include insights into the structure and relationships within the database.

Additionally, this might draw the conclusions about the potential for data integrity, ease of use and the overall efficiency of data retrieval and manipulation based on the UI Design.

3. All seven Nielsen Heuristics produced extremely positive findings in the prototype's usability study. The system's visibility of system state, resemblance to the real world, user control and freedom, consistency and standards, preference for recognition over recollection, ease of use and flexibility, and elegant yet simple design were all rated highly by users. With a mean rating of 4.33 (highly agree), the prototype effectively communicated feedback and system status in an understandable and straightforward manner. Even at this early stage of development, users believed that the state of the system was easily discernible.

High ratings were also given to the interface's intuitive and user-friendly design. Users perceived the interactions and behaviors as being comfortable and natural, and the terminology and everyday language as being simple to understand. This heuristic had an overall mean rating of 4.47 (highly agree). Users felt in control of their interactions and had a sense of autonomy thanks to the prototype. The freedom to explore features, the ease of undoing mistakes, and the clear options and pathways were all praised by the participants. This heuristic had a mean rating of 4.30 (highly agree).

The prototype complied closely with industry norms and recognized design patterns. Users perceived the vocabulary and symbols as being familiar, and the interface as being predictable and consistent. This heuristic had a mean rating of 4.33 (highly agree). Users largely felt that the interface was effective in fostering recognition, even if the total mean rating for recognition rather than recall was slightly lower at 3.48 (agree). They didn't need to recall specifics; they could quickly recognize and select possibilities.

The prototype's adaptability and usability were excellent. The ability to swiftly and simply finish activities and tailor the UI to their tastes was well-liked by users. This heuristic had a mean rating of 4.36 (highly agree). Lastly, many were struck by how elegant and

simple the prototype's design was. They thought the UI was aesthetically pleasing, simple to use, and clear. This heuristic had a mean rating of 4.55 (highly agree).

Overall, the findings of the usability test show that the prototype is headed toward becoming a successful and intuitive solution. Positive user feedback offers insightful information that may be applied to enhance the interface's functionality and design even more.

4. Regarding data integrity and normalization, the database design's usability assessment produced favorable findings. Users gave the database's capacity to retain correct and consistent data excellent marks; the mean rating for data integrity was 4.31 (very agree), while the mean rating for normalization was 4.19 (agree). The findings show that suitable data integrity-ensuring strategies, including unique and foreign keys, are incorporated into the database design to assist avoid mistakes and inconsistencies. Additionally, users thought that the level of normalization that had been applied was appropriate, indicating that the database is effective and well-organized.

The overall mean rating of 4.1 (agree) indicates that users found the prototype to be generally supportive and straightforward to use, indicating the possibility for good user experience in a fully built system, even though the evaluation of scalability was not directly applicable in the context of the prototype. The database design appears to be headed toward success in terms of data integrity, normalization, and user experience, according to the usability evaluation overall. Still, there are certain things that could be done better, such making data integrity limitations more understandable and optimizing for better scalability in a finished system.

Recommendations

Based on the conclusions discussed above, the researchers recommend the following:

1. Use this study as a basis to develop an Online Appointment System.
2. Develop a secure and well-organized centralized database to store applicant information and appointment schedules.
3. Offer clear instructions and support for both applicants and staff using the appointment management to enhance the user experience and mitigate potential issues.
4. To further validate the findings of the issue observed, future researchers can carry out a comparable study in other pertinent research environments.
5. For future researchers prioritize a user-centered design to enhance user experience.



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APPENDICES

APPENDIX A
EVALUATION TOOLS

INTERVIEW GUIDE

Dear Respondents,

Good day!

We, the researchers of Bachelor of Science in Information System (BSIS), conducting a study entitled “**Analysis and Design for Online Appointment Management in Camarines Sur Polytechnic Colleges Admission Office**”. In line with this we would like to ask for a few minutes of your time to answer the following questions as truthfully as possible. Your participation is highly appreciated and a great help in our study. Rest assured that all personal information and necessary data from this survey will be treated with utmost confidentiality.

Thank you so much for your time and cooperation, God Bless

Researchers
Mary Joy E. Tuyay
Pablo R. Talagtag Jr.
Renela Rose A. Garo

Interview Questions:

1. What is the existing process of application for CSPC College Entrance Examination? Do you have flowchart for the existing process?

2. Did you find that the online application process saved your time compared to the traditional method of scheduling for entrance examination?



3. What are the problems or issues you have encountered when using the existing process?

4. How many applicants come to your office every day to submit the application form and requirements, and how many applicants can take the entrance examination per day?

5. How did the online application process affect the communication between the staff members and applicants?

SURVEY QUESTIONNAIRE

Dear Respondents,

Good day!

We, the researchers of Bachelor of Science in Information System (BSIS), conducting a study entitled “**Analysis and Design for Online Appointment Management in Camarines Sur Polytechnic Colleges Admission Office**”. In line with this we would like to ask for a few minutes of your time to answer the following questions as truthfully as possible. Your participation is highly appreciated and a great help in our study. Rest assured that all personal information and necessary data from this survey will be treated with utmost confidentiality.

Thank you so much for your time and cooperation, God Bless

Researchers
Mary Joy E. Tuyay
Pablo R. Talagtag Jr.
Renela Rose A. Garol

I. PROFILE OF THE RESPONDENT:

Name (Optional): _____
Age: ____

Course: _____
Gender: _Male _Female

Thank you for participating in this survey. Your feedback is essential in helping us improve the design of our online appointment management. Please answer the following questions by selecting the option that best represents your opinion.

II. INSTRUCTIONS:

Please read each statement and put a check on the column of the scale that best suits your answer using the scale values below.

SCALE	RANGE	DESCRIPTION
Strongly Agree	5	Extremely content, with the quality of service prompt response times and overall performance.
Agree	4	Generally find the prototype of Online Appointment Management satisfactory.
Neutral	3	The respondent has neither a positive response nor a negative response, but undecided denotes a state of confusion of the respondent.
Disagree	2	Not satisfied with the design prototype of Online Appointment Management
Strongly Disagree	1	Strongly Disagree that the design of the prototype is helpful and satisfied the user.



I. USER INTERFACE

A. Visibility of system status	5	4	3	2	1
1. The prototype effectively conveys the simulated status and feedback of the visualize system.					
2. I can easily interpret and understand the status indicators presented in the prototype.					
3. The prototype provides clear and concise information about the simulated processes and activities.					
4. The status updates within the prototype align well with what I would expect from a fully developed system.					
5. Even in this prototype stage, the system status is visibly represented, contributing to a sense of user awareness.					
B. Match between system and the real world	5	4	3	2	1
1. The prototype uses terminology that aligns well with real-world language and concepts.					
2. The actions and interactions in the prototype are closely similar to real-world scenarios.					
3. The prototype's interface effectively copies real-world objects or processes, creating a sense of familiarity.					
4. I find that the prototype's commands and functionalities are consistent with what I would expect in a real-world context.					
5. Interacting with the prototype feels natural, and it accurately reflects how I would perform tasks in the real world.					
C. User control and freedom	5	4	3	2	1
1. The prototype provides clear options and pathways, giving me a sense of control over my interactions.					
2. I feel that the prototype allows me the freedom to navigate and explore features according to my preferences.					
3. The prototype offers sufficient user guidance and feedback, empowering me to make informed choices.					



4. I find it easy to backtrack actions within the prototype, enhancing my perceived control and freedom.					
5. I feel a sense of freedom and flexibility in how I can interact with the simulated system.					
D. Consistency and Standards	5	4	3	2	1
1. The elements and interfaces within the prototype follow consistent design patterns and standards.					
2. I find that the prototype sticks to established conventions, making it easy to predict the behavior of different components.					
3. The use of terminology and symbols in the prototype aligns with industry or user interface standards.					
4. The interactions and workflows within the prototype maintain a high level of consistency and conformity.					
5. The design and structure demonstrate a commitment to consistency and adherence to established standards.					
E. Recognition rather than Recall	5	4	3	2	1
1. The prototype presents information in a way that allows me to recognize and select options without having to recall specific details.					
2. I find that the prototype's interface includes visible signs that aid in recognizing available actions or features.					
3. The prototype's design minimizes the need for me to remember complex sequences of steps; instead, options are easily recognizable.					
4. I can easily identify and understand icons, labels, or symbols within the prototype, reducing the cognitive load on recall.					
5. The prototype's layout and structure promote recognition of relevant information, minimizing the need for users to remember specific details					
F. Flexibility and Efficiency of use	5	4	3	2	1
1. The prototype allows for customization and adapts well to					



different user preferences, enhancing flexibility.					
2. I feel that the prototype's design enables me to perform tasks efficiently, with minimal unnecessary steps or interactions.					
3. The prototype provides shortcuts or advanced features for experienced users, contributing to efficiency.					
4. I find that the prototype supports a range of user abilities and skill levels, offering a flexible and inclusive experience.					
5. This prototype stage, I recognize the balance between flexibility and efficiency, making it adaptable to various user needs.					
G. Aesthetic and Minimalist design	5	4	3	2	1
1. The prototype exhibits a visually appealing and aesthetically pleasing design.					
2. I appreciate the use of minimalist elements in the prototype, contributing to a clean and uncluttered interface.					
3. The color scheme and visual elements in the prototype are thoughtfully chosen, enhancing overall aesthetics.					
4. The prototype's design prioritizes simplicity and clarity, aligning with a minimalist approach.					
5. I find the design to be both visually pleasing and functionally minimalist.					

I. DATABASE

A. Data Integrity	5	4	3	2	1
1. The database design includes proper mechanisms to ensure data integrity (e.g., unique keys, foreign keys).					
2. Data integrity constraints are clearly defined and well-implemented in the database.					
3. Data accuracy and consistency are maintained effectively in the database.					



B. Normalization	5	4	3	2	1
1. The database design applies appropriate normalization techniques to minimize data redundancy.					
2. Normalization has been performed without sacrificing data usability.					
3. The level of normalization applied is suitable for the database's purpose and performance requirements.					
C. Scalability	5	4	3	2	1
1. The prototype provides clear options and pathways, giving me a sense of control over my interactions.					
2. I feel that the prototype allows me the freedom to navigate and explore features according to my preferences.					
3. The prototype offers sufficient user guidance and feedback, empowering me to make informed choices.					

Thank you for taking the time to complete this questionnaire. Your feedback is invaluable in enhancing the design for online appointment management.



**TALLY SHEET
USER INTERFACE**

Criteria	Equivalent					Weighted Mean
Visibility of the system status	5	4	3	2	1	
Q1	16	9	5	0	0	4.3
Q2	15	13	2	0	0	4.73
Q3	13	10	7	0	0	4.2
Q4	12	15	3	0	0	4.3
Q5	10	15	5	0	0	4.16
Average Weighted Mean						4.33

Criteria	Equivalent					Weighted Mean
Match between the system and the real world	5	4	3	2	1	
Q1	19	9	2	0	0	4.56
Q2	13	15	2	0	0	4.36
Q3	18	11	1	0	0	4.56
Q4	18	12	0	0	0	4.6
Q5	12	15	3	0	0	4.3
Average Weighted Mean						4.47

Criteria	Equivalent					Weighted Mean
User control and freedom	5	4	3	2	1	
Q1	10	15	5	0	0	4.16
Q2	12	16	2	0	0	4.33
Q3	5	17	8	0	0	3.9
Q4	18	12	0	0	0	4.6
Q5	16	14	0	0	0	4.53
Average Weighted Mean						4.30

Criteria	Equivalent					Weighted Mean
Consistency and standard	5	4	3	2	1	
Q1	18	12	0	0	0	4.6
Q2	10	15	5	0	0	4.16
Q3	20	5	5	0	0	4.5
Q4	12	15	3	0	0	4.3
Q5	9	15	6	0	0	4.1
Average Weighted Mean						4.33

Criteria	Equivalent					Weighted Mean
Recognition rather than recall	5	4	3	2	1	
Q1	8	19	3	0	0	4.16
Q2	11	15	4	0	0	4.23
Q3	16	9	5	0	0	4.36
Q4	18	12	0	0	0	4.6
Q5	12	15	3	0	0	4.3
Average Weighted Mean						3.48



Criteria	Equivalent					Weighted Mean
Flexibility and efficiency of use	5	4	3	2	1	
Q1	19	11	0	0	0	4.63
Q2	13	9	8	0	0	4.16
Q3	16	9	5	0	0	4.66
Q4	9	16	5	0	0	4.13
Q5	11	15	4	0	0	4.23
Average Weighted Mean						4.36

Criteria	Equivalent					Weighted Mean
Aesthetic and minimalist design	5	4	3	2	1	
Q1	16	14	0	0	0	4.53
Q2	18	12	0	0	0	4.6
Q3	19	10	1	0	0	4.6
Q4	18	11	1	0	0	4.56
Q5	15	15	0	0	0	4.5
Average Weighted Mean						4.55

DATABASE

Criteria	Equivalent					Weighted Mean
Data integrity	5	4	3	2	1	
Q1	17	12	1	0	0	4.53
Q2	14	8	8	0	0	4.2
Q3	11	14	5	0	0	4.2
Average Weighted Mean						4.31

Criteria	Equivalent					Weighted Mean
Normalization	5	4	3	2	1	
Q1	16	9	5	0	0	4.36
Q2	8	18	3	0	0	4.03
Q3	11	14	5	0	0	4.2
Average Weighted Mean						4.19

Criteria	Equivalent					Weighted Mean
Scalability	5	4	3	2	1	
Q1	5	17	8	0	0	3.9
Q2	9	15	6	0	0	4.1
Q3	12	15	3	0	0	4.3
Average Weighted Mean						4.1

APPENDIX B

DOCUMENTATION

The photos attached below are the documentation of the researchers during interview in specific objective 1 and also the documentation for evaluation of User Interface and Database Design.



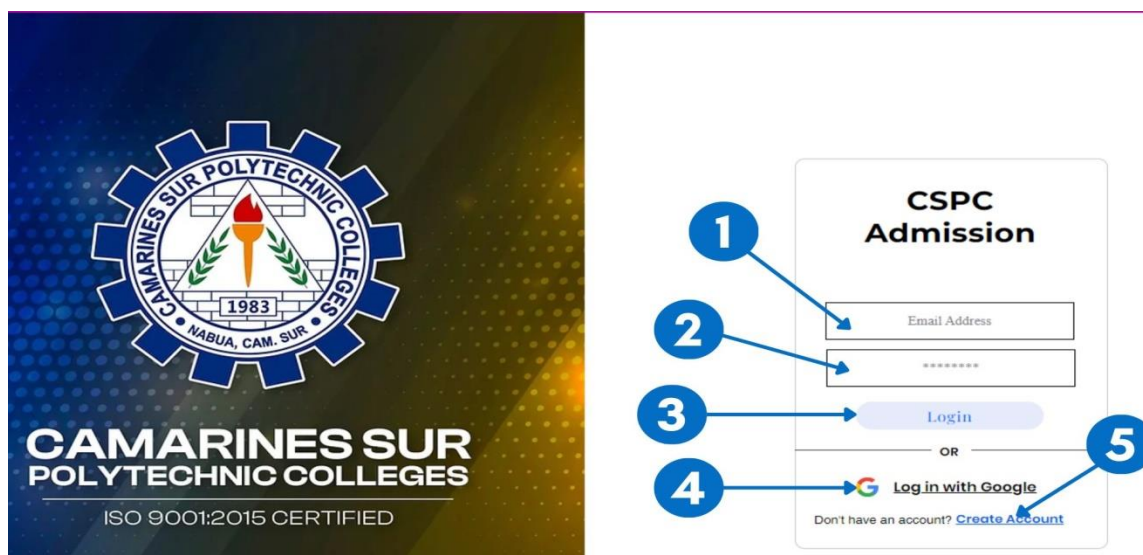
APPENDIX C USER'S GUIDE

1. Landing Page - landing page of online appointment management. In this figure it will show the apply button, but if the applicants wants to see the instruction first they can scroll down to see the instructions, for them to be familiar with the online appointment management.



1. To begin with the online appointment management, the applicants must click the apply button.

2. Login Page (Applicant) - log in page for the students, this is the first step that the students need to fill up to proceed in to the next step, if the students click the apply button, this figure below shows the login page for students.



1. Input Email Address.
2. Input Password.
3. Click Login button.

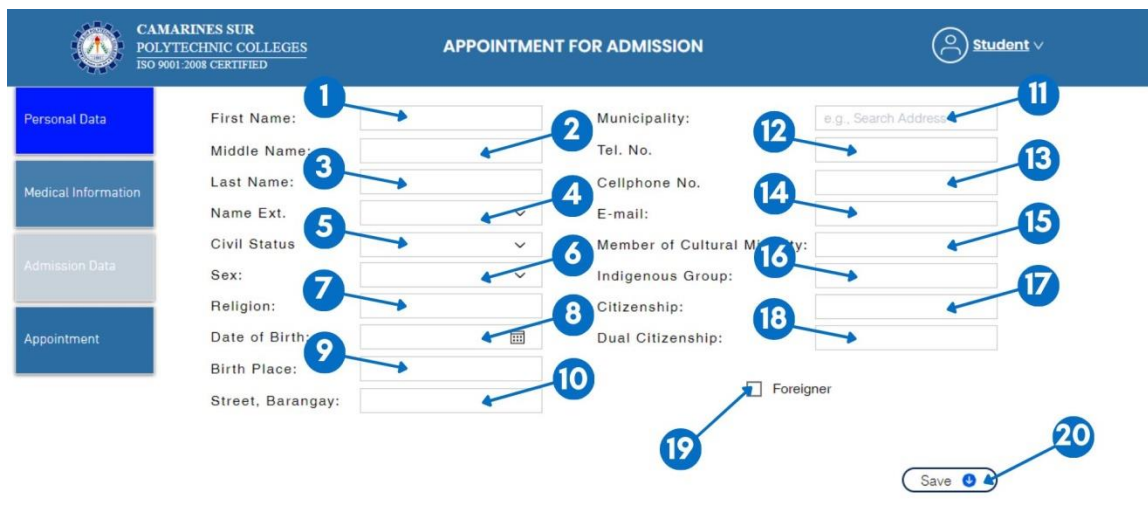
4. The applicant can click login with Google if they have an existing account.
5. If the applicants doesn't have an existing account, they can click the create account button to create account.

3. Main Page – the main page will appear after the applicants click the login button. In this interface the students will see the main menu which is the personal information, medical information, admission data and appointment.



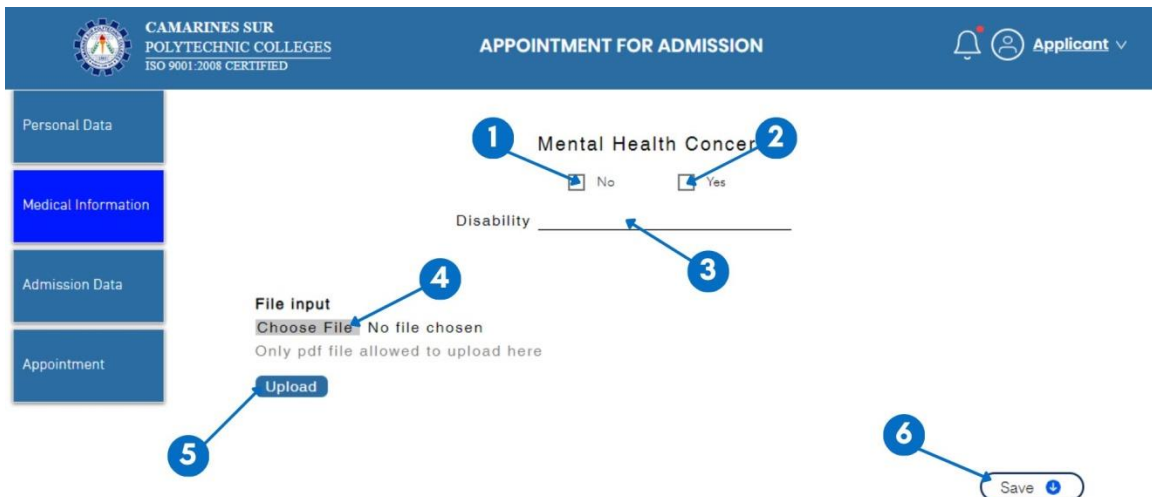
1. Click Personal Data.
2. They can click Medical Information if they want.
3. They can also click the Admission Data
4. Click the Appointment button to schedule their examination.
5. The applicants can click the notification bell.

4. Personal Data - the applicants will provide the needed information for the admission, this is necessary to fill out the form and complete to serve as the personal profile of the students and will help the admissions committee evaluate their personal data, wherein it will be submitted to the administrator and verify the information.



1. Input first name.
2. Input middle name.
3. Input last name.
4. Input name extension (if they have).
5. Input civil status.
6. Input sex.
7. Input religion.
8. Input date of birth.
9. Input birth place.
10. Input Street, barangay
11. Input municipality.
12. Input telephone number (if they have).
13. Input cellphone number.
14. Input e-mail.
15. Input if member of cultural minority.
16. Input if belong to indigenous group.
17. Input citizenship.
18. Input if you are dual citizen.
19. Click if you are foreign.
20. Click save button to successfully save your information.

5. Medical Information - The medical information form is not mandatory to fill out, if the student doesn't have a medical issues or certificate they can also proceed to the next step of the admission.



The screenshot shows the 'APPOINTMENT FOR ADMISSION' form. On the left is a sidebar with four tabs: 'Personal Data', 'Medical Information' (highlighted in blue), 'Admission Data', and 'Appointment'. The main form area contains the following elements with numbered annotations:

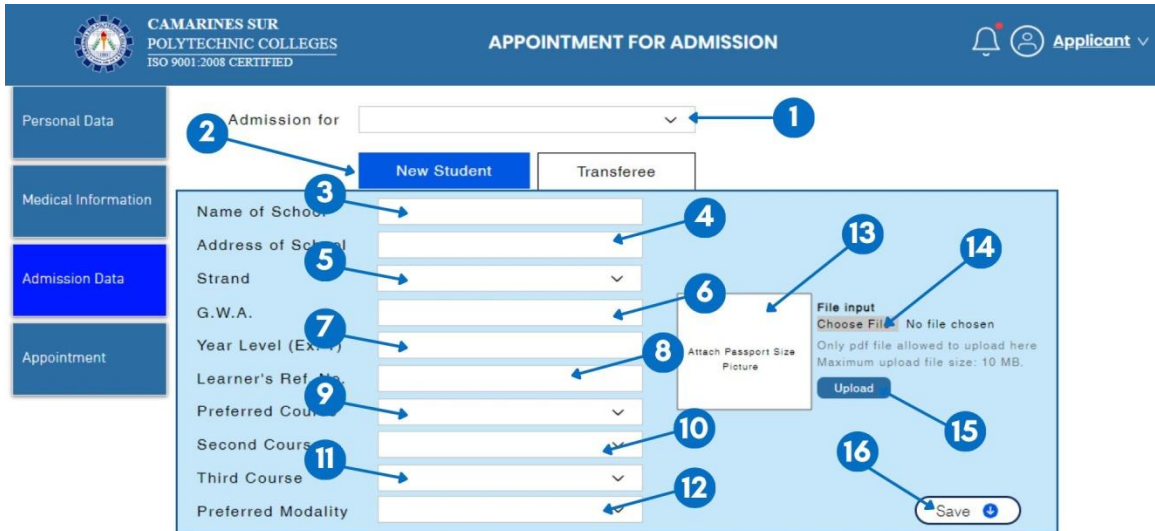
- 1**: Points to the 'Mental Health Concern' label.
- 2**: Points to the 'Yes' radio button.
- 3**: Points to the 'Disability' text input field.
- 4**: Points to the 'Choose File' button in the 'File Input' section.
- 5**: Points to the 'Upload' button.
- 6**: Points to the 'Save' button at the bottom right.

Other visible text includes 'CAMARINES SUR POLYTECHNIC COLLEGES ISO 9001:2008 CERTIFIED' in the top left, 'APPOINTMENT FOR ADMISSION' in the top center, and 'Applicant' with a dropdown arrow in the top right.

1. If the applicant don't have mental health concern click "No" if you have mental concern click "Yes".
2. Input disability.
3. Upload file for medical certificate.

4. Click “Upload”.
5. Click save to successfully update your medical information.

6. Admission Data for New Students - the admission data the students must need to fill out all the information that is needed. This will be their academic information during their last school year.



The screenshot shows the 'APPOINTMENT FOR ADMISSION' form for a 'New Student'. The form is divided into sections: Personal Data, Medical Information, Admission Data, and Appointment. The 'Admission Data' section is highlighted. The form contains the following fields and buttons:

- 1**: Dropdown button to select degree.
- 2**: 'New Student' button.
- 3**: 'Name of School' input field.
- 4**: 'Address of School' input field.
- 5**: 'Strand' dropdown menu.
- 6**: 'G.W.A.' input field.
- 7**: 'Year Level (Ex. 1)' dropdown menu.
- 8**: 'Learner's Ref. No.' input field.
- 9**: 'Preferred Course' dropdown menu.
- 10**: 'Second Course' dropdown menu.
- 11**: 'Third Course' dropdown menu.
- 12**: 'Preferred Modality' dropdown menu.
- 13**: 'Attach Passport Size Picture' button.
- 14**: 'Attach scanned requirements' button.
- 15**: 'Upload' button.
- 16**: 'Save' button.

1. Click dropdown button to select degree.
2. Click “New Student”.
3. Input name of school.
4. Input address of school.
5. Input strand.
6. Input G.W.A.
7. Input year level.
8. Input LRN.
9. Input preferred course.
10. Input second course.
11. Input third course.
12. Input preferred modality.
13. Attach passport size picture.
14. Attach scanned requirements.
15. Click “Upload”.
16. Click “Save” to successfully update your admission data.

7. Admission Data for Transferee - This interface will not appear in the new student button, their interface is separated to avoid confusions to the students, and this interface is the continuation of admission data, but in this figure is applicable only for transferee students.

The screenshot shows the 'APPOINTMENT FOR ADMISSION' form. On the left is a sidebar with buttons: Personal Data, Medical Information, Admission Data (highlighted), and Appointment. The main form has a top header with the college logo, name, and 'APPOINTMENT FOR ADMISSION' title. Below the header are two tabs: 'New Student' and 'Transferee'. The form contains several input fields and a file upload section. Numbered callouts point to specific elements: 1 points to the degree dropdown, 2 to the 'Transferee' tab, 3 to the 'Name of School' field, 4 to the 'Address of School' field, 5 to the 'Strand' field, 6 to the 'Last Year Attended' field, 7 to the 'Last Year Level' field, 8 to the 'Year Level (Expected)' field, 9 to the 'Learner's Ref. No.' field, 10 to the 'Preferred Course' dropdown, 11 to the 'Second Course' dropdown, 12 to the 'Third Course' dropdown, 13 to the 'Preferred Modality' dropdown, 14 to the 'Attach Passport Size Picture' field, 15 to the 'Attach Scanned Requirements' field, 16 to the 'Upload' button, and 17 to the 'Save' button.

1. Click dropdown button to select degree.
2. Click “Transferee”.
3. Input name of school.
4. Input address of school.
5. Input strand.
6. Input last year attended.
7. Input year level.
8. Input year level.
9. Input LRN.
10. Input preferred course.
11. Input second course.
12. Input third course.
13. Input preferred modality.
14. Attach passport size picture.
15. Attach scanned requirements.
16. Click “Upload”.
17. Click “Save” to successfully update your admission data.

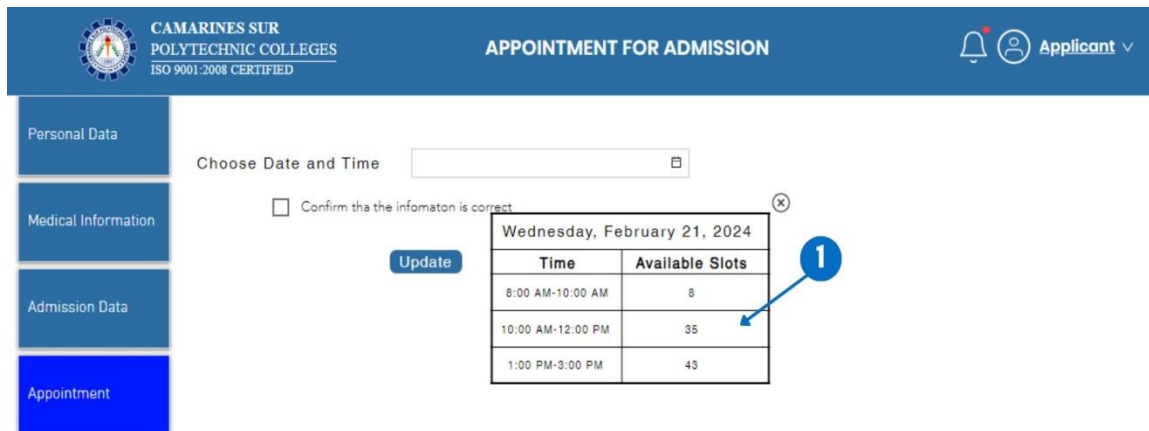
8. Appointment Selection - the appointment of the Online Appointment Management for Camarines Sur Polytechnic Colleges. It includes the date and time that are unavailable and available and the remaining slot for the entrance examination.

The screenshot shows the 'APPOINTMENT FOR ADMISSION' form with the 'Appointment' tab selected. The 'Choose Date and Time' section is active, displaying a calendar for February. A 'Confirm' checkbox is present. Numbered callouts point to specific elements: 1 points to the 'Choose Date and Time' dropdown, and 2 points to the date '14' in the calendar grid.

February						
Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29		

1. Click calendar button.
2. Choose available date for entrance examination.

9. Time Selection - is the time selection wherein the students will choose their preferred time that aligns with their personal preferences for examination.

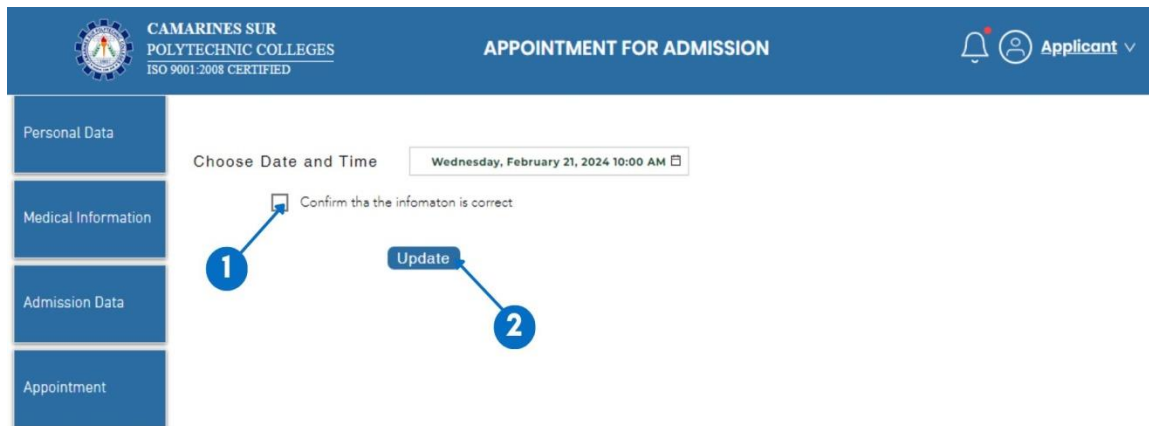


The screenshot shows the 'APPOINTMENT FOR ADMISSION' form. On the left is a sidebar with buttons for 'Personal Data', 'Medical Information', 'Admission Data', and 'Appointment'. The main area has a 'Choose Date and Time' dropdown menu. Below it is a checkbox labeled 'Confirm the information is correct'. An 'Update' button is visible. A modal window is open, displaying the date 'Wednesday, February 21, 2024' and a table of available slots. A blue circle with the number '1' and an arrow points to the 'Available Slots' column in the table.

Time	Available Slots
8:00 AM-10:00 AM	8
10:00 AM-12:00 PM	35
1:00 PM-3:00 PM	43

1. Choose available time for entrance examination.

10. Confirmation of Appointment – the applicants must double check their appointment for entrance examination.



The screenshot shows the 'APPOINTMENT FOR ADMISSION' form with the date and time selected as 'Wednesday, February 21, 2024 10:00 AM'. The 'Confirm the information is correct' checkbox is now checked. A blue circle with the number '1' and an arrow points to the checkbox. The 'Update' button is still present, with a blue circle with the number '2' and an arrow pointing to it.

1. Click box to confirm that the information is correct.
2. Click update button to successfully save your appointment for entrance examination.

11. Applicant's Profile - the profile of the students after they fill out all the requirements needed such as, personal data, medical information, admission data, and last appointment scheduling. This interface will also appear on the admin interface.



The screenshot shows the 'APPOINTMENT FOR ADMISSION' interface for an applicant. It features a sidebar with 'Profile Background' and 'Educational Background' sections. The main area displays 'Admission' and 'Appointment' details. A user profile card on the right includes a 'Settings' button (4), a 'Log Out' button (3), and an 'Edit Profile' button (1). A 'Download PDF' button (2) is located below the profile card. The user's name is Tuyay, Mary Joy Erasga, and their email is tuyaymaryjoy1@gmail.com.

Profile Background	
Name	Tuyay, Mary Joy Erasga
Sex	Female
Religion	Born Again Christian
Date of Birth	July 24, 2002
Address	Balatan, Camarines Sur
Phone No.	09052498592
Email	tuyaymaryjoy1@gmail.com

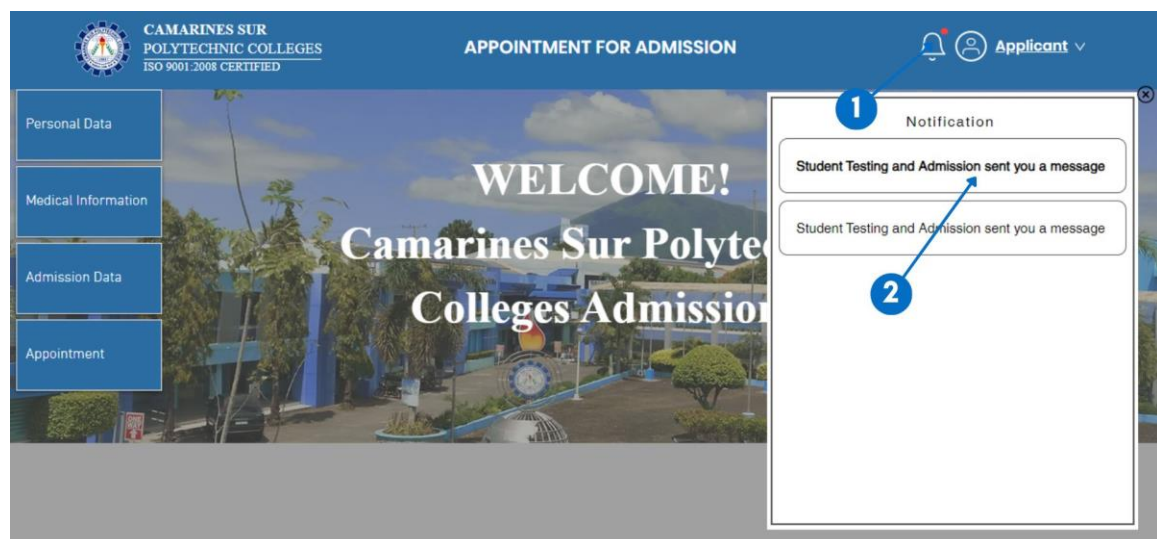
Educational Background	
Name of School	Don Gregorio O. Balatan ...
Address of School	Balatan, Camarines Sur
Strand	T.V.L.
Year Graduated	2019

Admission	
Admission for	College
☒ Freshmen	☐ Transferee
Preferred Course	BS Nursing
Second Course	BS Information Technology
Third Course	BS Information System

Appointment	
Transaction No.	1
Date	February 21, 2024
Time	10:00 AM-12:00 PM
Status	Not Complete
Result	N/A

1. Click "Edit Profile" if you want to edit your previous actions.
2. Click "Download PDF" to download your appointment slip.
3. Click Log out after using the Online Appointment Management.
4. Click settings if they want to delete their account.

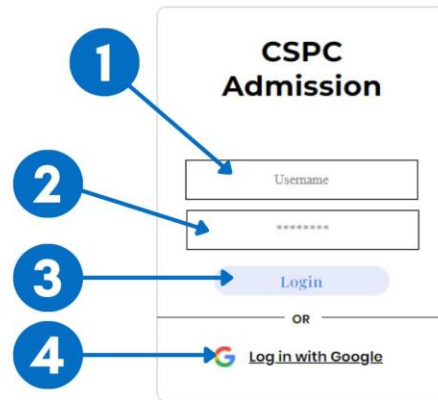
12. Notification for Admission Result - the notification for admission result, in this figure the applicants will be notified through message from STA by clicking the notification bell, this message will be a big help for the applicants to be updated about their admission result.



The screenshot shows the 'APPOINTMENT FOR ADMISSION' interface with a notification bell icon (1) in the top right corner. A notification box is open, displaying a message from 'Student Testing and Admission' (2). The background features a 'WELCOME! Camarines Sur Polytechnic Colleges Admission' banner.

1. Click the notification bell.
2. Click the message sent by the STA and check the update for approval or exam result.

13. Login Page (Admin) - this is the login page wherein the admin will input their cspc email and input password, after that the admin will click the login button, they can also login with google.

The login form is titled "CSPC Admission". It contains four numbered steps: 1. Username input field, 2. Password input field (masked with dots), 3. Login button, and 4. OR Log in with Google button.

1. Input username.
2. Input password.
3. Click login button.
4. The admin can also click the login with google.

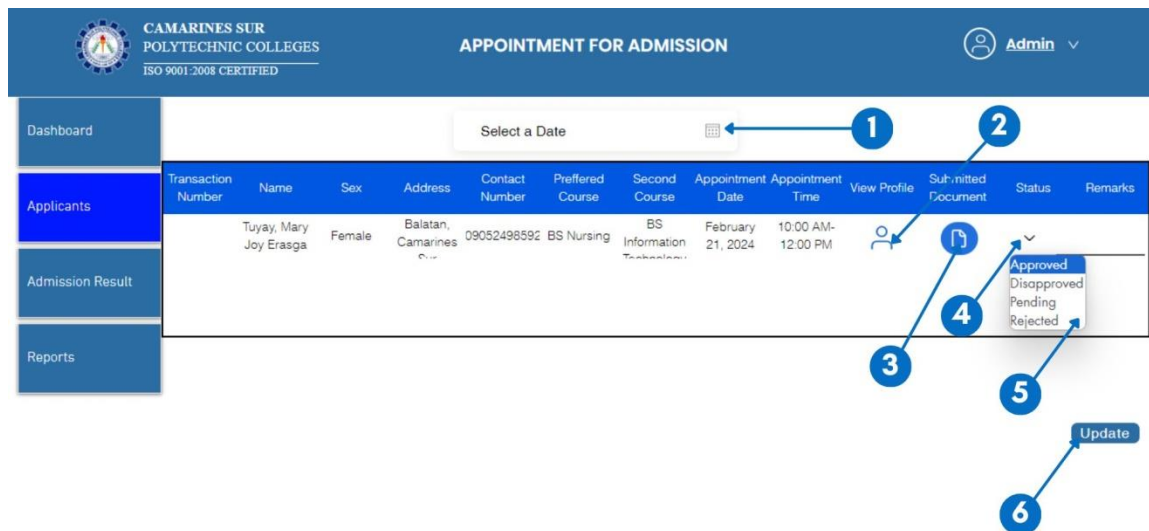
13. Dashboard (Admin) - the total examinees of the students throughout the admission date and the percentage of the program that the students applied, this will show the difference of number of the students who applied for the admission for the upcoming entrance examination within the



organizations.

1. Total Examinees.
2. Examinees this month.
3. Incoming Examinees.

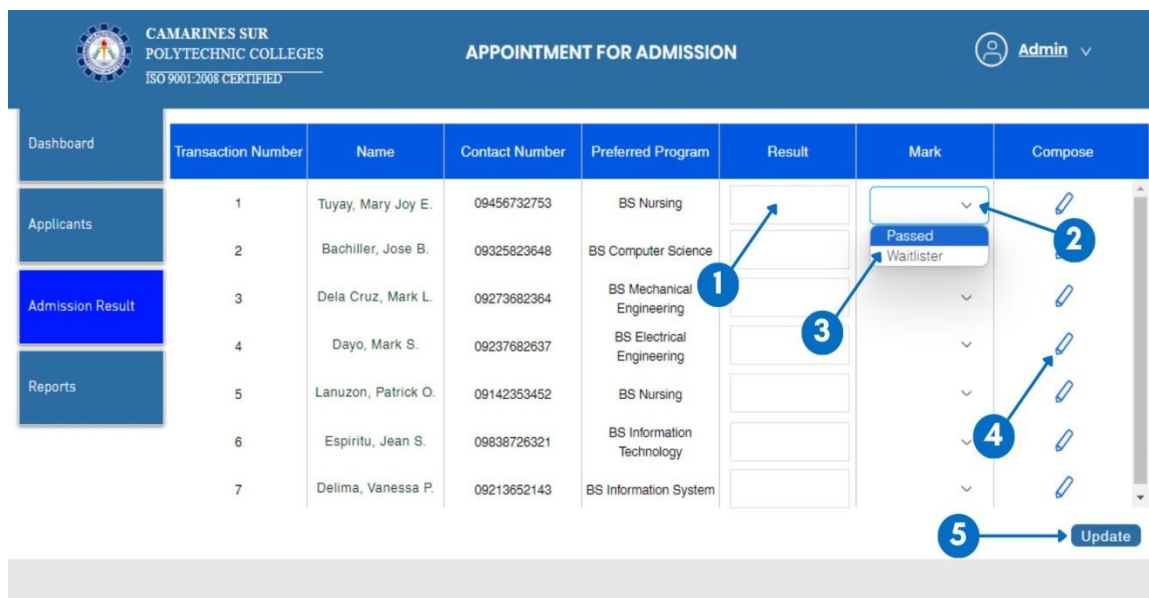
14. Applicants List - shows the applicant's list, wherein the admin can monitor, recording, change the status of the students and also evaluate the requirements of the applicant, this will endure to address reducing downtime and improving overall user satisfaction.



The screenshot shows the 'APPOINTMENT FOR ADMISSION' interface. The left sidebar contains 'Dashboard', 'Applicants', 'Admission Result', and 'Reports'. The main table has columns: Transaction Number, Name, Sex, Address, Contact Number, Preferred Course, Second Course, Appointment Date, Appointment Time, View Profile, Submitted Document, Status, and Remarks. A callout menu for the Status column shows options: Approved, Disapproved, Pending, and Rejected. Numbered callouts indicate: 1. Calendar icon for selecting a date; 2. View Profile icon; 3. Submitted Document icon; 4. Status dropdown menu; 5. Status menu options; 6. Update button.

1. Click calendar button and choose specific date.
2. The admin can also view the profile of the applicant by clicking the icon.
3. Click the documents button to evaluate the requirements of applicants.
4. Click dropdown button.
5. Choose status for approval and click.
6. Click Update.

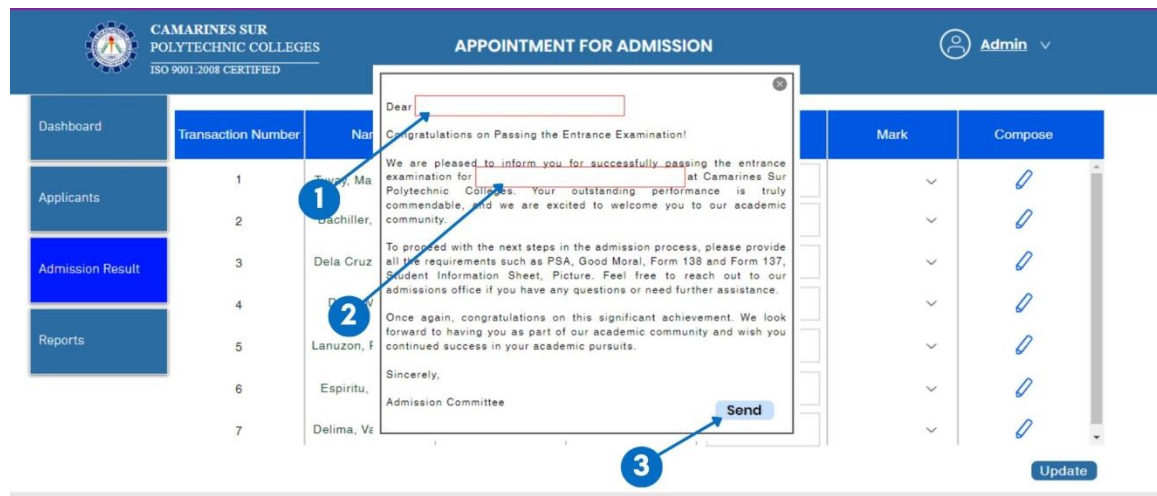
15. Admission Results - the admission result, the figure shows the list of applicants. The admin can manage the examination result of the applicant who has already taken the entrance examination.



The screenshot shows the 'APPOINTMENT FOR ADMISSION' interface with the 'Admission Result' tab selected. The table has columns: Transaction Number, Name, Contact Number, Preferred Program, Result, Mark, and Compose. Numbered callouts indicate: 1. Result column; 2. Mark dropdown menu; 3. Mark menu options (Passed, Waitlist); 4. Compose column; 5. Update button.

1. Input entrance examination result.
2. Click dropdown to choose status for examination.
3. Choose the status and click.
4. Click the compose button to send message to the applicants.
5. Click Update.

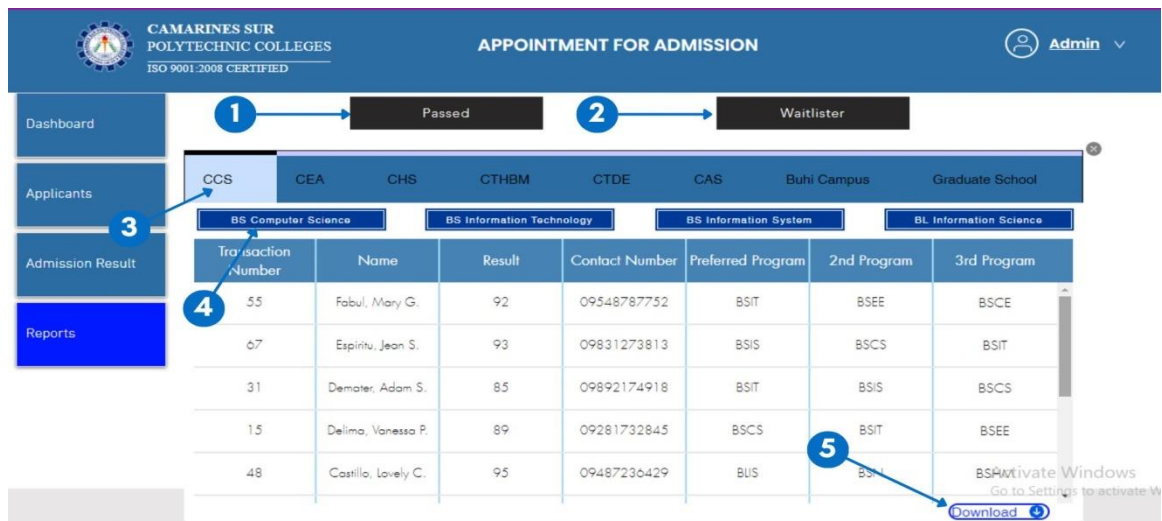
16. Admission Letter – the admission letter that the admin will send to the applicants for them to be notified regarding their examination result if there passed the entrance examination or they are belong to waitlisted and also indicated in the letter all the requirements that needed to submit in STA offices.



The screenshot shows the 'APPOINTMENT FOR ADMISSION' interface. On the left is a sidebar with 'Dashboard', 'Applicants', 'Admission Result', and 'Reports'. The main area has a table with columns: Transaction Number, Name, Mark, and Compose. A modal window titled 'Dear [Name]' is open, showing a congratulatory message and a 'Send' button. The 'Compose' column in the table has a 'Compose' button. The 'Update' button is at the bottom right.

1. Input the name of the applicants who passed for entrance examination.
2. Input the preferred program.
3. Click send

17. Admission Reports - the admission reports that the admin can view the list of the applicants who already taken the entrance examination, this figure shows the Camarines Sur Polytechnic Colleges academic and the programs.



The screenshot shows the 'APPOINTMENT FOR ADMISSION' interface. The 'Admission Result' section is active, showing a table of applicants. The table has columns: Transaction Number, Name, Result, Contact Number, Preferred Program, 2nd Program, and 3rd Program. The table is filtered by 'Passed' status. The 'Download' button is at the bottom right.

Transaction Number	Name	Result	Contact Number	Preferred Program	2nd Program	3rd Program
55	Fabul, Mary G.	92	09548787752	BSIT	BSEE	BSCE
67	Espiritu, Jean S.	93	09831273813	BSIS	BSCS	BSIT
31	Demater, Adam S.	85	09892174918	BSIT	BSIS	BSCS
15	Delima, Vanessa P.	89	09281732845	BSCS	BSIT	BSEE
48	Castillo, Lovely C.	95	09487236429	BUS	BSIT	BSCS



1. Click passed to see the list of applicants who passed the entrance exam.
2. Click waitlister to see the list of applicants who is waitlister.
3. Click CCS to see the list of all the passers under CCS Department.
4. Click BS Computer Science to see the list of applicants who passed under the BS Computer Science program.
5. Click download to have the copy of the reports.

APPENDIX D
ACM FORMAT

**ANALYSIS AND DESIGN FOR ONLINE APPOINTMENT MANAGEMENT IN
CAMARINES SUR POLYTECHNIC COLLEGES ADMISSION OFFICE**

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Abstract

An Online Appointment Management is a software application to well organize the admission process for educational institutions such as schools, colleges, and universities. The Camarines Sur Polytechnic Colleges Admission Office already has an existing process where applicants will submit application form and schedule entrance examination. This study was conducted to analyze and design Online Appointment Management in Camarines Sur Polytechnic Colleges Admission Office. In general, the objective is to design user interface of online appointment management and database. To address the objectives of the study, the researchers used a mixed method research design and the design thinking framework. Data were obtained through interviews and survey questionnaires. The study respondents were the students of Camarines Sur Polytechnic Colleges and administrators in Admission Office. The data was evaluated and interpreted using a weighted mean. Due to problems encountered with the time consuming, duplication of applicant's name, lack of automation in appointment confirmations, delays and errors because of high volume of applicants and complex application process. To address this, the

study designed the user interface and Database. Thus, the overall interpretation revealed that the designed user interface and database satisfied the respondents. As a result, designing of the User Interface and Database will speed up the appointment process, avoid duplication of applicant's name and provide better experience for applicants and admission staffs.

CCS CONCEPTS

Information system- Information System
Application- Information Processing System

KEYWORDS

Analysis, Design, Online appointment management, Admission, User Interface, Database.

1.INTRODUCTION

Efficient and simplified procedures are critical in admission offices of educational institutions given the continually changing landscape of educational technology. This project, "Analysis and Design for Online Appointment Management in Camarines Sur Polytechnic Colleges Admission Office," aims to solve the problems caused by the protracted appointment process. The project intends to design an online appointment



management that improves the overall experience for students, parents, and admission personnel in order to introduce the newly design appointment process in an era marked by digital change.

This project's main goal is to research, evaluate, and create an online appointment management system that is specifically suited to the requirements of the admission office of Camarines Sur Polytechnic Colleges. Through the identification of current bottlenecks in the online application process, namely related to arranging appointments and receiving and disseminating results, the project seeks to facilitate the development of a more adaptable, effective, and user-friendly system. The study also includes a review of the database design based on data integrity, normalization, and scalability, as well as a usability evaluation of the user interface using Nielsen's Heuristic Evaluation. The expected advantages include higher precision, lower expenses, more convenience, real-time tracking, and the timely production of reports, all of which will eventually help to improve the admissions procedure in the educational institution.

2. BACKGROUND

The necessity for streamlined and effective administrative procedures in admission offices has grown in light of the ever-changing landscape of educational technology. Inefficiencies, a lack of communication, and a worse than ideal user experience for both administrative personnel and potential students are frequently caused by the current process and its laborious procedures. One ground-breaking approach to overcoming these obstacles is the incorporation of online appointment scheduling. A global trend in the use of these systems is revealed by the literature study, which highlights how they can improve communication, lessen administrative work, and offer an easy-to-use interface for appointment scheduling.

The main objective is to replace antiquated procedures with a modern online appointment scheduling method, which will increase educational institutions' general efficacy and efficiency, especially in admission offices.

Furthermore, the synthesis of literature underscores the multifaceted benefits of implementing online appointment management in the context of admission offices. Notably, these systems provide effective record keeping, prompt communication, improved user experiences, and heightened institutional competitiveness. The proposed online appointment management system for the Camarines Sur Polytechnic Colleges Admission Office seeks to close current gaps and address serious flaws in the admissions process by utilizing insights from both domestic and international research. The synthesis lays the groundwork for the upcoming phases of analysis and design, highlighting the potential of digital solutions to transform administrative procedures, enhance student experiences, and help educational institutions become more flexible in the age of technological advancement.

3. METHODOLOGY

3.1 Design Thinking Framework

The benefit of applying design thinking in the Analysis and Design for Online Appointment Management in Camarines Sur Polytechnic Colleges Admission Office is that it puts the needs of the user first. It is critical to understand the needs of the students, parents, and staff who will use an online appointment management while building one. This can be accomplished by user interviews, monitoring users in their environment, or developing personas. You may begin to develop a system that satisfies the needs of the user once you have a thorough grasp of them [1].

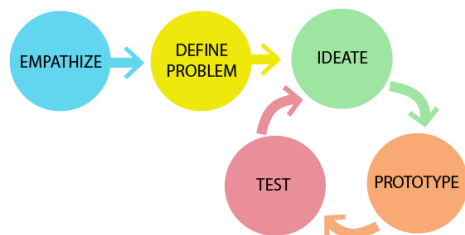


Figure 1: Design Thinking Framework

Figure 1, further presents components of the framework to be adopted

3.2 Materials and Statistical Tools

Detailed information on the research design and research method in reference to the objective of the study will be discussed in this section.

3.2.1 Research Design

The researchers will employ mix method research design. This research design is applicable to this study as it specifies objectives that suggest both descriptive and quantitative forms of research. The study will also try to assess the level of performance of existing and newly designed systems in the record section in terms of speed and accuracy which are both quantitative and descriptive data. Further, this study sought to compare the existing and new proposed system in the record section and determine its significant difference in terms of speed and accuracy. To implement this mix method approach, the researcher will conduct an interview to the students and staff of the admission office on the problems encountered in terms of document storage and retrieval, and on the evaluation of the existing online application on its level of performance in terms of speed and accuracy.

3.2.2 Research Method

This section will discuss the detailed implementation of the mixed-method approach to achieve the specific objective of the study. To conduct specific research or find a reliable information, it involves using

a set of instruments and approaches. In this part, the proponents employed instruments, selected respondents and utilize data collection. One of the primary sources of information is the staff, who include the admission office such as the admission staff, and students. The study's respondents will be selected base on their knowledge and purposive sampling methods to ensure diverse perspectives.

3.2.2.1 Instruments

The researchers will employ an interview guide and survey questionnaires to gather data from the respondents and evaluate the proposed designs for future users. The interview guide will be used to address specific objective 1 of the study, while the survey questionnaire will address specific objectives 3. These instruments were prepared in this study to align with the research model, focusing on user satisfaction regarding the proposed system's content, instructional information, interaction, adaptation, technology quality, usefulness, and overall satisfaction with the proposed design system.

The table 1, present the Likert scale allows respondents to indicate their level of agreement or preference using the provided scale.

Table 1
5-POINT LIKERT SCALE FOR EVALUATING USER INTERFACE AND DATABASE DESIGN

Scale	Range	Level of Agreement	Description
5	4.21 - 5.00	Strongly Agree	The participant strongly supports or agrees with the statement that the User Interface and Database is well-designed
4	3.21 - 4.20	Agree	Implies a positive stance toward the statement or question. The respondent agrees with the presented User Interface and Database Design.
3	2.61 - 3.20	Neutral	The respondent has neither a positive response nor a negative response, but undecided denotes a state of confusion of the respondent.
2	1.81 - 2.60	Disagree	Suggests a level of disagreement with the statement or question, but not as strong as Strongly Disagree.
1	1.00 - 1.80	Strongly Disagree	Indicates a strong and definitive disagreement with the statement or question. The respondent strongly opposes or disagrees with the presented User Interface and Database Design.

Table 1, shows the provided the Likert scales, indicating the level of quality or

performance of the proposed design system along with its corresponding description.

3.2.2.2 Data Gathering and Procedure

Interview, and Survey Testing will be used to gathered information. Through interview the researchers engaging to the admission staff to analyzing the specific objective 1 to determine the current appointment process that exist in their organizations to understand the study process and identify areas for improvements. To validate the observation the interviews with admission staff in the field will be conducted to observe the journey of beneficiaries in the program seeking a comprehensive response to gain a clear understanding about their appointment. Additionally, Survey questionnaire used to evaluate the proposed solution to the main clients of the study, researchers distribute a survey questionnaire after evaluating the prototype design to the IT experts. Moreover, we used concept testing to identify flows or improvements, and ensure that the prototypes align with the needs and preferences of the intended users. Gathering data to main respondents will be use to improve the design system that will help in their organization particularly in their current appointment process.

3.2.2.3 Sampling techniques

A convenience sample strategy will be utilized to collect comments and insights for the Analysis and Design at Student Testing and Admission. a non-probability sampling method where units are selected for inclusion in the sample because they are the easiest for the researcher to access [3].

3.2.2.4 Statistical Test

The researchers will utilize the following statistical measures to analyze the gathered data which is the weighted means and thematic analysis.

4. RESULTS AND DISCUSSION

This presented on how the project was performed, delivered and completed in

reference to the specific objectives enumerated in this study. And also presented the results and findings of this study.

4.1 Design Thinking

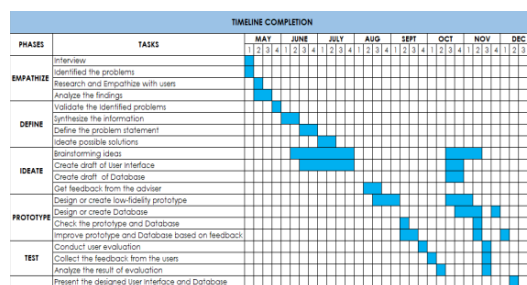
In Design thinking, we present a problem-solving framework for Admission staffs regarding the appointment process. Base on the data gathered and requirements specifications, the researchers discovered that the existing process is time consuming, Lack of automation in appointment confirmations. The goal of this study is to identify and address the challenges faced by the admission staffs in their current appointment through the application of design thinking principles.

4.1.1 Empathize

In this section, the researchers comprehensively discuss and elucidate the structure and content of the study, outlining both the form and substance of the proposed system design tailored for the target end-users and clients of the systems.

4.1.1.1 Gantt Chart

The researchers employed a Gantt chart to establish the project's time schedule, effectively managing the allocated time required to complete the proposed design. The phases of the Design Thinking framework, namely empathize, define,



ideate, prototype, and testing, were incorporated into the schedule, aligning with the researchers' approach.

Figure 2 shows the graphical presentation of the schedule time and duration for each task to be done. Gantt chart used as a reference

for scheduling tasks that must be completed and submitted on time. It depicts the researcher's objectives for the period allocated which must be met to meet the criteria.

4.1.2 Define

In this phase, it shows the table and interpretation of the identified problems that the researchers gathered during the interview to achieve the specific objective 1: To determine the problems of the existing process in Online Application for all incoming students.

4.1.2.1 Identified Problems and Solutions

In these sections, researchers have identified the challenges face by the admission staff during the appointment process, particularly focusing on issues arising from their current operations, which serves as the primary obstacles in collecting data information about appointment.

Table 2

IDENTIFIED PROBLEMS AND PROPOSED SOLUTIONS

Identified Problems	Description	Proposed Solutions
Time consuming	Delays in appointment scheduling, reduced efficiency	Design an easy-to-use online appointment scheduling system that simplifies the booking process for applicants and automates scheduling tasks.
Duplication of Applicant's Name	Data inconsistencies, potential confusion	To resolve the issue of duplicate applicant data, we will set up a centralized database. This database will securely store applicant information and eliminate any unnecessary duplication.
Lack of automation in appointment confirmations	The risk of missed appointments arises when students are uncertain about their schedules due to a delay in administrators confirming them promptly.	Integrate email or SMS reminders into the system to reduce the need for manual follow-ups and confirmations.
Handling a High Volume of Applicants	The existing manual appointment process faces challenges in handling a high volume of applicants, leading to delays and the possibility of errors in the scheduling of appointments.	Design an automated appointment scheduling system capable of effectively handling a substantial influx of applicants. This system will assign appointment slots by considering both staff availability and applicant preferences, thereby reducing scheduling conflicts and minimizing delays.
Complex Application Process	The current application process is intricate and burdensome, resulting in confusion among applicants and potentially causing some to abandon the process.	Simplify the application process by designing a user-friendly online portal that features straightforward instructions and easy-to-navigate menus. Offer real-time assistance to applicants, incorporate a progress tracking feature, and ensure seamless uploading and verification of necessary documents directly within the system.

Table 2 highlights significant problems within the present appointment scheduling and application procedure that can have a negative impact on system efficiency and effectiveness. The first issue, time-consuming appointment scheduling that causes delays and lower efficiency, can be

solved by using an easy-to-use online appointment scheduling system. This proposed solution intends to streamline the appointment process for applicants and automate scheduling chores, hence speeding up the process and lowering the possibility of human error.

4.1.2.2 Fishbone Diagram

The Fishbone Diagram serves as an analytical tool, offering a systematic approach to examining the causes that lead to specific effects. Due to its role in illustrating cause-and-effect relationships, it is commonly known as cause-and-effect diagram.

In this section fishbone diagram as a root cause of the specific problem researchers illustrate it in a graphical representation.

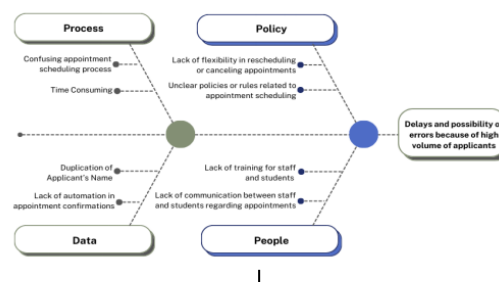


Figure 3: Fishbone Diagram

Figure 3: Fishbone Diagram

The fishbone diagram presented focuses on systematically identifying and categorizing challenges associated with managing a high volume of applicants that caused delays and errors within an online appointment management. Within the outlined categories, specific issues are brought to light. In the field of Process, issues include a Confusing Appointment Scheduling Process, where the lack of clarity leads to confusion among applicants, potentially resulting in errors, missed appointments, or delays. Additionally, the process is noted to be Time Consuming, posing a risk of frustration for both applicants and staff due to prolonged procedures.

4.1.3 Ideate

Our vision is to create a low-fidelity prototype for an online appointment management specific to the demands of the Admission Office.

4.1.3.1 Flowchart of Existing Process

This section presented the existing appointment process.

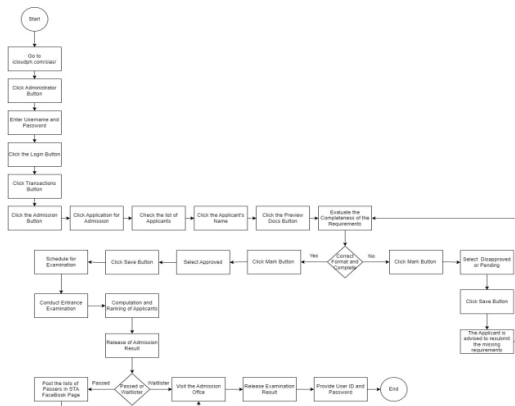


Figure 4: Flowchart of Existing Process (Admin)

Figure 4 shows the flowchart of the existing process for staff in the admission office. The outlined flowchart describes a systematic and structured admission process for prospective staff.

The figure below shows the flowchart of the current process for the applicant interface of online application in Admission Office. This figure shows the step-by-step process.

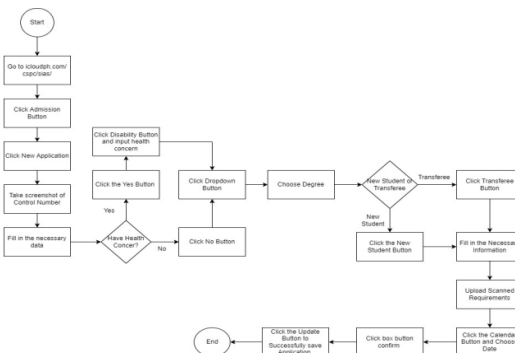


Figure 5: Flowchart of Existing Process (Applicant)

Figure 5 shows the flowchart of the existing process for student in the admission office.

4.1.3.2 Flowchart of New Process

This section presented the new appointment process

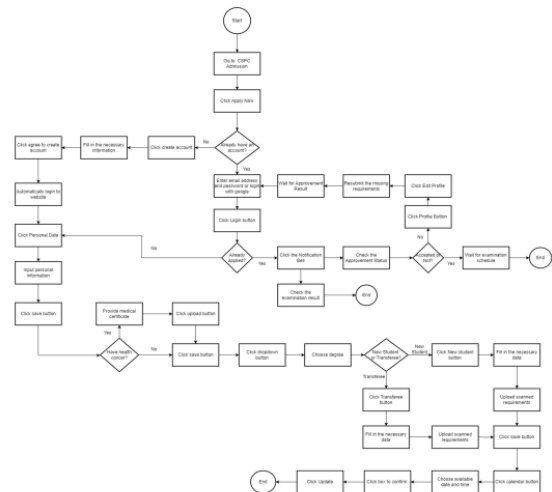
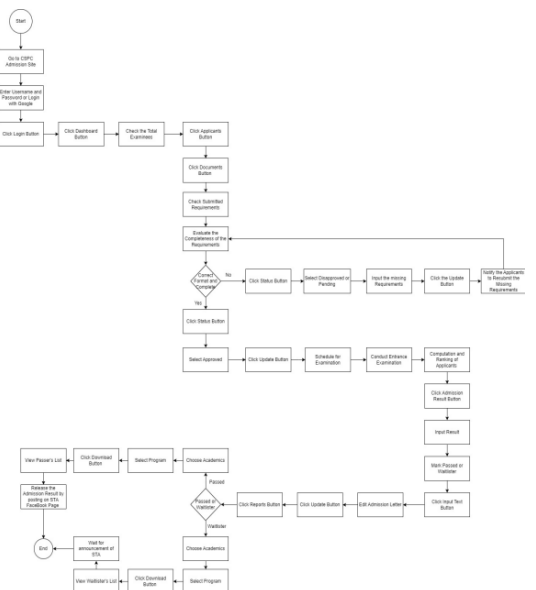


Figure 6: Flowchart of Online Appointment Process (Applicant)

Figure 6: Flowchart of Online Appointment Process (Applicant)

Figure 6 represents the flowchart of the proposed online appointment management process in CSPC Admission Office.



The next figure represents the flowchart of the proposed online appointment management in process in Camarines Sur Polytechnic Colleges Admission Office. This flowchart is for the staffs' interface, this is separated to the student interface.

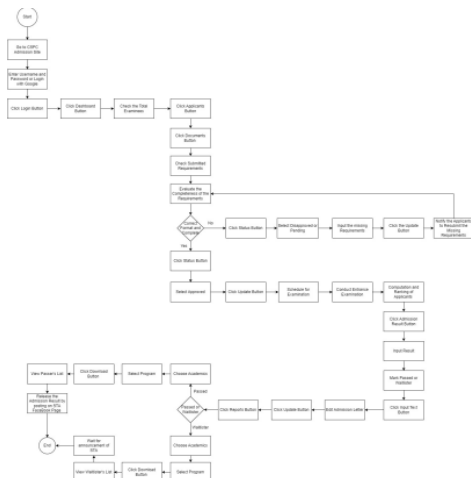


Figure 7: Flowchart of Online Appointment Process (Admin)

In this figure, it shows the flowchart of the proposed online appointment management in Camarines Sur Polytechnic Colleges Admission Office.

4.1.4 Prototype

In this phase, the researchers move on to the visualization and testing of the proposed solution's design and functionality.

4.1.4.1 Entity Relationship Diagram

This section presented the Entity relationship Diagram of the User Interface design application.

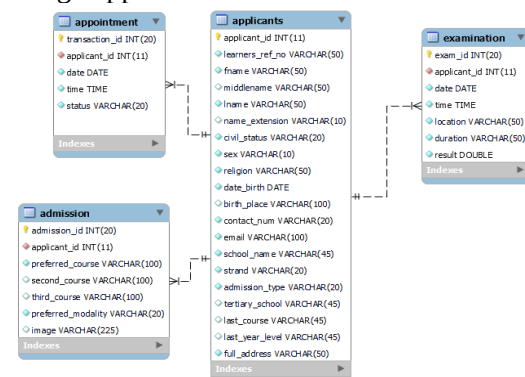


Figure 26: Entity Relationship Diagram

Figure 26 shows the entity relationship diagram which shows the relationship of entities that can help to visualize the structure and logic of a database of the low-fidelity prototype. also known as an entity relationship model, is a graphical representation that depicts relationships among people, objects, places, concepts or events within an information technology (IT) system [4].

4.1.4.2 Data Dictionary

Data dictionary systems serve as databases responsible for storing and overseeing data related to the configuration and utilization of information resources.

Table 3

APPLICANT DATA DICTIONARY

FIELD NAME	DESCRIPTION	FIELD TYPE	FIELD SIZE	CONSTRAINTS
applicant_id	Primary Key: Stores the ID assigned for every student	int	11	Null
learners_ref_no	Stores the L.R.N of the student	varchar	50	Null
fname	Stores the first name of the student	varchar	50	Not null
middlename	Stores the middle name of the student	varchar	50	Not null
lname	Stores the last name of the student	varchar	50	Not null
name_extension	Stores the extension name of the student	varchar	10	Null
civil_status	Stores the civil status of the student	varchar	20	Not null
sex	Stores the sex of the student	varchar	10	Not null
religion	Stores the religion of the student	varchar	50	Not null
date_birth	Stores the date of birth of the student	date		Not null
birth_place	Stores the birth place of the student	varchar	100	Not null
contract_num	Stores the contract number of the student	varchar	20	Unique key
email	Stores the email of the student	varchar	100	Unique key
school_name	Stores the school name of the student	varchar	45	Not null
strand	Stores the strand of the student	varchar	20	Not null
admission_type	Stores the admission type of the student	varchar	20	Not null
tertiary_school	Stores the tertiary of the student	varchar	45	Not null
last_course	Stores the last course of the student	varchar	45	Not null
last_year_level	Stores the last year level of the student	varchar	45	Not null
full_address	Stores the address of the student	varchar	50	Not null

The applicant data table data dictionary was shown in Table 3. The table serves a distinct purpose in organizing and managing comprehensive information related to applicants.

Table 4

APPOINTMENT DATA DICTIONARY

FIELD NAME	DESCRIPTION	FIELD TYPE	FIELD SIZE	CONSTRAINTS
transaction_id	Primary Key: Stores the transaction ID	Int	20	Not null
applicant_id	Foreign Key: Stores the student ID	Int	11	Not null
date	Stores the date of appointment	Date		Not null
time	Stores the time of appointment	time		Not null
status	Stores the status of appointment	Varchar	20	Not null

The appointment table data dictionary was shown in Table 4. The table serves a distinct purpose in organizing and managing comprehensive information related to appointment.

Table 5

EXAMINATION DATA DICTIONARY

FIELD NAME	DESCRIPTION	FIELD TYPE	FIELD SIZE	CONSTRAINTS
exam_id	Primary Key. Stores the exam ID	Int	20	Notnull
applicant_id	Foreign Key. Stores the student ID	Int	11	Notnull
date	Stores the date of examination	Date		Notnull
time	Stores the time of examination	time		Notnull
location	Stores the location of examination	Varchar	50	Notnull
duration	Stores the duration of examination	varchar	50	Notnull
result	Stores the result of examination	double		Notnull

The appointment table data dictionary was shown in Table 5. The table serves a distinct purpose in organizing and managing comprehensive information related to examination.

Table 6

ADMISSION DATA DICTIONARY

FIELD NAME	DESCRIPTION	FIELD TYPE	FIELD SIZE	CONSTRAINTS
admission_id	Primary Key. Stores the admission ID	Int	20	Notnull
applicant_id	Foreign Key. Stores the student ID	Int	20	Notnull
preferred_course	Stores the preferred course of the student	Varchar	100	Notnull
second_course	Stores the second course of the student	Varchar	100	Notnull
third_course	Stores the third course of the student	Varchar	100	Notnull
preferred_modality	Stores the preferred modality of the student	varchar	20	Notnull
image	Stores the image of the student	varchar	225	Notnull

The admission table data dictionary was shown in Table 6. The table serves a distinct purpose in organizing and managing comprehensive information related to admission.

4.1.4.3 User Interface Design

The user interface will discuss the design system of the study. The developers describe the interface stages of the system designs to be the easiest to discover and use, as the most significant advancement in the system development. The various forms of data collection will be utilized to illustrate the user interface design of the online appointment management system. In the study they have two modules, the admission staff and applicants that plays a pivotal role

in the design system ensuring seamless operation and efficient management.

4.1.5 Test phase

In the test phase of the project, a crucial step involved evaluating a low fidelity prototype. The prototype was thoughtfully crafted to reflect the essential features and user interface envisioned for the online appointment management of admission office, deliberately keeping a simplified design.

This table presented the result of evaluation of the proposed prototype to the respondents of CSPC.

Table 7

USER INTERFACE EVALUATION RESULT

STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
I. USER INTERFACE		
A. Visibility of system status		
1. The prototype effectively conveys the simulated status and feedback of the visualizing system.	4.3	Strongly Agree
2. I can easily interpret and understand the status indicators presented in the prototype.	4.73	Strongly Agree
3. The prototype provides clear and concise information about the simulated processes and activities.	4.2	Agree
4. The status updates within the prototype align well with what I would expect from a fully developed system.	4.3	Strongly Agree
5. Even in this prototype stage, the system status is visibly represented, contributing to a sense of user awareness.	4.16	Agree
OVERALL MEAN	4.33	Strongly Agree

Table 7 shows the result of evaluation for user interface in terms of visibility of system status. The study of the prototype's system status visibility yielded very favorable results regarding user response.

Table 8

USER INTERFACE EVALUATION RESULT

STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
B. Match between system and the real world		
1. The prototype uses terminology that aligns well with real-world language and concepts.	4.56	Strongly Agree
2. The actions and interactions in the prototype are closely similar to real-world scenarios.	4.36	Strongly Agree
3. The prototype's interface effectively copies real-world objects or processes, creating a sense of familiarity.	4.56	Strongly Agree
4. I find that the prototype's commands and functionalities are consistent with what I would expect in a real-world context.	4.6	Strongly Agree
5. Interacting with the prototype feels natural, and it accurately reflects how I would perform tasks in the real world.	4.3	Strongly Agree
OVERALL MEAN	4.47	Strongly Agree

Table 8 shows the result of evaluation for user interface in terms of match between system and the real world. The results of the evaluation of how well the system matches the real-world point to a very positive user experience.

Table 9
USER INTERFACE EVALUATION RESULT

STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
C. User control and freedom		
1. The prototype provides clear options and pathways, giving me a sense of control over my interactions.	4.16	Agree
2. I feel that the prototype allows me the freedom to navigate and explore features according to my preferences.	4.33	Strongly Agree
3. The prototype offers sufficient user guidance and feedback, empowering me to make informed choices.	3.9	Agree
4. I find it easy to backtrack actions within the prototype, enhancing my perceived control and freedom.	4.6	Strongly Agree
5. I feel a sense of freedom and flexibility in how I can interact with the simulated system.	4.53	Strongly Agree
OVERALL MEAN	4.30	Strongly Agree

Table 9 shows the result of evaluation for user interface in terms of user control and freedom. An evaluation of the prototype's user freedom and control indicates a satisfying experience for the user.

Table 10
USER INTERFACE EVALUATION RESULT

STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
D. Consistency and Standards		
1. The elements and interfaces within the prototype follow consistent design patterns and standards.	4.6	Strongly Agree
2. I find that the prototype sticks to established conventions, making it easy to predict the behavior of different components.	4.16	Agree
3. The use of terminology and symbols in the prototype aligns with industry or user interface standards.	4.5	Strongly Agree
4. The interactions and workflows within the prototype maintain a high level of consistency and conformity.	4.3	Strongly Agree
5. The design and structure demonstrate a commitment to consistency and adherence to established standards.	4.1	Agree
OVERALL MEAN	4.33	Strongly Agree

Table 10 shows the result of evaluation for user interface in terms of consistency and Standards. The results of the evaluation of the prototype's consistency and standards show that it adheres closely to accepted design patterns and industry standards.

Table 11
USER INTERFACE EVALUATION RESULT

STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
E. Recognition rather than Recall		
1. The prototype presents information in a way that allows me to recognize and select options without having to recall specific details.	4.16	Agree
2. I find that the prototype's interface includes visible signs that aid in recognizing available actions or features.	4.23	Strongly Agree
3. The prototype's design minimizes the need for me to remember complex sequences of steps; instead, options are easily recognizable.	4.36	Strongly Agree
4. I can easily identify and understand icons, labels, or symbols within the prototype, reducing the cognitive load on recall.	4.6	Strongly Agree
5. The prototype's layout and structure promote recognition of relevant information, minimizing the need for users to remember specific details.	4.3	Strongly Agree
OVERALL MEAN	3.48	Agree

Table 11 shows the result of evaluation for user interface in terms of recognition rather than Recall. Users generally agreed on the effectiveness of the interface in promoting

recognition, according to the findings from the assessment of recognition rather than recall in the prototype, indicating a favorable user experience.

Table 12
USER INTERFACE EVALUATION RESULT

STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
F. Flexibility and efficiency of use		
1. The prototype allows for customization and adapts well to different user preferences, enhancing flexibility.	4.63	Strongly Agree
2. I feel that the prototype's design enables me to perform tasks efficiently, with minimal unnecessary steps or interactions.	4.16	Agree
3. The prototype provides shortcuts or advanced features for experienced users, contributing to efficiency.	4.66	Strongly Agree
4. I find that the prototype supports a range of user abilities and skill levels, offering a flexible and inclusive experience.	4.13	Agree
5. This prototype stage, I recognize the balance between flexibility and efficiency, making it adaptable to various user needs.	4.23	Strongly Agree
OVERALL MEAN	4.36	Strongly Agree

Table 12 shows the result of evaluation for user interface in terms of flexibility and Efficiency of use. The results of the assessment of the prototype's adaptability and usability point to a very favorable user impression.

Table 13
USER INTERFACE EVALUATION RESULT

STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
G. Aesthetic and Minimalist design		
1. The prototype exhibits a visually appealing and aesthetically pleasing design.	4.53	Strongly Agree
2. I appreciate the use of minimalist elements in the prototype, contributing to a clean and uncluttered interface.	4.6	Strongly Agree
3. The color scheme and visual elements in the prototype are thoughtfully chosen, enhancing overall aesthetics.	4.6	Strongly Agree
4. The prototype's design prioritizes simplicity and clarity, aligning with a minimalist approach.	4.56	Strongly Agree
5. I find the design to be both visually pleasing and functionally minimalist.	4.5	Strongly Agree
OVERALL MEAN	4.55	Strongly Agree

Table 13 shows the result of evaluation for user interface in terms of aesthetic and Minimalist design. The results of the evaluation of the prototype's minimalistic and beautiful design show that users have responded very well to it.

Table 14
SUMMARY OF THE EVALUATION RESULT OF USER INTERFACE

STATEMENTS	OVERALL WEIGHTED MEAN	VERBAL INTERPRETATION
1. Visibility of the system status	4.33	Strongly Agree
2. Match between the system and the real world	4.47	Strongly Agree
3. User control and freedom	4.30	Strongly Agree
4. Consistency and standard	4.33	Strongly Agree
5. Recognition rather than recall	3.48	Agree
6. Flexibility and efficiency of use	4.36	Strongly Agree
7. Aesthetic and minimalist design	4.55	Strongly Agree
OVERALL MEAN	4.26	Strongly Agree

Presented in table 14, the summary of results gathered upon the evaluation of students and

admission staff in Camarines Sur Polytechnic Colleges Admission Office on the User Interface of Online appointment management.

Table 15
DATABASE EVALUATION RESULT

STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
II. DATABASE		
A. Data Integrity		
8. The database design includes proper mechanisms to ensure data integrity (e.g., unique keys, foreign keys).	4.53	Strongly Agree
9. Data integrity constraints are clearly defined and well-implemented in the database.	4.2	Agree
10. Data accuracy and consistency are maintained effectively in the database.	4.2	Agree
OVERALL MEAN	4.31	Strongly Agree

Table 15 shows the result of evaluation for database in terms of data integrity. The results of the assessment of the database design's data integrity show that users have a high degree of confidence.

Table 16
DATABASE EVALUATION RESULT

STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
B. Normalization		
1. The database design applies appropriate normalization techniques to minimize data redundancy.	4.36	Strongly Agree
2. Normalization has been performed without sacrificing data usability.	4.03	Agree
3. The level of normalization applied is suitable for the database's purpose and performance requirements.	4.2	Agree
OVERALL MEAN	4.19	Agree

Table 16 shows the result of evaluation for database in terms of normalization. Positive user perception is indicated by the evaluation of normalization in the database design findings.

Table 17
DATABASE EVALUATION RESULT

STATEMENTS	WEIGHTED MEAN	VERBAL INTERPRETATION
C. Scalability		
1. The prototype provides clear options and pathways, giving me a sense of control over my interactions.	3.9	Agree
2. I feel that the prototype allows me the freedom to navigate and explore features according to my preferences.	4.1	Agree
3. The prototype offers sufficient user guidance and feedback, empowering me to make informed choices.	4.3	Strongly Agree
OVERALL MEAN	4.1	Agree

Table 17 shows the result of evaluation for database in terms of scalability. Participants rated user assistance and feedback slightly higher than other features, but overall, they agreed that the prototype gives users a sense of power and freedom in their interactions.

Table 18
SUMMARY OF THE EVALUATION RESULT OF DATABASE

STATEMENTS	OVERALL WEIGHTED MEAN	VERBAL INTERPRETATION
1. Data integrity	4.31	Strongly Agree
2. Normalization	4.19	Agree
3. Scalability	4.1	Agree
OVERALL MEAN	4.2	Agree

Presented in table 18, the summary of results gathered upon the evaluation of students and admission staff in Camarines Sur Polytechnic Colleges Admission Office on the Database of Online appointment management.

5. CONCLUSIONS

The researchers identified key issues in CSPC's existing admission process, designed and assessed a potential online appointment management solution, and concluded that it holds significant promise for reducing redundancy, and improving user experience for both applicants and staff. The prototype received overwhelmingly positive feedback on its usability, data integrity, and normalization, highlighting its potential for successful implementation. The researchers recommend further development based on these findings, emphasizing secure data storage, user support, and user-centered design for continued optimization. While some limitations remain regarding scalability, the overall outlook for the online appointment management is positive, suggesting it could significantly improve the admission process at CSPC.

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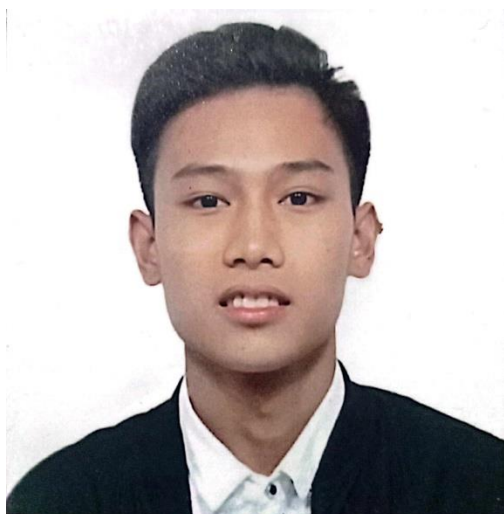
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